

Diffusion and Flow-Based Models for Controllable Image Generation: A Comprehensive Survey of Theory, Mechanisms, and Frontiers

1 Theoretical Foundations and Unified Frameworks

1.1 Core Formulations of Diffusion and Score-Based Generative Models

1.1 Core Formulations of Diffusion and Score-Based Generative Models

The theoretical landscape of modern generative modeling has been fundamentally reshaped by the convergence of Denoising Diffusion Probabilistic Models (DDPMs) and Score-Based Generative Models (SGMs). While initially developed as distinct methodologies, rigorous mathematical analysis has revealed that these approaches are effectively equivalent, differing primarily in their discretization strategies and training objectives. At the heart of this unification lies the concept of transforming complex data distributions into simple prior distributions through a forward noising process, and subsequently recovering the data via a learned reverse process. A comprehensive understanding of these core formulations requires examining their discrete-time origins, their continuous-time limits governed by Stochastic Differential Equations (SDEs), and the critical role of score functions in reversing diffusion dynamics. This foundational framework sets the stage for the deterministic interpretations discussed in the following section, where stochastic trajectories are recast as ordinary differential equations.

In the discrete-time formulation, DDPMs define a Markovian forward process that gradually corrupts data with Gaussian noise over a fixed number of steps. The reverse process is then learned to denoise the data, typically by predicting the added noise at each timestep. Early work demonstrated that with specific modifications, such as learning the variance of the reverse process, DDPMs could achieve competitive log-likelihoods while maintaining high sample quality [1]. However, the computational cost of simulating this Markov chain remained a significant bottleneck, often requiring hundreds or thousands of sequential function evaluations to generate a single high-fidelity sample [2]. To address the inefficiency inherent in standard sampling procedures, researchers have explored various acceleration techniques, including the development of bilateral denoising diffusion models which parameterize both forward and reverse processes to achieve high-quality synthesis in significantly fewer steps [3]. Furthermore, the introduction of non-Markovian diffusion processes allowed for the construction of Denoising Diffusion Implicit Models (DDIMs), which share the same training objective as DDPMs but enable much faster deterministic sampling by skipping intermediate steps [2]. These developments in discrete time foreshadowed the transition to continuous-time formulations that would later unify diffusion with normalizing flows.

Parallel to the development of DDPMs, Score-Based Generative Models emerged with a focus on estimating the gradient of the log-density of the perturbed data, known as the score function. The seminal work by Song et al. established a unified framework by demonstrating that both DDPMs and SGMs can be viewed as discretizations of a continuous-time stochastic process [4]. In this continuous-time perspective, the forward noising process is described by a Stochastic Differential Equation (SDE) that smoothly transforms the data distribution into a known prior, such as a Gaussian distribution. Crucially, the reverse-time SDE, which generates data from noise, depends entirely on the time-dependent score function of the perturbed data distribution. This theoretical insight bridges the gap between the discrete variational lower bound optimization used in DDPMs and the score matching objectives used in SGMs, showing that minimizing the score matching loss

is equivalent to maximizing a lower bound of the likelihood of the reverse SDE [5].

The equivalence between these frameworks is further illuminated by the analysis of the underlying probability flow. The reverse-time SDE provides a stochastic path for generation, but it also implies a corresponding deterministic Ordinary Differential Equation (ODE), often referred to as the Probability Flow ODE. This ODE shares the same marginal probability densities as the SDE at every time step, enabling exact likelihood computation and more efficient sampling strategies [4]. However, empirical observations have long noted a performance gap between samples generated via stochastic SDE solvers and those from deterministic ODE solvers. Recent theoretical investigations attribute this discrepancy to the approximation errors in the neural network’s estimation of the score function and the resulting deviations in the associated Fokker-Planck equations. Specifically, it has been shown that reducing the Fokker-Planck residual through regularization can help close the gap between ODE and SDE induced distributions, although this may involve trade-offs in sample quality [6]. Moreover, the presence of noise in the reverse process has been proven to offer superior approximation capabilities, acting as a regularizer that allows neural SDEs to track trajectories more accurately than their deterministic ODE counterparts, particularly when the underlying score function is not Lipschitz continuous [7]. Understanding these nuances in the stochastic versus deterministic dichotomy is essential for appreciating the rigorous unification presented in the subsequent discussion on Probability Flow ODEs.

The mathematical rigor of these formulations extends to convergence guarantees and error analysis. Theoretical results have established that, assuming accurate score estimates, SGMs can efficiently sample from essentially any realistic data distribution with polynomial-time complexity, even without restrictive assumptions on the data’s log-concavity [8]. Additionally, the asymptotic behavior of these models has been analyzed, proving that specific denoising strategies within DPMs converge to the mean square error-optimal conditional mean estimator as the number of diffusion steps increases [9]. Despite these advances, challenges remain in handling singular data distributions, such as those concentrated on lower-dimensional manifolds, where the analytical mean drift function can asymptotically blow up. To mitigate this, novel target functions and loss formulations have been derived to ensure boundedness and improve approximation stability [10].

Finally, the flexibility of the SDE framework allows for the exploration of diverse forward processes beyond standard Gaussian diffusion. This includes the adaptation of diffusion models to discrete state spaces using Continuous Time Markov Chains, providing a natural approach for modeling intrinsically discrete data like language and graphs [11]. Furthermore, the integration of metric functions and the development of cascaded architectures have enabled the effective incorporation of perceptual losses, leading to state-of-the-art image quality [12]. The formulation of diffusion models as a sequence of restoration tasks also offers an alternative interpretation, linking generative modeling closely with image restoration literature and enabling flexible forward processes grounded in Maximum A Posteriori (MAP) estimation [13]. Collectively, these core formulations provide a robust mathematical foundation that unifies discrete and continuous perspectives, driving the rapid advancement of controllable image generation and paving the way for the deterministic flow-based interpretations that follow.

1.2 Deterministic Sampling via Probability Flow ODEs and Normalizing Flows

1.2 Probability Flow ODEs and the Deterministic Unification with Continuous Normalizing Flows

Building upon the stochastic foundations and SDE formulations discussed in the previous section, the transition to deterministic generative trajectories represents a pivotal theoretical advancement that bridges score-based diffusion models with Continuous Normalizing Flows (CNFs). At the heart of this unification lies the Probability Flow Ordinary Differential Equation (ODE), a deterministic dynamical system that preserves the exact marginal probability density path of its stochastic counterpart, the Stochastic Differential Equation (SDE). While standard Denoising Diffusion Probabilistic Models (DDPMs) rely on iterative noise injection and removal governed by stochastic dynamics, the probability flow ODE offers a rigorous mathematical framework for deterministic sampling. Crucially, this formulation addresses a longstanding limitation of early diffusion models by enabling exact likelihood evaluation, a capability previously exclusive to flow-based architectures. This deterministic perspective not only recasts diffusion as a specific instance of CNFs but also sets the stage for the efficiency-driven Optimal Transport paradigms explored in the subsequent section.

The derivation of the probability flow ODE stems from the observation that the time evolution of the probability density $p_t(x)$, defined by a forward diffusion SDE, can equivalently be described by a continuity equation associated with a specific deterministic velocity field. By equating the Fokker-Planck equation governing the SDE with the continuity equation of an ODE, one can isolate a unique vector field that transports probability mass along the same trajectory as the stochastic process but without random fluctuations. This theoretical insight establishes that diffusion models are, in essence, a specialized form of Continuous Normalizing Flows, where the transformation is parameterized by the score function of the perturbed data distribution. Recent work has solidified this connection, demonstrating that the probability flow ODE of diffusion models is a particular case of CNFs, thereby enabling deterministic inference and exact likelihood evaluation [14]. This equivalence allows researchers to leverage the rich theoretical toolkit developed for normalizing flows, such as change-of-variable formulas, to analyze and optimize diffusion-based generators.

A primary advantage of formulating diffusion generation as a probability flow ODE is the ability to perform exact likelihood estimation, a task that is computationally prohibitive for standard stochastic samplers due to the intractability of the path integral over noise realizations. However, achieving state-of-the-art likelihood scores with diffusion ODEs has historically been challenging due to approximation errors in the learned score functions. To address this, recent advancements have introduced improved techniques for maximum likelihood estimation, including novel parameterizations of the velocity field and variance reduction strategies that accelerate convergence during training. Furthermore, the development of error-bounded high-order flow matching objectives has been instrumental in finetuning these models, smoothing the ODE trajectories, and significantly improving likelihood scores on benchmark datasets like CIFAR-10 and ImageNet [14]. These improvements demonstrate that with appropriate architectural and objective function refinements, diffusion ODEs can rival or surpass traditional likelihood-based generative models while retaining the superior sample quality characteristic of diffusion processes.

The theoretical robustness of the probability flow ODE framework has also been subjected to rigorous convergence analysis. Unlike stochastic samplers where convergence guarantees often rely on the contractive properties of the underlying dynamics, deterministic ODE solvers require different analytical tools. Recent studies have provided the first polynomial-time convergence guarantees for the probability flow ODE implementation of score-based generative modeling. By incorporat-

ing a carefully designed corrector step based on underdamped Langevin diffusion, researchers have achieved better dimension dependence in convergence rates compared to prior works on DDPM, highlighting the potential efficiency advantages of the ODE framework in high-dimensional spaces [15]. Additionally, non-asymptotic analyses have offered new operational interpretations of deterministic sampling, decomposing a single step of the probability flow ODE into a restoration step involving gradient ascent on the conditional log-likelihood and a degradation step, thereby extending the applicability of these samplers to general non-linear forward processes [16].

Beyond Euclidean spaces, the principles governing probability flow ODEs and normalizing flows have been successfully extended to data residing on complex manifolds. Traditional flows often assume a flat geometry, which can lead to misspecification when modeling data with non-trivial topologies, such as spherical or hyperbolic structures. The generalization of Neural ODEs and continuous normalizing flows to arbitrary smooth manifolds allows for the parameterization of flexible probability measures on these spaces by defining flows as solutions to ordinary differential equations constrained to the manifold’s geometry [17]. This extension utilizes fundamental concepts from differential geometry and geometric control theory to define vector fields and compute divergences in a generalized setting, ensuring that the generated samples respect the intrinsic structure of the data domain [18]. Moreover, the introduction of Neural Manifold Ordinary Differential Equations provides a unified methodology for constructing Manifold Continuous Normalizing Flows (MCNFs) that require only local geometry, thus overcoming the design constraints of previous manifold-specific approaches [19].

The intersection of optimal transport theory and probability flow ODEs has further enriched the theoretical landscape, serving as a direct precursor to the Flow Matching paradigms detailed in the following section. Flow Matching proposes a simulation-free approach for training CNFs by regressing vector fields of fixed conditional probability paths, subsuming existing diffusion paths as specific instances. This framework not only offers a more robust alternative for training diffusion models but also enables the use of Optimal Transport (OT) displacement interpolation to define probability paths. These OT-based paths result in straighter trajectories, reducing the number of function evaluations required for sampling and improving generalization capabilities [20]. Theoretical bounds on the approximation error of flow matching methods under deterministic sampling regimes have also been established, providing a rigorous foundation for the observed empirical successes [21]. Furthermore, the formulation of discrete probability flows through optimal transport principles has extended these deterministic concepts to discrete domains, proving that continuous probability flows can be viewed as Monge optimal transport maps under certain conditions and establishing equivalent evidence for discrete cases [22].

In summary, the derivation and analysis of Probability Flow ODEs have fundamentally transformed the understanding of diffusion models, recasting them as deterministic continuous normalizing flows. This perspective not only unlocks the capability for exact likelihood evaluation and efficient deterministic sampling but also facilitates the integration of advanced mathematical concepts from optimal transport, differential geometry, and dynamical systems. By resolving the dichotomy between stochastic diffusion and deterministic flows, this framework paves the way for the next evolution in generative modeling: the explicit optimization of transport paths for maximal efficiency, as discussed in the upcoming exploration of Optimal Transport and Flow Matching.

1.3 Optimal Transport Perspectives and Flow Matching Paradigms

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Building upon the theoretical unification of diffusion models and Continuous Normalizing Flows (CNFs) established via Probability Flow ODEs, the integration of Optimal Transport (OT) theory has catalyzed a paradigm shift in how continuous-time dynamics are learned for image synthesis. While the probability flow framework provides a deterministic route for generation, standard diffusion processes often follow curved, inefficient trajectories that necessitate numerous function evaluations. Optimal Transport offers a geometric solution to this inefficiency by defining the most direct path between a source noise distribution and a target data distribution. The core objective within this perspective is to learn a vector field that transports probability mass while minimizing a specific cost function, typically the squared Euclidean distance, thereby “straightening” the sampling trajectories. This theoretical grounding has given rise to Flow Matching (FM) and Rectified Flow methodologies, which construct deterministic Ordinary Differential Equations (ODEs) that are computationally superior for inference, bridging the gap between the robustness of diffusion models and the efficiency of optimal transport maps.

Flow Matching represents a significant departure from the simulation-based training methods inherent in early CNFs and the stochastic sampling of traditional diffusion. Instead of simulating the entire trajectory during training, Flow Matching formulates the learning problem as a regression task on conditional vector fields defined by fixed probability paths. As detailed in foundational work, this approach allows for the training of CNFs at unprecedented scale by regressing vector fields of fixed conditional probability paths [20]. A critical innovation of this framework is its compatibility with a general family of Gaussian probability paths, which subsumes standard diffusion paths as specific instances while also enabling the use of Optimal Transport displacement interpolation. By employing OT displacement interpolation to define these conditional paths, models can achieve faster training and sampling speeds, resulting in better generalization and more efficient sample generation using off-the-shelf numerical ODE solvers [20]. The theoretical robustness of this approach is further supported by convergence analyses, which demonstrate that the distribution of samples generated via estimated ODE flows converges to the target distribution in Wasserstein-2 distance under mild assumptions, particularly when leveraging transformer architectures in latent spaces [23].

Building upon the general Flow Matching framework, Rectified Flow introduces a specifically designed mechanism to learn ODE models that transport data along straight lines. The central premise is that straight paths are the shortest connections between two points and can be simulated exactly without time discretization errors, yielding computationally efficient models [24]. The method achieves this by solving a straightforward nonlinear least squares optimization problem, effectively learning the ODE to follow straight paths connecting points drawn from the source and target distributions. A key theoretical property of Rectified Flow is the process of “rectification,” which transforms an arbitrary coupling of distributions into a new deterministic coupling with provably non-increasing convex transport costs [24]. By recursively applying this rectification procedure, one can obtain a sequence of flows with increasingly straight paths, enabling high-quality results even with a single Euler discretization step during inference. This marginal-preserving approach distinguishes itself by traversing inside the set of valid couplings to monotonically reduce transport costs, offering a single-object variant tailored to minimize specific convex cost functions [25].

Despite the efficacy of standard Flow Matching, challenges remain in finding optimal coupling

strategies that guarantee straight trajectories without relying on heuristic approximations or multi-round training. To address this, recent advancements have introduced methods like Multisample Flow Matching, which utilizes non-trivial couplings between data and noise samples within mini-batches. This generalization reduces gradient variance during training and yields straighter flows, allowing for high-quality sample generation with fewer function evaluations [26]. Furthermore, novel approaches such as Optimal Flow Matching have been developed to recover the straight OT displacement for quadratic costs in just one flow matching step, bypassing the need for complex iterative procedures that often accumulate errors [27]. Another strategy, Straighter trajectories of Flow Matching (StraightFM), employs a coupling strategy guided by diffusion models from an entire distribution level to create couplings between image and noise samples, jointly optimizing mutually complementary directions to enable effective one-step and few-step generation [28].

The pursuit of straighter trajectories is also driven by the need to minimize the global truncation error of numerical ODE solvers, which explodes in the low-number-of-function-evaluations (NFE) regime due to high curvature. Methods like Sequential Flow Straightening (SeqRF) explicitly target this issue by learning techniques that straighten the probability flow, thereby reducing truncation errors and accelerating sampling while improving synthesis quality [29]. Similarly, techniques that minimize trajectory curvature directly during training have shown that reducing the curvature of generative trajectories leads to decreased sampling costs without compromising performance [30]. Additionally, the Bellman Optimal Stepsize Straightening (BOSS) technique utilizes dynamic programming to optimize stepsizes for pretrained networks, refining the velocity network to match optimal step sizes and further straightening generation paths under computational budget constraints [31].

Beyond Euclidean spaces, the principles of Flow Matching and Optimal Transport have been extended to general geometries and complex scientific domains, mirroring the manifold extensions discussed for probability flows. Riemannian Flow Matching (RFM) provides a simulation-free framework for training continuous normalizing flows on manifolds, overcoming limitations related to expensive simulations and biased training objectives in non-Euclidean settings [32]. In scientific applications, such as molecular design, equivariant flow matching with hybrid probability transport has been introduced to handle both categorical and continuous features, regularizing coordinate paths with equivariant optimal transport to ensure physical symmetries are preserved [33]. Moreover, the integration of entropic regularization has led to frameworks like Generative Entropic Neural OT (GENOT), which models conditional distributions of optimal entropic couplings, capable of handling outliers and transporting points across heterogeneous spaces [34]. These developments collectively illustrate how Optimal Transport perspectives not only provide a rigorous theoretical foundation for generative modeling but also drive practical innovations in efficiency, stability, and applicability across diverse domains, setting the stage for even more generalized transport frameworks such as Schrödinger Bridges.

1.4 Schrödinger Bridges and Diffusion Bridge Models for Generalized Transport

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While the Optimal Transport (OT) perspectives and Flow Matching paradigms discussed in the previous section successfully “straighten” trajectories to enhance computational efficiency, they primarily operate within deterministic or specific stochastic frameworks that often assume fixed priors or rely on asymptotic convergence to Gaussian noise. To further generalize the transport mechanism and overcome the rigid structural constraints of standard diffusion processes, the field

has increasingly turned to the Schrödinger Bridge (SB) problem. The SB framework provides a rigorous mathematical foundation for entropy-regularized optimal transport, offering a mechanism to define stochastic processes that interpolate between *any* two given marginal distributions in finite time. This approach effectively bridges the gap between the geometric efficiency of OT and the stochastic robustness of diffusion models, enabling flexible mapping beyond the simple noise-to-data paradigm.

At its core, the Schrödinger Bridge problem seeks the most likely stochastic evolution of particles between two observed empirical marginal distributions at distinct time points, minimizing the Kullback-Leibler divergence relative to a reference process [35]. This formulation can be viewed as a dynamic extension of the Monge-Kantorovich optimal mass transport (OMT) problem, where the addition of an entropy regularization term renders the problem computationally tractable and connects it to the celebrated Sinkhorn algorithm [36]. Unlike traditional OMT, which often requires solving complex variational problems to find deterministic maps, or the straight-path assumptions of Rectified Flow, the SB approach yields a stochastic differential equation (SDE) that naturally balances the minimization of transport cost with the maximization of entropy. This effectively acts as a “temperature-controlled” version of optimal transport, providing a smoother interpolation that is particularly advantageous when dealing with complex, multi-modal data distributions [35].

A critical advantage of the SB framework over standard diffusion models is its ability to operate over finite time horizons without requiring the target distribution to be approximately Gaussian. Standard score-based approaches typically necessitate running the forward SDE for a prolonged duration to “wash out” data structure into pure noise, which can be inefficient and introduce approximation errors. In contrast, solving the SB problem yields diffusions that generate samples from the data distribution in finite time by directly learning the time-inhomogeneous drift of the reverse-time SDE [37]. This capability allows for the construction of generative models that are not only more sample-efficient but also theoretically grounded in the principles of stochastic control and large deviations theory, setting the stage for more direct and controlled transportation of probability mass [36].

Building upon this theoretical foundation, recent advancements have led to the development of Diffusion Bridge Models, which generalize the standard diffusion paradigm to map between arbitrary endpoint distributions. These models leverage the concept of diffusion bridges, a family of processes conditioned to start at one distribution and end at another, thereby unifying various generative approaches including score-based diffusion and optimal transport flow matching [38]. For instance, the Denoising Diffusion Bridge Models (DDBMs) framework learns the score of the diffusion bridge directly from data, enabling the solution of image-to-image translation tasks where the source is not random noise but a specific domain distribution [38]. This approach circumvents the need for cumbersome guidance mechanisms or projected sampling often required when adapting standard diffusion models to non-noise inputs, offering a more natural formulation for controllable generation tasks.

The computational landscape for solving SB problems has also seen significant innovation to address the challenges of scalability and stability. Early methods relied on iterative procedures such as Iterative Proportional Fitting (IPF), which can be slow and prone to error accumulation. To overcome these challenges, novel algorithms like Diffusion Schrödinger Bridge Matching (DSBM) and Iterative Markovian Fitting (IMF) have been introduced. These methods significantly improve numerical stability and scalability, recovering various recent transport methods as special or limiting cases while providing a robust tool for continuous state-space optimal transport [39]. Furthermore, the integration of Forward-Backward Stochastic Differential Equations (FBSDEs) theory has en-

abled likelihood training of SB models, establishing a direct link between the optimality conditions of SB and modern deep generative training objectives [40]. This theoretical bridge ensures that SB models can be trained using standard gradient-based optimization while inheriting the optimality properties of entropy-regularized transport.

The versatility of SB models extends to specialized applications where the assumption of a Gaussian prior is particularly limiting, such as unpaired image-to-image translation. The Unpaired Neural Schrödinger Bridge (UNSB) framework addresses this by formulating the SB problem as a sequence of adversarial learning tasks, allowing for scalable translation between high-resolution images without paired data [41]. Additionally, the framework has been adapted for scientific domains, such as modeling the evolution of single-cell genomics data, where closed-form expressions for SBs between Gaussian measures have been derived to ensure numerical stability and physical consistency [42].

Further generalizations include the handling of unbalanced marginals, where the total mass of the source and target distributions may differ. Traditional SB formulations assume mass conservation, but recent work has extended the framework to account for particle creation and annihilation, modeling the “killing” of particles via probabilistic laws to reconcile observed marginals in lossy transport scenarios [43]. This extension is crucial for applications in physics and biology where mass is not preserved. Moreover, the concept of Soft-constrained Schrödinger Bridge (SSB) relaxes the hard constraint on the terminal distribution, penalizing the divergence instead, which leads to robust generative models capable of handling noisy or imperfect target specifications [44].

In summary, Schrödinger Bridges and Diffusion Bridge Models represent a powerful generalization of modern generative modeling that logically extends the OT and Flow Matching principles discussed previously. By framing generation as an entropy-regularized optimal transport problem between arbitrary distributions, these models overcome the rigid structural constraints of standard diffusion processes and the deterministic limitations of straight flows. Through advancements in algorithmic efficiency, such as simulation-free score matching [45] and refined iterative solvers [39], the SB framework has emerged as a unified theoretical and practical toolkit. This toolkit not only facilitates controllable, efficient, and high-fidelity image generation but also provides the conceptual bridge necessary to connect explicit density models with implicit generative frameworks, a synthesis that will be explored in the following section on unified frameworks connecting SDEs, GANs, and implicit modeling.

1.5 Unified Frameworks Connecting SDEs, GANs, and Implicit Modeling

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Building upon the generalized transport mechanisms established by Schrödinger Bridges, which enable flexible mapping between arbitrary distributions, the field has further progressed toward a holistic theoretical synthesis. Historically, deep generative modeling has been stratified into distinct paradigms: explicit density models, such as score-based diffusion models and normalizing flows, versus implicit generative models, exemplified by Generative Adversarial Networks (GANs). Explicit models offer tractable likelihood evaluation and stable training dynamics but often suffer from slow sampling speeds due to the iterative nature of solving stochastic or ordinary differential equations. Conversely, implicit models like GANs enable rapid, one-step generation but are frequently plagued by training instability, mode collapse, and the inability to compute exact likelihoods. Recent theoretical advancements have sought to bridge this dichotomy by constructing unified frameworks that describe the learning dynamics of both model classes within a single mathematical formalism, typically grounded in stochastic differential equations (SDEs) or deterministic flow theories. These

unified perspectives not only elucidate the underlying connections between seemingly disparate algorithms but also pave the way for hybrid architectures that can simultaneously achieve high sample fidelity, comprehensive mode coverage, and computational efficiency, effectively extending the versatility of the bridge concepts discussed previously.

A pioneering effort in this unification is the proposal of the Discriminator Denoising Diffusion Flow (DiffFlow), which establishes a rigorous theoretical link between score-based diffusion models (SDMs) and GANs. This framework posits that the learning dynamics of both SDMs and GANs can be described by a novel SDE where the drift term is determined by a weighted combination of the scores of the real data distribution and the generated data distribution [46]. The profound implication of the DiffFlow formulation is that by adjusting the relative weights assigned to these different score terms, one can induce a smooth, continuous transition between the behavior of a standard SDM and that of a GAN, all while maintaining the invariance of the marginal distribution of the SDE [46]. Furthermore, this unified view allows for the proof of asymptotic optimality and provides a maximum likelihood training scheme for the resulting dynamics, offering new algorithmic instantiations that go beyond the capabilities of traditional GANs or SDMs alone [46]. This suggests that the perceived trade-off between sample quality and sampling speed is not fundamental but rather a consequence of specific parameter choices within a broader, unified dynamical system.

Complementing the DiffFlow perspective, the concept of the Score-Difference (SD) Flow offers another powerful lens through which to view implicit generative modeling, deeply resonating with the entropy-regularized transport principles of Schrödinger Bridges. This approach formulates the generation process as a flow driven by the score difference between an arbitrary target distribution and a source distribution, which optimally reduces the Kullback-Leibler divergence between them [47]. Remarkably, this formulation solves the Schrödinger bridge problem and demonstrates formal equivalence to denoising diffusion models under specific conditions [47]. More critically, it reveals that the training objective of GANs inherently contains a hidden data-optimization sub-problem that induces the SD flow when the discriminator is optimal and specific loss functions are employed [47]. By identifying this common mechanism, the Score-Difference Flow effectively links model classes that individually address the three challenges of the “generative modeling trilemma”: high sample quality, mode coverage, and fast sampling [47]. This theoretical synthesis sets the stage for developing unified approaches that can leverage the strengths of both diffusion processes and adversarial training.

Further extending these unification efforts, other frameworks have explored the connection between generative modeling and optimal transport theory, reinforcing the geometric foundations introduced in the context of bridge models. For instance, the UnifiGem framework integrates approaches from optimal transport, numerical ODEs, and density-ratio estimation to formulate implicit generative learning as the problem of finding an optimal transport map characterized by the Monge-Ampère equation [48]. By interpreting the infinitesimal linearization of this equation as gradient flows in measure spaces, leading to the McKean-Vlasov equation, such frameworks provide theoretical guarantees for implicit modeling that were previously absent [48]. Similarly, the equivalence between GAN training and gradient flows in the space of probability density functions has been rigorously established, showing that the unknown data distribution emerges as the long-time limit of a distribution-dependent ODE [49]. This equivalence provides a “cooperative” view of GANs, revealing that their divergence issues often stem from implicitly minimizing the mean squared error between sample sets, a insight derived from analyzing the associated non-linear Fokker-Planck equation [49].

These unified frameworks collectively suggest that the boundaries between explicit and implicit

modeling are increasingly porous. The recognition that GANs can be viewed as performing a form of score matching or flowing along a specific vector field defined by density differences allows researchers to import stability techniques from diffusion models into adversarial settings. Conversely, understanding diffusion models through the lens of implicit flows encourages the development of accelerated sampling strategies akin to those found in GANs. For example, the cooperative learning of Langevin flows and normalizing flows toward Energy-Based Models demonstrates how two distinct flow models can be iteratively updated based on jointly synthesized examples to achieve moment matching estimators [50]. Such hybrid strategies leverage the amortized sampling of normalizing flows to initialize MCMC chains, thereby addressing the mixing issues inherent in pure energy-based models while retaining their expressive power [50].

In summary, the emergence of unified frameworks like DiffFlow and Score-Difference Flow represents a paradigm shift in generative modeling theory, serving as a conceptual bridge between the transport-focused methods of the previous section and the domain-specific expansions to follow. By describing the learning dynamics of both explicit density models and implicit models within a single stochastic or deterministic flow formalism, these approaches dissolve the traditional silos separating diffusion models, GANs, and flow-based methods. They provide a cohesive mathematical language to analyze convergence, optimality, and the trade-offs between likelihood estimation and sampling efficiency. As the field progresses, these unified perspectives will likely drive the design of next-generation generative architectures that are not constrained by the limitations of their predecessors but instead harness the full spectrum of capabilities offered by the unified theory of generative flows, setting the theoretical groundwork for applying these dynamics to complex function spaces, phase spaces, and discrete domains as explored in the subsequent section.

1.6 Advanced Dynamics in Function Spaces, Phase Spaces, and Discrete Domains

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Building upon the unified theoretical foundations established in the previous section, which dissolve the boundaries between explicit and implicit generative paradigms, the field has expanded its scope beyond finite-dimensional Euclidean spaces. While unified frameworks like DiffFlow and Score-Difference Flow provide a cohesive language for standard image generation, many scientific and structured domains involve continuous fields, high-dimensional dynamical systems, or inherently discrete structures that defy vector-based representations. To address these complexities, recent advancements have generalized the mathematics of diffusion and flow-based models into infinite-dimensional function spaces, augmented phase spaces, and discrete domains. This expansion not only overcomes the resolution and structural limitations of traditional approaches but also unlocks new degrees of controllability and fidelity essential for complex generative tasks, setting the stage for the rigorous error analysis discussed in the subsequent section.

A primary frontier in this generalization is the formulation of diffusion processes directly within infinite-dimensional function spaces. Traditional methods often resort to discretizing continuous data, such as fluid dynamics fields or time-series signals, prior to modeling, introducing resolution-dependent artifacts and obscuring intrinsic functional semantics. To circumvent this, researchers have introduced Denoising Diffusion Operators (DDOs), a rigorous framework extending denoising score matching to infinite-dimensional Hilbert spaces [51]. In this paradigm, the forward process perturbs input functions via Gaussian processes rather than finite-dimensional noise, while the reverse process integrates function-valued Langevin dynamics, ensuring resolution-invariant generation critical for applications like solving the Navier-Stokes equation [51]. Complementing this,

the theory of Diffusion Generative Models in Infinite Dimensions establishes foundations using Gaussian measures on Sobolev spaces, enabling explicit specification of function spaces for both unconditional and conditional generation [52]. Furthermore, Continuous-Time Functional Diffusion Processes (FDPs) leverage infinite-dimensional versions of Girsanov’s theorem to compute evidence lower bounds, demonstrating that high-quality generation is achievable with simple MLP architectures, thereby bypassing the need for specialized grid-based backbones [53]. The versatility of these approaches is epitomized by Diffusion Probabilistic Fields (DPF), which learn distributions over continuous functions on arbitrary metric spaces, effectively managing non-Euclidean domains without latent vector bottlenecks [54].

Beyond spatial generalizations, the internal dynamics of the diffusion process have been re-engineered through the lens of statistical mechanics to enhance sampling efficiency and trajectory quality. Standard Score-Based Generative Models (SGMs) often rely on simplistic diffusion mechanisms that can induce unnecessarily complex denoising landscapes. A significant breakthrough in this regard is Critically-Damped Langevin Diffusion (CLD), which operates in an extended phase space by coupling data variables with auxiliary “velocity” variables, analogous to Hamiltonian dynamics [55]. This formulation shifts the learning objective from estimating the data score directly to learning the score of the conditional velocity distribution, a task that is frequently computationally more tractable and yields superior synthesis quality [55]. Although theoretical analysis suggests that CLD may not reduce the fundamental complexity of sampling under minimal data assumptions compared to standard Langevin dynamics, it offers substantial practical performance gains in high-resolution synthesis [8]. To further exploit these augmented spaces, advanced integrators such as Conjugate Integrators and Splitting Integrators have been developed; these generalize DDIM by alternating updates between data and auxiliary variables to reduce numerical simulation errors [56]. Additionally, the generalized DDIM (gDDIM) framework extends deterministic sampling to non-isotropic diffusions like CLD, achieving drastic reductions in function evaluations while preserving high fidelity [57].

Parallel to these continuous advancements, there has been a surge in adapting diffusion models for discrete and categorical data, where gradients and continuous noise addition are not naturally defined. Early discrete diffusion approaches relied on uniform transition probabilities, but recent work has generalized this to Discrete Denoising Diffusion Probabilistic Models (D3PMs), which utilize structured transition matrices mimicking Gaussian kernels or introducing absorbing states to better model discrete sequences [58]. A major theoretical challenge has been unifying discrete-time and continuous-time formulations for such data. The Unified Simplified Discrete Denoising Diffusion (USD3) framework addresses this by deriving simplified variational lower bounds and exact backward denoising formulations, enabling flexible noise distributions and accelerated sampling for language and graph generation [59]. Furthermore, the difficulty of defining gradients in discrete spaces has been tackled by generalizing Wasserstein gradient flows, leading to Discrete Langevin Monte Carlo (DLMC), which facilitates time-uniform sampling and larger jump distances in binary and categorical distributions [60]. Another innovative strategy, Continuous Diffusion for Categorical Data (CDCD), preserves the benefits of continuous diffusion by embedding categorical data in a continuous input space, effectively bridging continuous physical phenomena and discrete symbolic representations [61]. Theoretical convergence guarantees for these discrete models have also been established using uniformization techniques for Continuous Time Markov Chains, providing Total Variation distance bounds that align with state-of-the-art results in continuous settings [62].

Finally, the extension of these dynamics to constrained and geometric domains represents a critical step for ensuring physical and structural controllability. Diffusion models have been adapted to

Riemannian manifolds and domains defined by inequality constraints through novel noising processes based on logarithmic barrier metrics and reflected Brownian motion [63]. These methods enable the generation of data that strictly adheres to physical or geometric constraints, such as protein structures or robotic trajectories, by integrating constraint projection operators directly into the denoising architecture [64]. Moreover, efficient Metropolis sampling schemes have been introduced to handle non-convex constraints with greater computational efficiency than previous barrier methods [65]. Collectively, these advancements in function spaces, phase spaces, and discrete domains signify a maturation of diffusion theory, transforming it from a tool for simple image synthesis into a universal framework for modeling complex, structured, and high-dimensional data distributions. As these models are deployed in increasingly diverse and demanding applications, the need for rigorous theoretical guarantees regarding their convergence, error bounds, and calibration becomes paramount, a topic explored in depth in the following section.

1.7 Theoretical Analysis of Error Bounds, Convergence, and Calibration

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Building upon the expanded dynamical frameworks in function spaces, phase spaces, and discrete domains discussed previously, the rapid empirical success of diffusion and flow-based generative models has necessitated a rigorous theoretical examination of their convergence properties, error bounds, and calibration capabilities. While early developments relied heavily on heuristic performance, recent advancements have established a robust mathematical framework guaranteeing that these models can efficiently sample from complex data distributions under minimal assumptions. A cornerstone of this theoretical progress is the derivation of polynomial-time convergence guarantees for score-based generative models (SGMs). It has been proven that, assuming accurate score estimates, SGMs can sample from essentially any realistic data distribution with complexity that scales polynomially in all relevant problem parameters [8]. Crucially, these guarantees hold under L^2 -accurate score estimates rather than the more restrictive L^∞ -accurate conditions, and they do not require the data distribution to satisfy strong functional inequality conditions such as log-concavity, thereby broadening the scope of applicable distributions [8]. Further refining these bounds, recent work has demonstrated that convergence can be achieved for distributions with only a finite second moment, providing efficient guarantees for approximating variance-perturbed data distributions in reverse KL divergence [66].

A critical component of this theoretical landscape involves the precise quantification of errors arising from discretization and score approximation, bridging the gap between continuous-time formulations and their discrete-time implementations. Researchers have established non-asymptotic convergence rates for discrete-time diffusion models covering substantially larger classes of distributions than previously known, including those with Lipschitz scores and Gaussian mixtures [67]. Moreover, the impact of score estimation errors on the final sampling distribution has been rigorously bounded. For strongly log-concave data distributions, full theoretical guarantees have been provided that combine optimization error bounds for score approximation with sampling estimates, yielding the best known upper bounds for the Wasserstein-2 distance between the target and generated distributions [68]. In the context of accelerating sampling—a key concern for the complex dynamics introduced in Section 1.6—novel training-free algorithms have been designed with provable convergence rates, achieving $O(1/T^2)$ for deterministic samplers and $O(1/T)$ for stochastic samplers, significantly outperforming standard DDIM and DDPM rates respectively [69]. These acceleration techniques leverage higher-order approximations similar to advanced ODE solvers while maintaining theoretical validity under ℓ_2 -accurate score estimates [69].

The Fokker-Planck equation (FPE) plays a pivotal role in understanding the density evolution and self-consistency of these generative processes, serving as the governing law for the spatial-temporal evolution of the probability density function associated with the underlying stochastic differential equations. Recent studies have highlighted that scores learned through standard denoising score matching often fail to satisfy the underlying score FPE, an inherent self-consistency property of the ground truth score [70]. Enforcing this constraint via regularization has been shown to improve likelihood estimation and the degree of conservativity in the generated samples [70]. Furthermore, the discrepancy between samples generated via Ordinary Differential Equations (ODEs) and Stochastic Differential Equations (SDEs) can be rigorously linked to the Fokker-Planck residual; reducing this residual through regularization effectively closes the distributional gap between ODE and SDE induced dynamics [6]. The probability flow ODE, which enables deterministic sampling, has also been proven to be fast, with polynomial-time convergence guarantees that offer better dimension dependence ($O(\sqrt{d})$) compared to prior SDE-based analyses, highlighting the theoretical advantages of the deterministic framework [15].

Beyond convergence and error bounds, the calibration of generative models remains a vital area of inquiry for ensuring reliable uncertainty quantification, a prerequisite for trustworthy controllable generation. Standard maximum likelihood training often yields models that are poorly calibrated, meaning their confidence intervals do not accurately reflect the true frequency of outcomes. To address this, methods based on low-dimensional density estimation have been introduced to recalibrate models, ensuring distribution calibration without sacrificing overall performance [71]. Theoretical analysis in this domain has established uniform convergence bounds for such recalibration procedures, proving their correctness under the assumption that densities can be estimated in low dimensions [71]. Additionally, the statistical efficiency of score matching, the primary training objective for these models, has been analyzed through the lens of isoperimetry. This analysis reveals a tight connection between the efficiency of score matching and the isoperimetric properties of the data distribution, such as the Poincaré and log-Sobolev constants, indicating that score matching is statistically comparable to maximum likelihood only when the distribution possesses favorable mixing properties [72]. Finally, recent work on consistency models has extended these theoretical foundations to one-step generation paradigms, providing the first convergence guarantees that match state-of-the-art bounds for multi-step SGMs while operating under minimal smoothness assumptions [73]. Collectively, these theoretical advancements provide a solid foundation for the reliability, efficiency, and trustworthiness of diffusion and flow-based models, setting the stage for their application in diverse controllable image generation tasks.

2 Architectural Innovations and Efficiency Optimization

2.1 Evolution of Backbone Architectures: From U-Nets to Scalable Transformers and State Space Models

2.1 Evolution of Backbone Architectures: From U-Nets to Scalable Transformers and State Space Models

The landscape of generative modeling has undergone a profound architectural transformation, shifting from the dominance of Convolutional Neural Network (CNN)-based U-Nets to the emergence of highly scalable Transformer and State Space Model (SSM) backbones. Initially, the success of Denoising Diffusion Probabilistic Models (DDPMs) was inextricably linked to the U-Net architecture, which leveraged skip connections to preserve spatial details while enabling deep hierarchical feature extraction. However, as demands for higher resolution, improved fidelity, and greater scalability

intensified, the inherent limitations of convolutions—specifically their local receptive fields and the difficulty in modeling long-range dependencies without excessive depth—became apparent. This catalyzed a paradigm shift toward token-based architectures that treat image patches as sequences, facilitating global context aggregation and linear scaling with computational resources. This architectural maturation provides the essential foundation upon which advanced sampling strategies, discussed in the following section, can operate more effectively.

The pioneering work in this transition is exemplified by the introduction of Diffusion Transformers (DiT), which replaced the traditional U-Net backbone with a standard Transformer operating on latent patches. Research demonstrated that DiT models exhibit exceptional scalability, where performance, measured by Fréchet Inception Distance (FID), consistently improves as the model size (depth, width) or the number of input tokens increases [74]. This finding challenged the prevailing notion that specialized convolutional inductive biases were necessary for high-quality image synthesis. Further refining this approach, subsequent studies introduced U-ViT, an architecture that treats time, conditions, and noisy image patches uniformly as tokens while employing long skip connections between shallow and deep layers. Remarkably, U-ViT achieved record-breaking FID scores on class-conditional ImageNet generation, suggesting that the long skip connection is the critical component for generative fidelity, while the specific down-sampling and up-sampling operators of CNNs are not strictly necessary [75].

Building upon the Transformer foundation, researchers have explored various optimizations to enhance efficiency and adaptability. For instance, DiffiT introduced a Time-dependent Multihead Self Attention (TMSA) mechanism, allowing for fine-grained control over the denoising process and achieving state-of-the-art performance with significantly fewer parameters than earlier Transformer-based diffusion models [76]. The versatility of this architectural shift is further evidenced by GenTron, which successfully adapted DiT from class-conditional to text-conditional generation and scaled the model to over 3 billion parameters, demonstrating superior visual quality and text alignment compared to established CNN-based baselines [77]. Moreover, the application of Transformers has extended beyond 2D imagery; DiT-3D adapted the plain Transformer architecture for 3D shape generation on voxelized point clouds, incorporating 3D window attention to manage the increased computational cost of the additional spatial dimension [78].

Parallel to the rise of Transformers, another significant architectural evolution involves the adoption of State Space Models (SSMs) for diffusion. Addressing the quadratic complexity of self-attention in Transformers, which can be prohibitive for very high-resolution inputs or long sequences, Diffusion State Space Models (DiS) propose replacing the U-Net backbone with an SSM architecture. DiS treats all inputs, including time, conditions, and noisy patches, as tokens, leveraging the SSM’s ability to efficiently accommodate long-range dependencies with linear complexity [79]. Empirical evaluations indicate that DiS models achieve performance comparable to, or superior to, both CNN-based and Transformer-based U-Nets of similar sizes while significantly reducing the computational burden, particularly in terms of forward pass complexity measured in GFLOPs. This positions SSMs as a potent alternative for scaling diffusion models to regimes where Transformers might face memory or speed bottlenecks.

The exploration of hybrid and alternative token-mixing mechanisms further enriches this architectural landscape. For example, Diffusion-RWKV adapts the RWKV architecture, known for its efficiency in natural language processing, to diffusion models. This approach eliminates the need for windowing or grouped cached operations, offering a distinct advantage in processing high-resolution images with reduced spatial aggregation complexity [80]. Additionally, the integration of large kernel designs in CNNs, such as RepLKNet, attempts to bridge the gap between ConvNets

and Transformers by expanding the effective receptive field, although pure token-based models often retain an edge in scalability [81].

The shift toward these scalable backbones is driven by the fundamental requirement to model complex, global dependencies inherent in high-fidelity data. Traditional U-Nets, while effective, often struggle with the global coherence required for intricate scene generation without massive parameter counts. In contrast, token-based processing in DiT and DiS allows the model to attend to or propagate information across the entire image canvas simultaneously. This capability is crucial for tasks requiring precise structural consistency and semantic alignment. Furthermore, the modular nature of these architectures facilitates seamless integration with other advancements, such as flow matching frameworks like Scalable Interpolant Transformers (SiT), which utilize the DiT backbone to explore various interpolant paths between distributions, surpassing standard diffusion performance [82].

In summary, the evolution from U-Nets to Transformers and State Space Models represents a critical maturation in controllable image generation. By moving away from localized convolutional operations to global token mixing mechanisms, modern generative models achieve unprecedented scalability and fidelity. Whether through the global attention of DiT [74], the efficient long-range modeling of DiS [79], or the specialized adaptations for 3D and video [77][78], these architectural innovations provide the robust backbone necessary for the next generation of high-resolution, controllable, and efficient generative systems. With these powerful backbones established, the focus naturally shifts to optimizing the sampling process itself, addressing the computational costs inherent in solving the underlying differential equations.

2.2 Accelerating Sampling via Advanced ODE/SDE Solvers and Trajectory Reformulation

2.2 Accelerating Sampling via Advanced ODE/SDE Solvers and Trajectory Reformulation

With the establishment of scalable Transformer and State Space Model backbones detailed in the previous section, the primary bottleneck for controllable image generation shifts from architectural capacity to the computational latency inherent in the sampling process. Traditional diffusion and flow-based models rely on iterative procedures that necessitate hundreds or thousands of sequential function evaluations to solve underlying Stochastic Differential Equations (SDEs) or Ordinary Differential Equations (ODEs). To bridge the gap between these powerful backbones and real-time application requirements, recent research has pivoted toward fundamentally rethinking the mathematical solvers and trajectory formulations governing the generative process. This paradigm shift encompasses the development of extended reverse-time formulations, the design of bespoke and high-order numerical integrators, and the strategic reformulation of probability flow trajectories to minimize curvature and discretization errors, serving as a critical intermediate step before the complete elimination of iterative steps discussed in the following section.

A critical avenue for acceleration involves unifying and extending the theoretical framework of reverse-time dynamics. While traditional approaches often treat ODE and SDE solvers as distinct entities, recent work demonstrates that formulating the sampling process as an extended reverse-time SDE (ER SDE) can unify these perspectives. By leveraging the semi-linear structure of ER SDE solutions, researchers have derived exact and arbitrarily high-order approximate solutions for both variance-preserving (VP) and variance-exploding (VE) SDEs. This unified view not only elucidates why ODE solvers often outperform SDE solvers in terms of speed but also enables the

creation of training-free samplers that achieve state-of-the-art performance with significantly fewer function evaluations [83]. Furthermore, the distinction between stochastic and deterministic sampling is being revisited; while ODEs offer speed, they often plateau in quality due to accumulated discretization errors. Novel algorithms like the Restart sampler address this by alternating between adding substantial noise in forward steps and strictly following a backward ODE, effectively contracting accumulated errors and surpassing previous methods in both speed and accuracy [84].

Beyond unified formulations, the design of specialized numerical solvers tailored to specific model characteristics has yielded substantial gains. Generic solvers often fail to account for the unique properties of the learned vector fields in diffusion models. The concept of “Bespoke Solvers” introduces a framework for constructing custom ODE solvers optimized for a given pre-trained flow model. By optimizing a parameter-efficient solver with minimal training overhead, these bespoke approaches significantly improve approximation quality, achieving near-ground-truth fidelity with as few as 10 to 20 function evaluations [85]. Similarly, the assumption of a Gaussian transition kernel in standard SDE solvers is frequently violated in complex data distributions. To rectify this, Gaussian Mixture Solvers (GMS) estimate higher-order moments and optimize parameters of a Gaussian mixture transition kernel at each step, thereby overcoming the efficiency-effectiveness dilemma inherent in traditional SDE-based sampling [86].

High-order exponential integrators represent another frontier in solver innovation, pushing the limits of numerical stability. While exponential integrators (EI) have shown promise, existing high-order variants often suffer from degeneration due to missing order conditions, making them sensitive to timestep schedules. Through rigorous order analysis, refined exponential solvers (RES) have been proposed to fulfill all necessary order conditions, resulting in superior error bounds and stability. These refined solvers demonstrate that simply switching to an order-satisfied method can drastically reduce numerical defects and improve generation quality without retraining the underlying model [87]. Complementing this, Stochastic Explicit Exponential Derivative-free Solvers (SEEDS) generalize exponential integrator approaches to the stochastic case, analytically computing linear parts and variance terms to achieve optimal quality sampling three to five times faster than previous SDE methods [88].

Parallel to solver improvements, trajectory reformulation offers a complementary strategy by altering the path itself rather than just the integration method. The curvature of generative trajectories is directly linked to the truncation error of numerical solvers; highly curved paths necessitate smaller step sizes for accurate integration. Methods that minimize trajectory curvature during training allow for larger integration steps and faster sampling. For instance, techniques that train the forward process to minimize curvature without explicit ODE/SDE simulation have successfully reduced sampling costs while maintaining competitive performance [30]. Similarly, Sequential Flow Straightening (SeqRF) explicitly learns to straighten the probability flow, reducing global truncation error and enabling acceleration in flow-based models [29]. By smoothing the path that the advanced backbones traverse, these methods maximize the efficiency of the underlying architecture.

Finally, the optimization of sampling schedules and the hybridization of solver strategies have emerged as powerful tools to further refine this efficiency. The “Align Your Steps” methodology proposes a principled approach to optimizing sampling schedules, moving away from hand-crafted heuristics to schedules specifically tuned for different solvers and datasets, which proves particularly effective in few-step synthesis regimes [89]. Moreover, the idea of dynamically scheduling different solvers throughout the generation process—such as using stochastic solvers in early noisy steps and deterministic solvers in later refinement steps—has been validated to improve both efficiency and quality [90]. This is further supported by unified sampling frameworks that reveal the benefits

of taking different solving strategies at different timesteps to decrease truncation error [91]. Additionally, momentum-based techniques like Heavy Ball momentum have been incorporated into diffusion numerical methods to expand stability regions and mitigate divergence artifacts common in low-step sampling [92].

Collectively, these advancements in solver design and trajectory reformulation transform diffusion models from slow, iterative processes into highly efficient engines capable of high-fidelity generation in tens of steps. However, while these methods drastically reduce the number of evaluations, they do not entirely eliminate the serial dependency of solving differential equations. To break this barrier and achieve true one-step generation, a more radical paradigm shift is required, leading to the emergence of Consistency Models and Deep Equilibrium approaches discussed in the next section.

2.3 Consistency Models and One-Step Generation Paradigms

2.3 Consistency Models and One-Step Generation Paradigms

While Section 2.2 detailed how advanced solvers and trajectory reformulation can drastically reduce the number of sampling steps, the fundamental iterative nature of solving differential equations remains a latent bottleneck for real-time applications. Even with high-order integrators, generating high-fidelity images typically requires multiple sequential function evaluations, which prohibits deployment in ultra-low latency scenarios. To break this serial dependency, a distinct paradigm shift has emerged towards Consistency Models (CMs) and related one-step architectures. Unlike the solver-centric acceleration in the previous section, these approaches fundamentally alter the learning objective to map noise directly to data distributions in a single function evaluation, thereby enabling real-time generation without the need for adversarial training [93].

Consistency models operate on the principle of enforcing self-consistency along the probability flow ODE (PF-ODE) trajectories. Whereas standard diffusion models learn to predict the score function or noise at discrete time steps, consistency models are trained to ensure that the model’s output for any two time points along the same trajectory is identical. This unique property allows the collapse of the entire multi-step generative process into a single step by mapping any point on the trajectory directly to the endpoint at time zero. Theoretical analyses have established rigorous convergence guarantees for these models, demonstrating that under mild assumptions regarding score-matching errors and data smoothness, consistency models can sample from realistic distributions with small Wasserstein-2 errors in a single step [73]. Furthermore, this framework retains flexibility; multi-step sampling within consistency models can further reduce errors, offering a tunable trade-off between computational cost and sample quality that complements the solver strategies discussed previously [93].

The training methodology for consistency models has evolved through two primary pathways: distillation and standalone training. Initially, consistency models were predominantly trained by distilling knowledge from pre-trained diffusion teachers, minimizing the distance between the student’s one-step output and the teacher’s multi-step sampling result. However, recent advancements have transcended these constraints. Improved techniques now enable consistency models to be trained directly from data, eliminating the dependency on a pre-trained teacher and the associated quality ceiling [94]. These standalone approaches rectify previous instabilities caused by Exponential Moving Average (EMA) in teacher models and replace perceptual loss functions like LPIPS with robust statistical losses such as Pseudo-Huber losses. By employing optimized noise schedules, such

as log-normal distributions, and dynamic discretization strategies, standalone consistency models have achieved Fréchet Inception Distance (FID) scores that surpass their distilled counterparts, marking a significant milestone in one-step non-adversarial generation [94]. Additionally, investigations into noise scheduling suggest that polynomial distributions and sinusoidal-based curricula can further enhance stability and performance, challenging the dominance of standard log-normal schedules [95].

Parallel to the development of consistency models, Deep Equilibrium (DEQ) models have emerged as a powerful architectural innovation that aligns with the goal of accelerating generative processes. DEQ models treat the entire sampling chain as a joint, multivariate fixed-point system rather than a serial sequence of updates [96]. By modeling the generation process as finding the fixed point of a recurrent function, DEQs enable parallel computation during inference, significantly reducing latency compared to the strictly sequential nature of traditional solvers. This formulation also facilitates efficient model inversion, allowing for the rapid recovery of latent noise vectors from observed images, a task that is computationally prohibitive in standard unrolled diffusion chains [96]. The application of DEQ principles extends to specific distillation architectures, such as the Generative Equilibrium Transformer (GET), which leverages deep equilibrium layers to distill diffusion models into efficient one-step generators. This approach has demonstrated the ability to match the performance of models several times larger while maintaining a critical balance between computational cost and image fidelity [97].

The versatility of these one-step paradigms extends beyond unconditional image synthesis to complex manipulation and domain-specific tasks, bridging the gap towards practical controllability. Generalized Consistency Trajectory Models (GCTMs) have been proposed to translate between arbitrary distributions, not just from Gaussian noise to data, enabling applications in image-to-image translation and restoration with minimal function evaluations [98]. In the realm of video generation, the Video Latent Consistency Model (VideoLCM) adapts these principles to the temporal domain, achieving high-fidelity video synthesis with as few as four sampling steps, thus addressing the immense computational demands of spatiotemporal generation [99]. Similarly, Latent Consistency Models (LCMs) have been successfully applied to large-scale latent diffusion frameworks like Stable Diffusion, enabling swift few-step inference for high-resolution text-to-image generation while preserving the semantic alignment of the original models [100].

Despite their remarkable efficiency, challenges remain in ensuring the robustness and geometric fidelity of one-step generators. Research into monotone operator learning and equilibrium models highlights the importance of imposing constraints, such as monotonicity or Lipschitz continuity, to guarantee convergence and stability in inverse problems [101]. Relaxing these constraints locally while maintaining global convergence properties has shown promise in improving performance without sacrificing theoretical guarantees [102]. Furthermore, alternative one-step approaches, such as Score Mismatching, propose training standalone generators by mismatching fake data distributions against real ones, offering a distinct pathway to bypass the ill-posed problems sometimes associated with consistency losses [103]. As these technologies mature, they are increasingly being integrated with the quantization and compression strategies discussed in the following section to push the boundaries of edge deployment. The convergence of consistency models, deep equilibrium architectures, and advanced training paradigms represents a definitive move towards the democratization of real-time, controllable image generation, ensuring that high-fidelity generative capabilities become accessible even in resource-constrained environments.

2.4 Quantization Strategies for Low-Bit Diffusion and Flow Models

2.4 Quantization Strategies for Low-Bit Diffusion and Flow Models

Building upon the architectural innovations in consistency models and deep equilibrium approaches discussed in Section 2.3, which fundamentally reduce inference latency by collapsing sequential steps, the deployment of these high-fidelity generative models on resource-constrained edge devices faces a secondary yet critical bottleneck: immense memory footprints and computational costs associated with high-precision parameters. While one-step paradigms address the temporal dimension of efficiency, quantization targets the spatial and numerical dimensions, compressing large generative models into low-bit representations suitable for real-time inference on hardware with limited bandwidth and storage. The landscape of quantization for generative models is broadly categorized into Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT), each presenting distinct trade-offs between implementation efficiency, data requirements, and generation quality. While PTQ offers a rapid pathway to compression without extensive retraining, it often struggles to maintain fidelity in extremely low-bit settings due to the sensitivity of generative dynamics to quantization noise. Conversely, QAT simulates quantization effects during the training phase to optimize parameters for low-precision arithmetic, typically yielding superior accuracy at the cost of significant computational overhead and data dependency.

Post-Training Quantization (PTQ) has gained traction as a practical solution for compressing large-scale models, including Latent Diffusion Models (LDMs), by converting full-precision weights and activations into lower-bit data types using only a small calibration set. However, applying standard PTQ techniques to diffusion models introduces unique challenges arising from their temporal evolution and structural complexity. Recent studies propose efficient quantization strategies that leverage the Signal-to-Quantization-Noise Ratio (SQNR) as a pivotal metric to identify and protect sensitive modules within the diffusion backbone. By treating quantization discrepancy as relative noise, these approaches employ global strategies to initiate higher-precision quantization on sensitive blocks while applying aggressive local treatments to time-sensitive modules, thereby achieving an effective balance between compression and generation quality [104]. Furthermore, specific components within transformer-based backbones, such as the softmax activation function, have been identified as highly sensitive to quantization errors. Naive quantization of softmax can introduce significant bias, degrading image synthesis quality. To address this, offline bias correction techniques have been developed to improve the quantizability of softmax operations without adding runtime overhead, allowing for fully 8-bit quantized inference in models like Stable Diffusion [105]. Advanced PTQ frameworks also explore the use of minifloats and integer formats across a design space of 3 to 8 bits, validating that while integer quantization often remains Pareto-optimal for hardware efficiency, low-precision floating-point formats can approach full-precision accuracy with reduced memory footprints [106].

Despite the efficiency of PTQ, pushing diffusion models to extreme low-bit precisions (e.g., 4-bit or below) often necessitates Quantization-Aware Training (QAT) to recover lost generative capabilities. Traditional QAT methods, however, are computationally prohibitive for large generative models due to the requirement for end-to-end fine-tuning on massive datasets. To bridge the gap between the efficiency of PTQ and the performance of QAT, novel parameter-efficient frameworks have been introduced. A notable advancement is the development of data-free, parameter-efficient fine-tuning methods such as EfficientDM, which utilizes a quantization-aware variant of Low-Rank Adapters (QALoRA). This approach distills the denoising capabilities of a full-precision model into its quantized counterpart by merging learnable low-rank adapters with model weights, which are

then jointly quantized. This strategy eliminates the need for original training data and significantly accelerates the quantization process—boasting speeds up to 16.2 times faster than conventional QAT methods—while maintaining comparable generation quality with minimal increase in Frechet Inception Distance (FID) scores [107].

The principles governing efficient QAT for generative models are increasingly informed by broader research in model compression for language and vision transformers. Techniques designed to handle outlier channels and activation distributions in Large Language Models (LLMs) are directly applicable to the attention mechanisms in Diffusion Transformers (DiTs). For instance, methods that employ adaptive channel reassembly to reallocate the magnitude of outliers or utilize learnable equivalent transformations to shift quantization challenges from activations to weights provide a theoretical foundation for stabilizing low-bit training in generative contexts [108; 109]. Additionally, the integration of consistency regularization into QAT frameworks has shown promise in pushing low-bit models toward full-precision performance by ensuring that augmented samples remain consistent in the latent feature space, a property crucial for the stability of diffusion trajectories [110].

Specialized optimization strategies further enhance the robustness of low-bit generative models. Approaches incorporating sharpness-aware minimization encourage the model to converge to flatter minima in the loss landscape, which inherently improves resilience to the perturbations caused by low-bit quantization [111]. Moreover, the exploration of mixed-precision quantization allows for dynamic bitwidth allocation, where critical layers or tokens are assigned higher precision based on entropy and distribution metrics, optimizing the trade-off between inference speed and generative fidelity [112]. Similarly, sparsity-aware quantization methods leverage dynamic activation sparsity to opportunistically allocate bit budgets, reducing accuracy degradation in ultra-low-bit regimes [113].

In summary, the field of quantization for diffusion and flow-based models is rapidly evolving from simple uniform compression to sophisticated, adaptive frameworks. By combining the data-efficiency of PTQ with the performance gains of parameter-efficient QAT methods like QALoRA, and drawing upon advanced techniques for outlier management and regularization from adjacent domains, researchers are successfully enabling the deployment of high-quality generative AI on edge devices. These advancements not only reduce latency and memory consumption but also lay the numerical groundwork for the structural optimizations discussed in the following section. While quantization optimizes the precision of existing parameters, the next frontier involves reshaping the model architecture itself through Knowledge Distillation and Neural Architecture Search to achieve even greater efficiency without compromising generative fidelity.

2.5 Knowledge Distillation and Neural Architecture Search for Model Compression

2.5 Knowledge Distillation and Neural Architecture Search for Model Compression

While quantization strategies discussed in the previous section effectively reduce the precision of model parameters, they often operate within a fixed architectural framework. To achieve further compression and efficiency gains, particularly for deployment on resource-constrained devices, it is essential to optimize the model structure itself. This necessitates advanced compression strategies that go beyond bit-width reduction, prominently featuring the synergistic integration of Knowledge Distillation (KD) and Neural Architecture Search (NAS). Whereas traditional pruning methods rely on heuristic criteria to remove redundant parameters, NAS offers a data-driven mechanism to automatically identify optimal lightweight sub-networks. The efficacy of NAS is significantly amplified when guided by the rich representational knowledge of a pre-trained teacher model, ensuring that

the compact student architecture retains the generative capabilities of its larger counterpart.

A primary bottleneck in NAS is the prohibitive computational cost associated with evaluating a vast search space of candidate architectures. To mitigate this, recent methodologies have shifted towards weight-sharing supernet, where a single over-parameterized network encompasses all possible sub-architectures. Within this framework, Knowledge Distillation serves as a critical regularization mechanism to stabilize the training of shared weights. For instance, the concept of “Inner Knowledge Distillation” has been proposed to optimize the final accuracy of pruned networks by distilling knowledge not just from the output layer, but from the intermediate layers of the over-parameterized parent network [114]. This approach formulates network pruning as a Knapsack Problem, optimizing the trade-off between neuron importance and computational cost while maintaining the high-level structural integrity of the network through distillation. Similarly, methods like [115] introduce automated algorithms to extract dependencies within connected layers, utilizing knowledge distillation to preserve trained weights even as channel sizes fluctuate during the search process. This ensures that the performance metrics of individual network components are accurately captured, leading to architectures that outperform baselines by significant margins.

The intersection of distillation and architecture search has also led to the development of specialized frameworks designed to search for student architectures that are inherently optimized for learning from a specific teacher. The [116] framework exemplifies this by employing Bayesian Optimization to explore a flexible graph-based search space. Instead of manually designing a student, AutoKD automatically learns the optimal distribution of student architectures and KD parameters, achieving sample efficiency improvements of up to 20x compared to state-of-the-art methods. This allows for the discovery of student models that can match teacher performance while utilizing drastically fewer parameters and memory. Furthermore, the [117] study reinforces this paradigm by proposing an L1-norm optimization approach to select a subgraph from the teacher’s topology as the student, minimizing the KL-divergence between their outputs to achieve superior generalization compared to manually specified architectures.

To address the inefficiency and inaccuracy often found in weight-sharing NAS, where shared parameters can lead to incorrect architecture ratings, block-wise supervision strategies have emerged. The [118] approach modularizes the large search space into smaller blocks, developing a family of models via “Distilling Neural Architecture” (DNA) techniques. By distilling neural architecture knowledge from a teacher model to guide block-wise search, this method resolves scalability and efficiency dilemmas, enabling the rating of all candidate architectures rather than relying on heuristic sub-space sampling. This principle is further refined in [119], which demonstrates that distilling architectural knowledge significantly improves the effectiveness of NAS, allowing searched architectures to even exceed the capacity of the teacher model in certain mobile settings.

In the specific context of generative models, such as diffusion architectures which often rely on U-Net backbones with inherent block-wise structures, specialized NAS techniques have been developed to tailor these compressions. The [120] (DiffNAS) proposes automatically removing structural redundancy by searching for the smallest architecture capable of matching a larger pretrained teacher. DiffNAS employs a block-wise local search strategy combined with a retraining phase utilizing a joint dynamic loss. This dynamic loss maintains consistency between supernet training and subnet retraining, providing informative objectives for each block and shortening gradient propagation paths, resulting in approximately 50% reduction in MACs and parameters for latent diffusion models.

Moreover, ensemble learning has been integrated into the NAS-KD pipeline to further boost the

performance of compact models. The [121] method constructs multiple branches over the primary model stream, transferring knowledge among sub-neural networks within an ensemble using adversarial learning strategies. This self-distillation approach enhances the performance of lightweight encoder-decoder architectures without incurring extra computational overhead during inference. Similarly, [122] introduces AdaNAS, which uses ensemble techniques to compose networks from smaller sub-networks, iteratively training them using the previous ensemble as a teacher via knowledge distillation.

Advanced search strategies continue to evolve to handle complex constraints and structural limitations. For example, [123] revisits layer-width optimization by adding depth-wise binary convolutions to learn pruning policies directly through gradient descent, combining structural reparameterization with search to enable arbitrary width flexibility. Additionally, [124] breaks structural limitations by learning the number of channels and layers through a parameterized distribution, aggregating feature map fragments from networks of different sizes and tuning them via knowledge transfer. These collective advancements illustrate that the fusion of knowledge distillation and neural architecture search provides a robust pathway for compressing complex generative models. By defining efficient, specialized architectures, these methods lay the foundational structure upon which the dynamic inference mechanisms and hardware-aware optimizations discussed in the following section can be effectively deployed, enabling high-quality image synthesis on edge devices while maintaining theoretical fidelity and computational efficiency.

2.6 Dynamic Inference and Hardware-Aware Optimization for Edge Deployment

2.6 Dynamic Inference and Hardware-Aware Optimization for Edge Deployment

Building upon the efficient, specialized architectures derived from Knowledge Distillation and Neural Architecture Search in the previous section, the deployment of advanced generative models—particularly diffusion and flow-based architectures—on resource-constrained edge devices necessitates a further shift from static execution to adaptive runtime strategies. While NAS defines an optimal structural baseline, the iterative nature and substantial computational demands of generative synthesis require dynamic inference mechanisms that adjust computational pathways in real-time based on input complexity, available resources, and hardware-specific constraints. A primary paradigm in this domain is dynamic weight slicing, which enables networks to adaptively adjust their capacity without incurring the memory access penalties associated with unstructured sparsity. For instance, the concept of dynamic weight slicing allows models to activate only a portion of network parameters for inputs of varying difficulty while keeping weights stored statically and contiguously in hardware. This method, exemplified by frameworks like DS-Net and DS-Net++, effectively prevents the overhead of sparse computation, achieving significant real-world acceleration on both CNNs and Transformers with minimal accuracy degradation [125]. Similarly, the Dynamic Slimmable Network (DS-Net) employs a double-headed dynamic gate to predictively adjust network width, utilizing techniques like in-place ensemble bootstrapping to ensure the generality of sub-networks [126].

Complementing weight adaptation is the strategy of input-dependent computation skipping, often realized through early exiting or layer skipping. This mechanism allows the inference process to terminate once a confident prediction is reached or to bypass redundant layers for “easy” samples, thereby conserving energy and reducing latency. The Dual Dynamic Inference (DDI) framework integrates both input-dependent and resource-dependent mechanisms, employing a multi-grained learning to skip (MGL2S) approach that facilitates simultaneous layer-wise and channel-wise skip-

ping to meet strict IoT resource constraints [127]. In the context of speech recognition, dynamic layer-skipping can be triggered by specific output signals, such as CTC blanks, to dynamically adjust encoder depth and accelerate inference [128]. Furthermore, resolution adaptive networks address spatial redundancy by routing low-resolution representations for easy inputs while reserving high-resolution processing for complex samples, effectively balancing accuracy and computational cost [129]. These dynamic strategies are particularly crucial for generative models, where the number of sampling steps can be adaptively reduced based on the convergence of the generated image quality, directly leveraging the structural flexibility established by prior compression techniques.

Beyond algorithmic adaptations, optimizing the underlying dataflow is critical for maximizing hardware utilization on heterogeneous edge platforms, ensuring that the compact architectures identified via NAS operate at peak efficiency. Traditional operator-centric optimization often fails to fully exploit the parallelism and memory hierarchy of modern edge processors. In response, dataflow-centric optimizations have emerged as a potent solution. The Xenos framework, for example, automatically conducts dataflow-centric optimization by restructuring inter-operator dataflows to improve data locality vertically and enabling higher parallelism across multiple DSP units horizontally. This approach has demonstrated substantial inference time reductions on edge devices, significantly outperforming conventional compiler stacks [130]. For attention-heavy architectures common in modern generative backbones, specialized dataflows like FLAT transform the quadratic memory growth of attention operations into linear patterns through unique fusion and tiling mechanisms, thereby mitigating memory bottlenecks on resource-scarce accelerators [131]. Additionally, formal dataflow models such as PRUNE provide a decidable and dynamic framework for deploying signal processing applications on heterogeneous CPU-GPU platforms, ensuring deadlock freedom and bounded memory while maximizing simultaneous device utilization [132].

Hardware-aware optimization also extends to managing the stochastic nature of edge environments, where network conditions and available compute power fluctuate. Adaptive resource management systems leverage reinforcement learning to dynamically allocate computation and communication resources. For instance, AutoScale employs a reinforcement learning engine to select the most energy-efficient execution target among CPUs, co-processors, or cloud offloading, adapting to stochastic runtime variances in mobile-cloud environments [133]. In distributed settings, frameworks like HexGen support asymmetric partitioning of generative inference across heterogeneous GPUs, using constrained optimization to schedule tasks effectively in decentralized networks [134]. Moreover, latency-aware unified dynamic networks integrate algorithmic design with scheduling optimization, utilizing latency predictors to guide dynamic computation paradigms such as spatial adaptation and channel skipping, ensuring that theoretical efficiency gains translate into actual latency reductions on specific hardware like GPUs and edge SoCs [135]. The synergy between dynamic algorithmic behavior and hardware-specific dataflow tuning, as seen in latency-aware spatial-wise dynamic networks, further bridges the gap between theoretical FLOPs reduction and practical end-to-end speedup by accounting for scheduling strategies and hardware properties [136]. Collectively, these advancements form a comprehensive toolkit for deploying controllable generative models on the edge, ensuring that high-fidelity synthesis remains feasible even under stringent power and latency budgets, thereby setting the stage for the application-specific controls and user-interaction mechanisms explored in subsequent sections.

3 Mechanisms for Precise Spatial, Structural, and Semantic Control

3.1 Adapter-Based Frameworks for Structural Guidance

3.1 Adapter-Based Frameworks for Structural Guidance

While natural language prompts offer semantic flexibility, they often lack the precision required to dictate specific geometric layouts, object poses, or scene depth. To bridge this gap between abstract semantic intent and concrete structural fidelity, the research community has increasingly turned to adapter-based frameworks. These methodologies introduce trainable or training-free modules that inject explicit spatial conditions—such as edge maps, depth estimations, segmentation masks, and human poses—directly into the generative backbone of pre-trained diffusion models. By decoupling the structural guidance mechanism from the core generative weights, these adapters enable fine-grained control without compromising the rich semantic knowledge acquired during the original pre-training phase. This approach serves as a foundational step in controllable generation, establishing the pixel-level and feature-level alignment necessary before advancing to higher-order semantic structuring discussed in subsequent sections.

The paradigm of trainable adapters was revolutionized by the introduction of ControlNet, a neural network architecture designed to lock the parameters of large, pre-trained diffusion models while learning a diverse set of conditional controls [137]. The core innovation of this approach lies in its “zero convolution” strategy, where convolutional layers are initialized with zero weights. This ensures that the initial state of the adapter introduces no noise or distortion to the frozen backbone, allowing the model to progressively learn spatial conditioning from scratch without degrading the original generation quality. ControlNet has demonstrated remarkable robustness across various conditioning types, including canny edges, depth maps, and human poses, effectively enabling the model to align generated images with complex spatial inputs even when trained on relatively small datasets [137]. Building upon this foundation, subsequent works have sought to optimize the efficiency and flexibility of such architectures. For instance, ControlNet-XS addresses the latency issues inherent in original controlling networks by redesigning the information flow to eliminate delays between the generation and controlling processes, resulting in a model that is significantly faster and requires fewer parameters while achieving higher fidelity control [138]. Similarly, SCEdit proposes an efficient tuning framework that focuses on editing skip connections within the U-Net architecture, integrating lightweight tuners to inject conditions while substantially reducing memory usage and computational expense compared to full backbone modifications [139].

Beyond single-condition control, the field has evolved towards unified frameworks capable of handling multiple simultaneous constraints. Uni-ControlNet introduces a unified architecture that allows for the composable utilization of diverse local controls (e.g., edges, depth) and global controls (e.g., image embeddings) within a single model, requiring only the fine-tuning of two additional adapters regardless of the number of conditions used [140]. This approach significantly reduces the deployment footprint and enhances the composability of different spatial cues. Furthermore, Meta ControlNet leverages meta-learning techniques to address the slow adaptation speed of standard ControlNets, enabling zero-shot adaptability for edge-based tasks and rapid fine-tuning for complex non-edge tasks like human pose estimation, thereby reducing the required learning steps from thousands to merely a handful [141]. For scenarios requiring instance-level appearance control alongside geometric constraints, FineControlNet extends these capabilities by spatially aligning instance-specific text prompts with 2D poses, allowing users to dictate not just the shape but also the visual appearance of specific entities within the scene [142].

Parallel to the development of trainable adapters, a significant body of work has emerged focusing on training-free alternatives that achieve structural alignment through attention manipulation. These methods are particularly valuable for their ability to work with any pre-trained model checkpoint without the need for additional parameter updates. FreeControl exemplifies this trend by designing structure and appearance guidance mechanisms that facilitate alignment with a guidance image across multiple architectures and checkpoints simultaneously, supporting challenging input conditions where other training-free methods often fail [143]. Another notable approach is Readout Guidance, which utilizes lightweight readout heads to extract signals such as pose or depth from the features of a frozen diffusion model; these signals are then compared to user-defined targets to guide the sampling process via back-propagation, offering a flexible recipe for conditional control with minimal added parameters [144]. Additionally, methods like LoCo introduce localized attention constraints that leverage semantic affinity in self-attention maps to ensure accurate object placement according to layout instructions, effectively addressing semantic failures in layout-to-image synthesis without any fine-tuning [145].

The integration of these adapter-based and attention-based mechanisms has also led to more sophisticated handling of conflicting or rough visual conditions. SmartControl, for example, modifies rough visual conditions by relaxing constraints in areas that conflict with text prompts, using a predicted control scale to balance the influence of spatial and semantic inputs [146]. Similarly, Decompose and Realign offers a training-free strategy to handle partially aligned conditions by separating conflicting pairs and realigning them through cross-attention mechanisms, thus improving controllability when perfect alignment between text and extra conditions is absent [147]. Moreover, the concept of local control has been expanded to allow users to manipulate specific regions of an image while leaving the rest governed by the text prompt, a capability achieved through training-free methods that mitigate local control dominance problems [148]. Collectively, these adapter-based frameworks provide the essential low-level structural scaffolding—defining edges, depths, and poses—that serves as a prerequisite for the more abstract, object-centric logical structuring explored in the following section on layout-to-image synthesis.

3.2 Layout-to-Image Synthesis via Scene Graphs and Bounding Boxes

3.2 Layout-to-Image Synthesis via Scene Graphs and Bounding Boxes

Building upon the pixel-level structural scaffolding established by adapter-based frameworks, the field advances to a higher order of semantic structuring through layout-to-image synthesis. While the preceding section detailed how edge maps and depth estimations enforce low-level geometric fidelity, this paradigm addresses the complex challenge of composing multiple interacting entities with explicit semantic logic. By leveraging structured inputs such as scene graphs, bounding boxes, and semantic masks, layout-to-image synthesis bridges the gap between abstract relational descriptions and concrete visual arrangements. This approach decomposes the generative process into distinct stages: first inferring or accepting a geometric and semantic layout that dictates object positioning and inter-object relationships, and subsequently rendering high-fidelity imagery conditioned on this rigorous scaffold. This transition from local feature alignment to global scene composition sets the stage for the fine-grained semantic binding challenges discussed in the following section.

Scene graphs serve as a powerful intermediate representation in this domain, encoding objects as nodes and their semantic or spatial relationships as edges. Early approaches recognized that directly mapping unstructured text to images often fails to capture complex relational structures, leading

to missing objects or incorrect interactions. To mitigate this, researchers proposed frameworks that generate images directly from scene graphs, enabling explicit reasoning about object connectivity [149]. These methods typically employ graph convolutional networks to process the input graph structure, predicting an intermediate layout of bounding boxes and segmentation masks before converting them into final images through cascaded refinement networks. However, generating coherent layouts from scene graphs remains challenging due to the diversity of object scaling and spatial configurations. Advanced frameworks have introduced spatial constraint modules specifically designed to fit reasonable scaling and spatial arrangements for object pairs by considering their inherent relationships [150]. Furthermore, to capture the nuanced dependencies between objects, contextual fusion modules have been developed to integrate pair-wise spatial information, ensuring that the generated layouts adhere to commonsense logic [150].

The fidelity of the final image is heavily dependent on the quality of this intermediate layout. Traditional methods often generated object layouts independently, resulting in significant overlap and low coverage. To address these issues, sequential approaches treat layout generation as a sequence-to-sequence learning task, where a scene graph is decomposed into semantic fragments and transduced into a series of “brick-action” code segments that dictate position and scale [151]. This sequential modeling allows the system to progressively construct the layout, thereby improving inter-object dependency and reducing collisions. Similarly, other works have adopted a hierarchical approach, constructing semantic layouts in a coarse-to-fine manner by first generating bounding boxes and then refining object shapes within those bounds [152]. By decoupling the layout inference from the image synthesis, these models not only improve image quality but also offer users the ability to modify the scene structure explicitly before rendering, providing a logical blueprint that precedes pixel generation.

Beyond scene graphs, direct control via bounding boxes and attributes provides a more granular level of manipulation for users who wish to specify exact spatial coordinates. Methods such as Layout2Im demonstrate that disentangling object representations into certain (category) and uncertain (appearance) parts allows for the generation of diverse images consistent with a specified coarse spatial layout [153]. To enhance instance-level control, attribute-guided generation frameworks accept tuples containing bounding boxes, categories, and specific visual attributes, enabling the synthesis of images where objects possess prescribed appearances at designated locations [154]. This capability is crucial for applications requiring precise stylistic consistency across multiple instances within a single scene. Moreover, object-centric generative adversarial networks have been designed to mitigate failure modes such as spurious object generation and merged objects caused by overlapping boxes, utilizing scene-graph similarity modules to learn robust spatial relationship representations [155].

The challenge of controlling multiple distinct objects at spatially distinct locations has also been tackled through dual-pathway architectures. These systems utilize an “object pathway” that focuses iteratively on individual objects at specified bounding box locations, alongside a “global pathway” that manages the background and overall image composition [156]. This separation ensures that local object details are preserved without compromising global coherence. In scenarios involving complex urban environments, semantic control can be further refined by conditioning layout generation on class proportions, allowing the model to produce layouts that match specific compositional requirements before image synthesis [157].

Recent advancements have integrated diffusion models into this domain to leverage their superior generation quality while adhering to layout constraints. For instance, diffusion-based approaches pre-train encoders to align scene graph embeddings directly with image features using masked

contrastive pre-training, thereby optimizing the semantic compliance between the graph structure and the generated output [158]. Additionally, training-free methods like LoCo have emerged, utilizing localized attention constraints within pre-trained diffusion models to enforce precise object placement based on layout instructions without requiring additional fine-tuning [145]. While these methods successfully enforce spatial boundaries, they primarily rely on hard constraints or external layout predictors, leaving the internal mechanism of how text tokens bind to these spatial regions less addressed—a gap that motivates the cross-attention manipulation strategies explored next.

The scope of layout-to-image synthesis extends beyond natural scenes to specialized domains such as document design and GUI creation, demonstrating the versatility of layout-guided generation. Frameworks like DocSynth adapt these principles to synthesize realistic document images from spatial layouts defined by bounding boxes and categories, facilitating data augmentation for layout analysis tasks [159]. Similarly, in the realm of graphical user interfaces, transformer-based models generate layouts that strictly adhere to positional constraints and design rules, demonstrating the versatility of layout-guided generation across diverse visual domains [160]. Ultimately, the integration of scene graphs and bounding boxes into generative pipelines provides a robust mechanism for bridging the gap between abstract semantic descriptions and concrete visual realizations. By enforcing structural constraints through intermediate layouts, these methodologies ensure that generated scenes are not only visually realistic but also semantically grounded and spatially logical. However, while these layouts define *where* objects should be, ensuring that the specific textual attributes correctly bind to these spatial regions requires a deeper intervention in the model’s attention mechanisms, as discussed in the subsequent section.

3.3 Cross-Attention Manipulation for Semantic Alignment and Binding

3.3 Cross-Attention Manipulation for Semantic Alignment and Binding

While the preceding section established how explicit spatial layouts—defined by scene graphs and bounding boxes—can enforce geometric structure, a distinct challenge persists even when such constraints are absent or loosely defined: the inherent ambiguity in how standard diffusion models associate textual tokens with visual regions. The remarkable fidelity of modern text-to-image diffusion models often masks a fundamental limitation in their compositional reasoning: the inability to strictly bind attributes to their corresponding entities. This phenomenon, frequently manifested as attribute leakage (e.g., assigning the color of one object to another) or entity leakage (merging distinct subjects), stems from the way cross-attention mechanisms distribute semantic information across the latent space. Unlike the rigid scaffolding of layout-based methods, standard softmax attention tends to blur the boundaries between tokens, especially in complex prompts involving multiple subjects or intricate spatial relationships. To address these deficiencies without the prohibitive costs of retraining massive foundation models, a significant body of research has emerged focusing on cross-attention manipulation. These strategies refine the attention maps during the inference process, enforcing stricter semantic alignment through masking, structural guidance, and dynamic optimization, effectively bridging the gap between the coarse control of layouts and the fine-grained feature disentanglement discussed in the following section.

A primary avenue for resolving attribute binding errors involves the direct modulation of attention maps using spatial priors, creating a soft counterpart to the hard constraints of bounding boxes. By predicting or specifying bounding boxes for entities mentioned in the prompt, models can constrain

the attention regions of specific tokens to their designated spatial locations. For instance, recent approaches have introduced novel attention mask control strategies where a dedicated network predicts object boxes, which are then used to apply unique mask controls to both cross- and self-attention maps [161]. This method effectively restricts the influence of each text token to its corresponding image region, thereby preventing the model from incorrectly attributing features to the wrong objects. Similarly, other training-free mechanisms have been developed to guide the model to concentrate on specific syntactic components of the prompt at distinct timesteps. These approaches often incorporate temperature control within early self-attention phases to mitigate entity leakage and utilize object-focused masking schemes in cross-attention modules to discern semantic affiliations more effectively [162].

Beyond simple masking, advanced techniques leverage the structural properties of linguistic inputs to enforce semantic fidelity, moving from geometric constraints to syntactic logic. The observation that keys and values in cross-attention layers possess strong semantic meanings associated with object layouts has led to the development of structured diffusion guidance. By manipulating cross-attention representations based on linguistic insights, such as syntactic dependency trees, models can better preserve compositional semantics during generation [163]. This structured approach ensures that the relationship between adjectives and nouns, or subjects and verbs, is maintained throughout the denoising process. Furthermore, focused cross-attention mechanisms have been proposed to control visual attention maps by utilizing syntactic constraints found directly in the input sentence. When combined with disentangled multimodal embeddings, these methods significantly improve attribute-object binding without requiring additional model training [164].

The challenge of semantic alignment is particularly acute in scenarios involving multiple subjects, where inadvertent information flow between subjects leads to visual blending. To counteract this, “Bounded Attention” methods have been introduced to strictly bound the information flow during the sampling process. By preventing detrimental leakage among subjects, these training-free techniques empower the generation of multiple distinct entities that align faithfully with complex prompts and layouts [165]. In multi-subject customization, the interference between concepts can be alleviated by rectifying activations in the cross-attention map using layout priors, which appoints and separates the locations of different subjects [166]. Additionally, dynamic prompt learning strategies have been formulated to address cross-attention leakage in image editing tasks. By updating dynamic tokens for nouns with specific leakage repairment losses, these methods force cross-attention maps to focus on the correct noun words, ensuring fine-grained editing while preventing unintended modifications to background or distractor objects [167].

Another critical dimension of cross-attention manipulation involves the adaptive adjustment of guidance scales across different semantic regions, refining the global control mechanisms often used in broader generation tasks. Standard Classifier-Free Guidance (CFG) applies a global scale, which can result in spatial inconsistencies where some semantic units are over-amplified while others are suppressed. Semantic-aware CFG addresses this by partitioning the latent image into independent semantic regions using renormalized cross-attention maps and adaptively adjusting the guidance degrees for each region [168]. This ensures a uniform level of text guidance across the entire composition. Furthermore, masked-attention guidance techniques have been developed to control generation according to given semantic masks by manipulating the noise images fed to the model, offering a more accurate spatial control than direct attention swapping [169].

The theoretical underpinnings of these manipulations often reveal that cross-attention outputs converge to a fixed point early in the inference process, suggesting that text conditions are primarily utilized for semantic planning in the initial stages. Exploiting this, methods like TGATE cache

cross-attention outputs once they converge, reducing computational complexity while maintaining performance [170]. Moreover, the efficacy of these manipulation strategies is often enhanced by understanding the max-margin nature of token selection in attention mechanisms, which formalizes attention as an optimal selection process [171]. In video editing contexts, similar principles apply, where spatial-temporal layout-guided attention mechanisms leverage cross-frame correspondences to boost attention scores for tokens within the same attribute while limiting interactions between different attributes, thus achieving accurate identity mapping across frames [172]. Collectively, these diverse strategies demonstrate that precise manipulation of cross-attention mechanisms serves as a potent, training-free lever for achieving high-fidelity semantic alignment. By resolving the ambiguity of token-to-pixel binding at the attention level, these methods lay the necessary groundwork for the subsequent disentanglement of content and style, where preserving structural integrity while altering appearance becomes paramount.

3.4 Disentanglement of Content and Style for Fine-Grained Control

3.4 Disentanglement of Content and Style for Fine-Grained Control

While the preceding section detailed how cross-attention manipulation resolves semantic ambiguity and enforces token-to-region binding, a complementary challenge arises once spatial alignment is secured: the independent control of *what* is depicted versus *how* it appears. Even with perfect semantic alignment, standard generative models often conflate structural content with artistic style, leading to unintended geometric distortions when altering appearance, or conversely, texture degradation when modifying structure. To achieve truly fine-grained controllability, it is essential to decouple these factors, enabling the transfer of complex artistic patterns while rigorously preserving the underlying scene geometry established by the layout and attention mechanisms discussed previously. The theoretical underpinning of this separation typically defines content as the spatial arrangement and semantic identity of objects, while style encompasses the statistical properties of textures, colors, and brushstrokes. This disentanglement serves as a critical bridge, transitioning from the semantic fidelity explored in Section 3.3 to the dense, region-specific conditioning required for complex scene composition in the subsequent section.

A primary strategy for achieving this separation involves the design of specialized attention mechanisms and normalization layers that explicitly isolate feature representations. For instance, the concept of diagonal attention has emerged as a powerful tool for hierarchical disentanglement. By employing hierarchical adaptive Diagonal spatial ATtention (DAT) layers, researchers have demonstrated the ability to separately manipulate spatial contents from styles in a coarse-to-fine manner [173]. This approach leverages a mathematical understanding of normalization and attention to ensure that the generator can flexibly control spatial features independently of stylistic attributes, outperforming traditional models where global content and style remain entangled. Similarly, the integration of adaptive attention normalization modules, such as AdaAttN, facilitates the adaptive performance of attentive normalization on a per-point basis by learning spatial attention scores from both shallow and deep features [174]. This ensures that local feature statistics are matched appropriately, preventing the unnatural distortions often caused by ignoring shallow feature distributions, thereby reinforcing the structural integrity maintained by the cross-attention controls described earlier.

Beyond attention mechanisms, unsupervised learning frameworks have been developed to distill geometry information into distinct latent spaces without requiring paired training data. A notable example is the two-branch Autoencoder framework that projects latent factors into a soft struc-

tured point tensor to constrain the structural content branch, while a complementary branch learns style information [175]. This method effectively disentangles object representations by distilling geometry information through prior knowledge constraints, enabling the generation of photo-realistic images where content and style are clearly separated. Furthermore, the Retriever framework models content-style representation as a token-level bipartite graph, utilizing cross-attention to retrieve permutation-invariant style information and vector quantization to produce interpretable content tokens [176]. This modal-agnostic approach has proven effective in both speech and image domains, verifying the interpretability of the decomposed representations and providing a robust foundation for manipulating specific attributes without affecting others.

In the realm of diffusion models, which have recently dominated the generative landscape, disentanglement is often achieved through feature injection strategies and prompt engineering that build upon the semantic bindings established in prior stages. The challenge here lies in leveraging the vast knowledge of pre-trained large-scale models while avoiding the degradation of content structure. To address this, frameworks like InstantStyle introduce a mechanism that decouples style and content within the feature space by injecting reference image features exclusively into style-specific blocks [177]. This strategy prevents style leakage into content regions and eliminates the need for cumbersome weight tuning, striking an optimal balance between style intensity and textual controllability. Similarly, the Adaptive Style Incorporation (ASI) method proposes a Siamese Cross-Attention structure to decouple single-track cross-attention into dual tracks, obtaining separate content and style features before blending them in a structure-consistent manner [178]. These approaches highlight a shift from prompt-level concatenation, which often causes structure distortions, to fine-grained feature-level incorporation, ensuring that the semantic alignments achieved via cross-attention manipulation are not compromised during stylization.

The extension of these disentanglement principles to three-dimensional and dynamic domains further underscores their versatility and necessity for consistent generation across views and time. In 3D scene stylization, maintaining multi-view consistency while transferring style requires robust separation of geometry and appearance. Methods such as StyleRF perform style transformation within the feature space of a radiance field, employing sampling-invariant content transformation to ensure that the stylization remains invariant to the holistic statistics of sampled 3D points [179]. This resolves the dilemma between accurate geometry reconstruction and high-quality stylization. Moreover, the ARF-Plus framework extends this controllability by offering manageable control over perceptual factors, including color preservation, style pattern scale, and spatial selection, allowing for the customized modification of stylization effects in 3D scenes [180]. For dynamic 4D scenes, StyleDyRF learns a linear style transformation matrix on a canonical feature volume, effectively separating the dynamic geometric prior from the static style pattern to achieve consistent 4D style transfer [181].

Loss function design also plays a critical role in enforcing disentanglement, complementing the architectural innovations with optimization objectives that penalize entanglement. Traditional Gram matrix-based losses often fail to capture fine-grained style details or preserve structure, leading to blurry results. Consequently, newer approaches utilize nearest neighbor-based losses that are highly effective at capturing style details while maintaining consistency, as seen in artistic radiance fields [182]. Additionally, the introduction of pattern repeatability metrics helps attention mechanisms learn precise semantic correspondence, avoiding the repetitive artifacts that arise from abusing specific local patches [183]. By combining these advanced architectural designs with sophisticated loss functions and feature space manipulations, modern generative models achieve a level of fine-grained control that allows for the independent and precise manipulation of scene geome-

try and visual appearance. This capability marks a significant evolution from early style transfer techniques, establishing the necessary precondition for the region-specific control and localized text conditioning discussed in the following section, where distinct textual descriptions must govern unique image regions without mutual interference.

3.5 Region-Specific Control and Localized Text Conditioning

3.5 Region-Specific Control and Localized Text Conditioning

Building upon the principles of disentangling global content and style discussed in the previous section, the evolution of text-to-image synthesis has progressed toward enabling fine-grained, region-specific control. While early generative models effectively captured the holistic semantics of a prompt, they often lacked spatial precision, failing to align specific textual attributes with designated areas of the canvas. This limitation necessitated the development of dense text-to-image synthesis frameworks, where distinct textual descriptions govern unique image regions, allowing for complex scene composition that mirrors human creative intent. The core challenge in this domain lies in effectively fusing localized textual features with corresponding visual regions without introducing semantic leakage or structural incoherence, a prerequisite for the multi-subject customization explored in the subsequent section.

One of the foundational approaches to addressing this challenge involves decomposing the generation process into hierarchical steps that infer and utilize semantic layouts. By first constructing a semantic layout from text—comprising object bounding boxes and shapes—and subsequently converting this layout into an image, models can achieve significantly higher semantic alignment [152]. This two-stage paradigm ensures that the spatial distribution of objects is explicitly modeled before pixel-level synthesis begins, effectively circumventing the difficulty of preserving cross-modality consistency in direct mapping methods [184]. Furthermore, advancements in this area have led to the formulation of dense text-to-image (DT2I) synthesis as a distinct task, moving beyond fixed class labels to allow for semantically rich region descriptions. Methods such as DTC-GAN introduce multi-modal region feature matching losses to encourage precise semantic image-text matching, thereby enabling the generation of plausible images for complex scenes defined by regional captions [185].

To enhance the granularity of control further, recent research has focused on manipulating the internal attention mechanisms of generative models. A pivotal insight is that the self-attention maps within diffusion models inherently contain spatial information that can be leveraged to guide object placement. Training-free approaches have emerged that modulate these attention maps to enforce layout constraints, allowing pre-trained models to handle dense captions without additional fine-tuning. For instance, methods utilizing attention modulation analyze the relationship between intermediate attention maps and desired layouts to guide objects into specific regions, achieving performance comparable to models specifically trained with layout conditions [186]. Similarly, the LoCo framework introduces Localized Attention Constraints (LAC) that exploit semantic affinity between pixels in self-attention maps to ensure accurate object placement, alongside Padding Token Constraints (PTC) to improve consistency between object appearance and layout instructions [145].

Beyond static layout adherence, the concept of “rich text” interfaces has been proposed to provide users with continuous control over region-specific attributes. By extracting word attributes from formatted text—such as font style, color, and size—models can enforce local style control and precise color rendering for specific regions. This is achieved through a region-based diffusion process where

region-specific detailed prompts and guidance are applied to maintain fidelity against plain-text generation [187]. This approach addresses the tedium of crafting detailed prompts for complex scenes and overcomes the limitations of standard text encoders in interpreting precise quantitative or spatial specifications.

The integration of Large Language Models (LLMs) has further revolutionized region-specific control by enabling the decomposition of intricate textual descriptions into coherent directives. The Progressive Text-to-Image Diffusion with Soft Latent Direction framework utilizes LLMs to break down protracted prompts into sequential operations such as insertion, editing, and erasing. Through a Stimulus, Response, and Fusion (SRF) framework, latent regions are stimulated in alignment with each operation, allowing for the systematic incorporation of multiple entities while adhering to strict spatial and relational constraints [188]. This progressive synthesis strategy effectively manages the complexity of scenes with numerous interacting objects, a scenario where standard diffusion models often falter.

Moreover, the disentanglement of content and style plays a crucial role in achieving fine-grained regional control, bridging the gap between the global style transfer techniques of Section 3.4 and local editing. Techniques that learn disentangled latent representations allow for the independent manipulation of attributes within specific spatial regions. For example, StyleT2I employs a Semantic Matching Loss and Spatial Constraint to identify latent directions for intended spatial region manipulations, facilitating compositional attribute adjustment [189]. This ensures that changes to one region’s attributes do not inadvertently affect the semantics of other parts of the image. Additionally, frameworks that incorporate global semantic fusion and self-similarity feature enhancement modules have been developed to guide object coherence, ensuring that both semantic relationships (e.g., a cat looking at flowers) and physical coherence (e.g., proper alignment of hand and racket) are maintained across different regions [190].

Finally, the flexibility of these region-specific controls extends to varying levels of precision, ranging from pure text to precise 2D semantic canvases. The SceneComposer framework exemplifies this by supporting any-level semantic image synthesis, where users can define regions with adjustable precision levels. This is achieved through a precision-encoded mask pyramid and text feature map representation that jointly encodes precision, semantics, and composition information, allowing the model to adapt to users with different levels of drawing expertise [191]. Collectively, these advancements demonstrate a robust trajectory toward fully controllable, dense text-to-image synthesis. By establishing rigorous mechanisms for localizing text conditioning and managing spatial layouts, these methods lay the essential groundwork for the more complex challenge of multi-subject customization and concept composition discussed in the following section.

3.6 Multi-Subject Customization and Concept Composition

3.6 Multi-Subject Customization and Concept Composition

Building upon the foundation of region-specific control and localized text conditioning established in the previous section, the field now confronts the more intricate challenge of multi-subject customization and concept composition. While Section 3.5 demonstrated how to align specific textual attributes with designated spatial regions, synthesizing complex scenes containing multiple *user-defined* subjects introduces profound new hurdles. The core difficulty shifts from mere spatial alignment to preserving distinct semantic boundaries between personalized entities. In multi-concept scenarios, standard diffusion models often suffer from “concept confusion,” where attributes of one subject bleed into another, or “concept vanishing,” where specific subjects fail to appear entirely.

Addressing these issues requires mechanisms that not only enforce spatial constraints via layout priors but also rigorously decouple subject representations within disentangled embedding spaces to prevent identity dilution.

A primary strategy for mitigating concept interference involves the explicit manipulation of attention mechanisms guided by the spatial layouts discussed previously. Recent advancements have demonstrated that abstract layout priors, such as bounding boxes, serve as effective spatial guidance to separate and appoint locations for different subjects, thereby significantly alleviating interference across them [166]. By rectifying activations in the cross-attention maps, these methods ensure that the text embeddings corresponding to specific subject tokens are attended to only within their designated regions. This approach is further refined in training-free frameworks like LoRA-Composer, which introduces Concept Isolation Constraints to refine self-attention computations and Concept Injection Constraints to enhance visibility, effectively preventing the merging of distinct conceptual features [192]. Furthermore, the integration of region-constrained cross-attention allows for the confinement of subject-related word impacts to desired regions, addressing unwanted artifacts in transition areas and improving overall scene coherence [193].

Beyond spatial masking, the architectural design of the attention mechanism itself plays a critical role in disentangling subject identity from the background context. The Mixture-of-Attention (MoA) architecture exemplifies this by distributing the generation workload between two distinct pathways: a personalized branch and a non-personalized prior branch [194]. Inspired by the Mixture-of-Experts paradigm, MoA utilizes a novel routing mechanism to manage the distribution of pixels across these branches, allowing the model to retain its original generative prior while minimally intervening to embed specific subjects. This separation enhances the distinction between pre-existing capabilities and augmented personalized interventions, facilitating the creation of high-quality images featuring multiple subjects with diverse compositions and interactions. Similarly, the challenge of occlusion in multi-concept generation has been addressed by frameworks like OMG, which employs a two-stage sampling solution [195]. The first stage focuses on layout generation and visual comprehension to handle occlusions, while the second stage utilizes noise blending strategies to integrate multiple concepts seamlessly, ensuring that identity preservation is maintained even when subjects overlap or interact physically.

The representation of subjects in the embedding space is another crucial factor, extending the disentanglement principles from regional style control to full concept modularity. Traditional methods that rely on fine-tuning shared parameter spaces often result in entangled content and style, making it difficult to compose multiple concepts harmoniously. To overcome this, researchers have proposed separating the parameter space to facilitate individual learning of content and style [196]. By breaking original adapters into divided sub-parameter spaces and employing a “break-for-make” learning pipeline, these methods enable the reconstruction of entity parameter spaces that support composable customization. Additionally, the use of residual learning on top of base text embeddings has proven effective for robustly shifting raw subjects to customized ones without extensive model tuning, supporting arbitrary combinations of subjects [166]. For scenarios requiring attribute-aware personalization, the Prompt Spectrum Space offers an expanded textual conditioning space where images are represented as collections of inverted token embeddings encoded from per-stage prompts, providing superior disentanglement compared to existing methods [197].

Furthermore, ensuring semantic coherence among multiple objects remains a persistent challenge, necessitating a move beyond simple co-presence to logical interaction. Methods such as those utilizing Global Semantic Fusion (GSF) argue that image captions contain rich information for defining semantic relationships within objects, which can be fused with layout restrictions to guide

the synthesis process [190]. This is particularly important when generating scenes where objects must interact logically, such as a person holding a specific tool or multiple entities engaging in a shared activity. The ability to control not just the presence but the interaction of subjects is also evident in works addressing persons in context synthesis, where separate neural branches attentively composite context and person inputs into a shared “compositional structural space” [198]. This disentangled encoding of shape, location, and appearance allows for the synthesis of diverse person instances in consistent contexts, even when input structures like bounding boxes and keypoints differ starkly.

Finally, the trade-off between subject fidelity and generative diversity is a key consideration in multi-subject customization. Approaches like Pick-and-Draw introduce training-free semantic guidance to boost identity consistency while maintaining generative diversity by constructing an appearance palette and inheriting strong image priors for diverse contexts [199]. This ensures that while multiple subjects are faithfully rendered, the surrounding environment and their interactions remain varied and natural. As the field progresses, the combination of layout-guided attention, architectural disentanglement, and advanced embedding strategies continues to push the boundaries of what is possible in multi-concept generation, moving closer to the human ability to compose complex, multi-entity scenes with precise control over each element’s identity and relationship.

4 Advanced Editing, Personalization, and Inverse Problem Solving

4.1 Training-Free and Optimization-Efficient Personalization

4.1 Training-Free and Optimization-Efficient Personalization

While the preceding discussions on foundational diffusion and flow models establish the theoretical capacity for high-fidelity synthesis, a critical bottleneck remains in translating these capabilities into responsive, user-centric applications. Traditional personalization paradigms, such as DreamBooth or Textual Inversion, rely heavily on computationally expensive test-time fine-tuning, often necessitating hundreds of gradient updates to adapt to a new concept or user preference. This latency creates a significant impediment for real-time interaction and deployment in resource-constrained environments. Consequently, the research community has pivoted towards training-free and optimization-efficient personalization strategies. These emerging methodologies aim to eliminate or drastically reduce iterative parameter updates during inference by leveraging in-context learning, hypernetwork-based weight prediction, and adapter-based instant personalization, thereby achieving high-fidelity customization with minimal computational overhead.

A primary avenue for achieving such efficiency is the adoption of in-context learning (ICL) principles within generative frameworks. Inspired by large language models where tasks are adapted via prompt examples rather than weight updates, recent works explore injecting visual concepts directly through context. The interplay between fine-tuning and in-context learning suggests that robust generalization can be achieved without modifying core model weights, provided the conditioning mechanism is sufficiently expressive. For instance, studies comparing few-shot fine-tuning against in-context learning have demonstrated that while fine-tuned models can generalize well out-of-domain when controlled for size, they exhibit higher variance, highlighting the stability advantages of context-based approaches [200]. Furthermore, the integration of natural language demonstrations with learned prompt embeddings, known as instruction prompt tuning, has shown that combining context with lightweight parameter adaptation can consistently reduce variance across tasks, although semantic similarity between the demonstration and the target input remains

a critical factor for success [201]. These findings underscore the potential of context-driven mechanisms to serve as a viable, stable alternative to heavy optimization loops.

Parallel to context-based methods, hypernetwork-based weight prediction has emerged as a powerful strategy for instant personalization. Instead of optimizing weights for each new task, a hypernetwork learns to generate task-specific parameters conditioned on an input description or reference image, transforming personalization from an iterative optimization task into a single forward pass. The efficacy of hypernetworks in generating conditional modules has been validated in multi-task learning scenarios, where prototype-based hyper adapters utilize instance-dense retrievers to generate modules in a sample-efficient manner, outperforming standard parameter-efficient fine-tuning methods especially in low-data regimes [202]. Similarly, unified frameworks employing shared hypernetworks to generate adapter parameters for transformers have demonstrated the ability to share knowledge across tasks while adapting to individual requirements with negligible parameter overhead [203]. In the domain of tabular data, the “HyperFast” framework exemplifies this paradigm by utilizing a meta-trained hypernetwork to generate task-specific neural networks in a single forward pass, effectively removing the need for training or tuning entirely [204]. Extending this to novel feature adaptation, Contextual HyperNetworks (CHN) have been proposed to generate parameters for extending base models to new features using metadata, offering significant speed-ups compared to re-training [205].

Adapter-based instant personalization represents another critical frontier, focusing on injecting trainable modules that can be rapidly configured without altering backbone weights. The “Instant-Booth” framework serves as a paradigmatic example, enabling instant text-guided image personalization without any test-time finetuning. By converting input images into textual tokens via a learnable encoder and introducing a few adapter layers to capture rich visual features, Instant-Booth achieves competitive identity preservation and language-image alignment while being orders of magnitude faster than test-time finetuning methods [206]. The efficiency of such adapter mechanisms is further supported by findings that tiny-attention adapters, which modify hidden states conditioned on all other positions, can outperform full fine-tuning while updating a minuscule fraction of parameters [207]. Moreover, the flexibility of adapter tuning allows for dynamic selection strategies; for example, adaptable weighting fusion methods dynamically adjust fine-tuning strategies for different samples, proving that adaptive multi-tuning can surpass standard fixed strategies in complex distributions [208].

Beyond architectural design, the efficiency of personalization is contingent upon robust adaptation protocols that mitigate distribution shifts without extensive re-optimization. Test-time adaptation (TTA) methods aim to adjust models to unseen test samples, but traditional optimization-based TTA can be computationally prohibitive on edge devices. To address this, optimization-free TTA frameworks have been developed, such as those utilizing exponential decay test-time normalization and prototype-based classifiers to adjust predictions without gradient descent, demonstrating superior efficiency for real-time deployment [209]. Additionally, forward-only adaptation methods that learn prompts via derivative-free strategies offer a viable path for quantized models where back-propagation is infeasible, achieving significant memory reductions while maintaining adaptation performance [210]. The challenge of balancing performance with calibration in these low-resource settings is also paramount; research indicates that self-ensembling techniques applied during adaptation can enhance both task performance and calibration, addressing the critical trade-off often observed in limited data setups [211].

In conclusion, the shift towards training-free and optimization-efficient personalization marks a transformative phase in generative AI, effectively dismantling the barriers of latency and computa-

tional cost. By leveraging the predictive power of hypernetworks, the flexibility of adapter modules, and the contextual reasoning of in-context learning, these methods enable high-fidelity personalization through single-forward-pass mechanisms and lightweight architectural modifications. This evolution paves the way for scalable, real-time generative systems capable of instant adaptation to user needs. Crucially, by decoupling the personalization process from heavy iterative optimization, these efficient frameworks lay the necessary groundwork for the subsequent challenge: ensuring that such rapid, flexible editing maintains strict identity preservation and allows for fine-grained attribute control, a topic explored in the following section.

4.2 Identity Preservation and Fine-Grained Attribute Editing

4.2 Identity Preservation and Fine-Grained Attribute Editing

Building upon the training-free and optimization-efficient personalization strategies discussed in Section 4.1, which successfully decouple adaptation from heavy iterative tuning, the focus now shifts to a more granular challenge: ensuring that such rapid, flexible editing maintains strict identity preservation while enabling precise attribute manipulation. The core dilemma in advanced image editing lies in the delicate trade-off between retaining the subject’s inherent identity and achieving high-fidelity attribute controllability. While generative models have achieved remarkable success in synthesizing realistic faces, applying surgical modifications—such as altering specific expressions, adjusting lighting conditions, or modifying hair textures—without inducing “attribute leakage” (where changes propagate to irrelevant regions) or “identity collapse” (where the subject becomes unrecognizable) remains a significant hurdle. To address these issues, recent research has evolved towards sophisticated frameworks that leverage staged noise guidance, hybrid attention mechanisms, and decoupled latent representations to achieve unprecedented precision in facial editing.

A prominent strategy to navigate this trade-off involves staged editing frameworks that disentangle the modification process into distinct phases, optimizing structural details and global consistency separately. For instance, the concept of staged noise-guided editing is effectively realized in frameworks like [212], which introduces a dual-stage mechanism comprising an editing stage and a boosting stage. By discerning these stages via the frequency and gradient of predicted noises, such methods first perform detailed manipulations on specific features guided by high-frequency information, ensuring fine-grained control. Subsequently, a stochastic denoising process in the boosting stage enhances overall image quality while rigorously preserving the identity-context of the source subject. This approach effectively mitigates the common pitfall where global regeneration leads to unintended drifts in facial identity. Similarly, [213] advances this paradigm by incorporating space-sensitive physical customization derived from 3D Morphable Models (3DMM). This ensures that attribute transfers, such as pose or lighting changes, adhere to physiological constraints, while a region-responsive semantic composition module learns decoupled features to preserve non-facial attributes like background and clothing, thereby enhancing both fidelity and generalization.

Beyond staged diffusion processes, the manipulation of latent spaces has evolved from simple linear traversals to complex, non-linear trajectories and region-specific inversions, drawing insights from both GANs and diffusion architectures. Early methods often assumed linearly separable attribute directions, an assumption that fails for complex, non-textured variations. [214] demonstrates that realizing latent code manipulations as flows of trainable Neural ODEs offers superior controllability for non-linear attribute changes compared to simple vector arithmetic. Furthermore, the fidelity of editing is heavily dependent on the quality of the initial mapping into the latent space. [215]

addresses the distortion-editability trade-off by introducing pivotal tuning, a technique that slightly alters the generator around an inverted latent pivot. This allows out-of-domain images to be faithfully mapped into the generator’s domain without sacrificing the ability to perform subsequent edits, effectively bridging the gap between reconstruction accuracy and editing flexibility. Complementing this, [216] rethinks face swapping as a fine-grained editing problem, utilizing a regional GAN inversion method to explicitly disentangle shape and texture. By projecting textures of facial components into regional style codes, this framework enables local swapping with high fidelity, preserving subtle geometric details that global methods often overlook.

Attention mechanisms have also emerged as a critical tool for enforcing spatial constraints and preventing attribute interference, serving as a bridge between efficient adaptation and precise control. Methods leveraging progressive or multi-attention structures ensure that edits are confined to semantically relevant regions. [217] employs a progressive attention mechanism that constrains editing within proper attribute boundaries at multiple feature levels, preventing undesired modifications to irrelevant regions from the outset. Similarly, [218] utilizes a symmetric U-Net structure with additive and self-attention mechanisms to adaptively transfer encoder representations, effectively balancing attribute editing ability with the preservation of exclusionary details. In the realm of diffusion models, attention manipulation is pivotal for zero-shot editing. [219] proposes correlating features from the inversion process with sampling features via self-attention keys and values, thereby mitigating inconsistencies in DDIM reconstruction and ensuring content-consistent editing without extensive fine-tuning.

The integration of natural language instructions has further refined the granularity of control, enabling users to dictate complex, multi-attribute changes through dialogue or chained instructions, thus enhancing the usability of these efficient models. [220] models a continuous semantic field in the latent space, allowing users to traverse curving trajectories determined by dialog interactions rather than fixed lines, facilitating smooth and fine-grained transitions like changing a slight smile to a broad laugh. Extending this to multi-attribute scenarios, [221] leverages Large Language Models to generate sequences of instructions, executing edits step-by-step to avoid the confusion and attribute entanglement often seen in single-prompt editing. Moreover, [222] introduces a decoupling training scheme with entropy constraints to edit latent codes separately for each attribute, ensuring that text-relevant attributes are manipulated while text-irrelevant ones remain intact.

Finally, the robustness of identity preservation under editing is critical for downstream applications like face recognition, validating the practical utility of these controlled generation techniques. Studies such as [223] highlight that even identity-preserving generative models can degrade recognition performance if not carefully constrained. Consequently, new techniques focus on local and global editing strategies that explicitly optimize for identity retention metrics alongside visual fidelity. By combining staged noise guidance, region-aware attention, and disentangled latent optimization, contemporary methods are successfully navigating the intricate landscape of fine-grained attribute editing. These approaches offer tools that are both powerful and precise, maintaining the essential identity of the subject while laying the groundwork for the next critical step: extending these controlled editing capabilities to arbitrary real-world images through advanced inversion techniques, as explored in the following section.

4.3 Advanced Inversion Techniques for Real Image Manipulation

4.3 Advanced Inversion Techniques for Real Image Manipulation

While Section 4.2 explored strategies to preserve identity and manipulate attributes within synthetic or well-aligned latent spaces, applying these capabilities to arbitrary real-world images introduces a fundamental bottleneck: the inversion process. The efficacy of high-fidelity editing on real images using pre-trained diffusion models hinges critically on accurately mapping a given image back into the model’s latent noise space. This mapping establishes the necessary starting point from which edited samples can be generated while maintaining the original content’s structural integrity. Early approaches relying on standard Denoising Diffusion Implicit Model (DDIM) inversion often suffer from significant reconstruction degradation, particularly when large classifier-free guidance (CFG) scales are employed to enhance editability. To address these limitations and bridge the gap between theoretical controllability and practical application, the field has evolved toward robust optimization-based strategies, accelerated proximal methods, and entirely inversion-free paradigms that leverage attention mechanisms.

A foundational advancement in this domain is Null-text Inversion (NTI), which optimizes the unconditional text embeddings at each timestep to align the reconstruction trajectory with the inversion path. This technique allows for precise editing using cross-attention control without altering the model weights or the conditional prompt [224]. By treating the null embedding as a learnable parameter, NTI effectively bridges the gap between the deterministic inversion of real images and the stochastic generation process, enabling intuitive text-based modifications. However, the computational cost of optimizing embeddings for every timestep remains a significant bottleneck. To mitigate this, Negative-prompt Inversion (NPI) was introduced as a training-free, closed-form approximation of NTI. NPI utilizes a negative prompt to represent the original image, achieving comparable reconstruction quality through forward propagation alone, thereby accelerating the inversion process by over thirty times compared to optimization-based methods [225]. Further enhancements to efficiency have been achieved through Wavelet-Guided Acceleration, which analyzes frequency characteristics to limit text optimization to specific timesteps, reducing editing time by more than 80% while maintaining performance [226].

Despite these accelerations, reconstruction fidelity can still degrade under strong guidance conditions, necessitating more sophisticated regularization. Proximal Guidance emerges as a solution to this problem by incorporating a regularization term and reconstruction guidance directly into the inversion framework. When integrated with NPI, proximal guidance significantly reduces artifacts and improves the stability of real image editing, even when extending the methodology to include mutual self-attention control for complex geometry and layout alterations [227]. Complementing this, Noise Map Guidance (NMG) offers an optimization-free alternative that richly encodes spatial context. Unlike NTI, which may fail to capture fine-grained spatial relationships, NMG preserves editing quality by leveraging noise maps that retain the spatial structure of the input image, demonstrating robustness across various DDIM inversion variants [228]. Similarly, Accelerated Iterative Diffusion Inversion (AIDI) employs a novel blended guidance technique to improve reconstruction accuracy with minimal computational overhead, proving particularly effective in fast editing regimes requiring fewer diffusion steps [229].

Beyond optimizing latent codes or embeddings, recent research has explored direct correlations between inversion features and sampling processes to ensure consistency. Tuning-free Inversion-enhanced Control (TIC) directly utilizes key and value features from the self-attention layers during inversion to enhance the subsequent sampling process. This approach mitigates inconsistencies

inherent in standard DDIM reconstruction and employs a mask-guided attention concatenation strategy to ensure content consistency during non-rigid edits [219]. Another variation, Prompt Tuning Inversion, encodes input image information into a learnable conditional embedding during a reconstruction stage, which is then linearly interpolated with the target embedding during editing. This method ensures a superior trade-off between editability and fidelity, allowing for specific attribute changes like color modification while preserving object shape and background [230]. Furthermore, the concept of “Direct Inversion” proposes disentangling the source and target diffusion branches, assigning distinct responsibilities for content preservation and edit fidelity. This theoretical insight leads to a remarkably simple implementation that achieves optimal performance with minimal code complexity, offering nearly an order of magnitude speed-up over traditional optimization-based techniques [231].

Pushing the boundaries further, completely inversion-free frameworks have emerged to eliminate the time-consuming reconstruction phase entirely, representing a paradigm shift toward real-time applicability. The Denoising Diffusion Consistent Model (DDCM) introduces a special variance schedule that reduces the denoising step to a form compatible with multi-step consistency sampling, effectively creating a virtual inversion strategy. When unified with attention control mechanisms, this leads to Inversion-Free Editing (InfEdit), which enables consistent and faithful editing for both rigid and non-rigid semantic changes in real-time, catering to intricate modifications without explicit inversion [232]. Additionally, the Zero-shot Inversion Process (ZIP) injects a fusion of generated visual references and text guidance into the semantic latent space of a frozen model using only a tiny neural network. ZIP demonstrates remarkable robustness for both in-domain and out-of-domain attribute manipulation, producing diverse content under intuitive text control without the need for full image inversion [233]. Collectively, these advanced techniques transition the field from costly, iterative optimization toward efficient, attention-driven, and sometimes inversion-free manipulation. This evolution not only broadens the practical applicability of diffusion models for real-world editing but also lays the groundwork for utilizing these powerful generative priors to solve broader ill-posed inverse problems, as discussed in the following section.

4.4 Generative Priors for Solving Ill-Posed Inverse Problems

4.4 Generative Priors for Solving Ill-Posed Inverse Problems

Building upon the advanced inversion techniques discussed in Section 4.3, which establish robust pathways for mapping real images into latent spaces for editing, this section explores a complementary application of generative models: solving ill-posed inverse problems. While the previous subsection focused on preserving structural integrity during semantic manipulation, the challenges here involve reconstructing high-quality images from severely degraded measurements—such as in super-resolution, inpainting, deblurring, and compressed sensing—where information is fundamentally missing or corrupted. Historically reliant on hand-crafted regularizers or supervised learning paradigms that struggle to generalize across diverse degradation operators, this landscape has been revolutionized by deep generative models. Both Generative Adversarial Networks (GANs) and Denoising Diffusion Models (DDMs) now serve as powerful, data-driven priors that encapsulate the manifold of natural images, effectively transforming ill-posed problems into well-constrained optimization tasks within the generator’s latent space.

A foundational approach to leveraging these priors formulates restoration as a Bayesian inference problem, aiming to sample from the posterior distribution of natural images given observed measurements. Early successes in this domain utilized GANs as explicit priors, exploiting pre-trained

StyleGAN generators to construct prior distributions by optimizing over style codes and latent inputs to achieve competitive results in inpainting and super-resolution [234]. To address the inherent representation error in GANs, where the learned distribution may not perfectly match the target image, hybrid approaches have emerged. These methods model images as a linear combination of a GAN prior and an unlearned signal model, such as the Deep Decoder, significantly improving reconstruction fidelity for both in-distribution and out-of-distribution samples [235]. Furthermore, the stability and adaptability of GAN-based restoration have been enhanced through robust unsupervised methods that utilize progressive latent space extensions, allowing a single set of hyperparameters to handle varying degradation levels without retuning [236].

While GANs offer computational efficiency, diffusion models have recently surpassed them in terms of sample diversity and perceptual quality, albeit often at a higher computational cost. A significant advancement is the development of zero-shot frameworks that utilize off-the-shelf pre-trained diffusion models without requiring task-specific fine-tuning. The Denoising Diffusion Null-Space Model (DDNM) exemplifies this by refining only the null-space components during the reverse diffusion process, ensuring data consistency while maintaining the realism imparted by the generative prior [237]. Building on this, the Denoising Diffusion Restoration Models (DDRM) framework introduces an efficient, unsupervised posterior sampling method that leverages the singular value decomposition of the degradation operator to solve linear inverse problems rapidly, outperforming previous unsupervised methods in both quality and runtime [238]. These concepts have been extended to handle non-linear problems, such as JPEG artifact correction, by generalizing the pseudo-inverse operator used in DDRM to broader measurement contexts [239].

To further enhance the applicability of these models, researchers have explored latent space formulations and adaptive strategies. The Generative Diffusion Prior (GDP) framework systematically explores conditional guidance protocols and optimizes degradation model parameters during the denoising process, enabling the solution of blind restoration problems and handling images of arbitrary resolutions through hierarchical guidance [240]. Similarly, solving inverse problems in the latent space of pre-trained models like Stable Diffusion has been addressed by frameworks that provide provable sample recovery, outperforming pixel-space algorithms in tasks ranging from random inpainting to destriping [241]. Addressing the challenge of content consistency in super-resolution, where stochastic diffusion processes might alter image semantics, recent works propose decoupling structure refinement via diffusion from detail enhancement via adversarial training, thereby stabilizing the output while preserving perceptual quality [242].

A critical theoretical perspective underpinning many of these methods is the decomposition of the solution space into range and null spaces relative to the forward operator. Rooted in classical inverse problem theory, this concept has been revitalized in the deep learning era to separate data fidelity (range space) from generative hallucination (null space), ensuring that restored images adhere strictly to measurements while filling in missing information with realistic textures [243]. This decomposition is not limited to diffusion models; it has also been integrated into deep learning architectures where neural networks are explicitly trained to learn decomposition functions, yielding superior performance in compressive sensing and super-resolution compared to standard residual learning [244]. Furthermore, adaptive diffusion strategies have been proposed to tailor pre-trained denoisers to specific inputs by retrieving nearest neighbors from diverse datasets, significantly boosting performance in super-resolution and deblurring tasks [245].

Beyond specific architectural innovations, variational perspectives have provided a unified theoretical grounding for these approaches. By framing the sampling process as stochastic optimization, variational methods approximate the true posterior distribution, allowing denoisers at different

timesteps to impose structural constraints weighted by signal-to-noise ratios [246]. This aligns with broader surveys indicating that diffusion-based methods now consistently outperform GAN-based counterparts in restoration tasks due to their superior modeling of complex data distributions and flexible conditioning mechanisms [247]. Additionally, the integration of prompt tuning in latent diffusion models allows for on-the-fly optimization of text embeddings, further refining the generative prior to produce images more faithful to the desired distribution while mitigating artifacts common in latent space manipulations [248]. Collectively, these advancements demonstrate that generative priors have become indispensable tools for solving ill-posed inverse problems, offering a robust balance between data consistency and perceptual realism. While the methods discussed here focus on restoring missing information within fixed structural constraints, the following section explores a more aggressive form of controllability: the active manipulation of scene geometry, layout, and semantic composition.

4.5 Structural Manipulation, Layout Control, and Semantic Composition in 2D

4.5 Structural Manipulation, Layout Control, and Semantic Composition in 2D

While the preceding section detailed how generative priors resolve ill-posed inverse problems by restoring missing information within fixed structural constraints, this subsection explores a more aggressive form of controllability: the active manipulation of scene geometry, layout, and semantic composition. The frontier of controllable image generation has rapidly expanded beyond simple attribute modification or style transfer to encompass complex structural transformations that fundamentally alter the spatial arrangement of a scene. This evolution addresses the critical need for precise layout control, subject repositioning, and the seamless integration of disparate visual elements within 2D images. Unlike earlier methods that often resulted in global image degradation or identity loss during significant edits, contemporary approaches leverage the rich latent spaces of diffusion models and Generative Adversarial Networks (GANs) to perform surgical structural changes while preserving high-fidelity details.

A primary challenge in this domain is subject repositioning, which involves relocating a specific object to a new coordinate within the image while simultaneously reconstructing the occluded background and the newly revealed parts of the subject. Traditional inpainting methods often fail to maintain the consistency of the moved object with its new context. To address this, recent frameworks have reformulated subject repositioning as a unified, prompt-guided inpainting task. For instance, the SEELE framework demonstrates that by employing task inversion techniques and integrating specific pre-processing and post-processing steps, a single diffusion model can effectively handle the dual sub-tasks of filling the void left by the subject and reconstructing obscured portions [249]. Similarly, geometric transformations provide a robust mechanism for structural editing. GeoDiffuser introduces a zero-shot optimization approach that unifies 2D and 3D object editing by viewing operations as geometric transformations directly incorporated into the attention layers of diffusion models. This method allows for object translation, 3D rotation, and removal while maintaining plausible lighting and shadows, effectively inpainting disoccluded regions without requiring model retraining [250].

Beyond moving existing elements, semantic composition often requires inserting new subjects or replacing existing ones with exemplar-based concepts, a task demanding precise harmonization to avoid visual artifacts. The integration of exemplar images with text prompts has emerged as a powerful paradigm for achieving this. DreamInpainter proposes a two-step approach that computes dense subject features for accurate replication and utilizes a discriminative token selection

module to eliminate redundant details, thereby balancing editability with subject fidelity [251]. Complementing this, the PhD framework introduces a “Paste, Inpaint, and Harmonize” strategy where an off-the-shelf segmentation model extracts a subject from a reference image, which is then inserted into a target scene. A dedicated harmonizing module guides the frozen diffusion model to blend the subject naturally, preserving both the scene context and the subject’s identity without extensive fine-tuning [252]. Furthermore, LAR-Gen advances this capability by adopting a coarse-to-fine manner that locates the target region, assigns multi-modal guidance through decoupled cross-attention, and refines subject details using a specialized network, ensuring seamless inpainting even with scarce training data [253].

Layout control represents another dimension of structural manipulation, enabling users to dictate the spatial arrangement of multiple objects within a generated or edited scene. This is particularly challenging in multi-subject customization where concept interference often leads to merged or distorted entities. Cones 2 addresses this by leveraging layout priors as abstract spatial guidance. By rectifying activations in cross-attention maps, the method appoints and separates the locations of different subjects, significantly alleviating interference and enabling robust composition of multiple customized concepts [166]. Extending this to existing images, recent work proposes the first framework for continuous layout editing of single images. This approach utilizes masked textual inversion to disentangle object concepts into separate tokens and employs a training-free optimization method to align regenerated images with user-specified layouts, preserving visual properties while altering spatial configurations [254]. Additionally, an item-prompt interaction framework named D-Edit disentangles comprehensive image-prompt interactions into specific item-prompt associations, allowing for versatile editing including item removal and mask-based editing within a unified system [255].

For scenarios involving complex scenes with multiple objects requiring simultaneous editing, localized control mechanisms are essential. LoMOE introduces a multi-diffusion process that enables zero-shot localized multi-object editing in a single pass. By leveraging foreground masks and simple text prompts, it exerts localized influences on target regions while using cross-attention and background preservation losses to ensure high-fidelity reconstruction and minimize artifacts [256]. In cases where masks are unavailable or difficult to define, attention-based guidance offers a flexible alternative. MAG-Edit optimizes noise latent features by maximizing mask-based cross-attention constraints, enabling localized editing in complex scenarios where traditional mask-free methods suffer from leakage and misalignment [257].

Finally, the harmonization of inserted or edited elements remains a critical component of semantic composition. Semantic-guided multi-mask image harmonization tackles the problem of adjusting multiple inharmonious regions perturbed by different methods without explicit input masks. By predicting operator masks that indicate the level and position of editing operations such as brightness or color adjustment, this approach provides a flexible, editable solution for complex scenes [258]. Collectively, these advancements signify a shift towards highly controllable, structurally aware editing systems capable of executing complex semantic compositions with unprecedented fidelity. However, as these tools empower users to manipulate identity and scene content with increasing ease, they inevitably raise profound questions regarding security, ethics, and the potential for misuse, a topic we address in the following section on defensive mechanisms and personalized generation safety.

4.6 Security, Ethics, and Defensive Mechanisms in Personalized Generation

4.6 Security, Ethics, and Defensive Mechanisms in Personalized Generation

The advancements in structural manipulation and semantic composition detailed in the preceding section empower users to alter scene geometry and integrate disparate visual elements with unprecedented fidelity. However, this same capability to precisely manipulate identity and context introduces profound security and ethical challenges. The rapid democratization of personalized image generation, particularly through diffusion-based models, has significantly lowered the barrier for malicious actors to engage in identity theft, deepfake creation, and non-consensual image synthesis. Consequently, the research community has pivoted toward developing defensive mechanisms designed to protect individual likenesses from unauthorized model training and inference. A primary avenue of defense involves the application of adversarial perturbations intended to disrupt the personalization process at its inception. Recent works propose adding subtle, imperceptible noise to reference images to prevent diffusion models from successfully learning a subject’s identity. For instance, methods like [259] explicitly aim to degrade the generation quality of any model trained on perturbed images, effectively rendering the personalization process futile even under adverse conditions such as prompt mismatches. Similarly, [260] advances this concept by exploring the inherent properties of diffusion time steps and feature layers, proposing an adaptive greedy search and feature-based optimization framework to significantly increase identity disruption and enhance user privacy.

Beyond preventing unauthorized learning, the tension between reconstruction quality and editability remains a central ethical dilemma in legitimate personalized generation. High-fidelity restoration and super-resolution tasks often struggle to maintain strict identity faithfulness, occasionally hallucinating features that do not belong to the subject. To address this, frameworks like [261] utilize independent trainable blocks and generative regularizers to balance detail generation with the degree of personalization, ensuring that restored faces remain true to the original identity while avoiding reliance on residual low-quality conditioning data. Furthermore, the challenge of “identity fine editing”—modifying specific attributes while preserving the core identity and context—has led to staged diffusion frameworks such as [212]. By discerning between editing and boosting stages via frequency analysis, such methods allow for precise manipulation of facial features without compromising the subject’s inherent identity, thereby offering a controlled environment for ethical editing that mitigates the risk of unintentional identity distortion.

As generative models become increasingly capable of synthesizing realistic media, the societal implications extend into the realm of accountability and attribution. The need to attribute deepfakes to their source models becomes critical for legal and ethical governance. Research into decentralized attribution, such as that presented in [262], proposes using binary classifiers parameterized by user-specific keys to distinguish model distributions from authentic data, providing a scalable solution for identifying the origin of maliciously generated content. This is complemented by efforts to establish plausible deniability for model developers, ensuring that creators can prove they did not generate specific media, a concept explored in [263]. These attribution mechanisms are essential for combating misinformation campaigns and holding bad actors accountable in an era where visual evidence can be easily fabricated.

Moreover, the intersection of personalization and privacy reveals complex trade-offs, particularly regarding the “personalization paradox,” where accurate user modeling often conflicts with adaptive system safety and privacy. As discussed in [264], feedback loops in personalized systems can lead to unintended consequences, necessitating robust auditing pipelines. The proposal of interactive

audit pipelines in [265] underscores the need for human-in-the-loop methodologies to assess fairness, privacy, and ownership issues throughout the machine learning lifecycle. This is particularly relevant given that standard regularization techniques may provide a false sense of security against membership inference attacks, as highlighted in [266], which demonstrates that training samples may remain distinguishable even when standard defenses are applied.

Finally, the broader ethical landscape requires a holistic approach to trustworthiness, encompassing bias mitigation, transparency, and regulatory compliance. Frameworks for auditing synthetic data, such as those introduced in [267], provide metrics to evaluate the trade-offs between utility, robustness, and privacy preservation, fostering collaboration among stakeholders to prevent discrimination. Additionally, the integration of ethical boundaries directly into model architectures, as seen in [268], suggests a future where security levels are differentiated to empower users to control their data disclosure. Together, these defensive strategies and ethical frameworks form a critical shield against the misuse of personalized generation, striving to balance the transformative potential of AI with the fundamental rights to privacy, identity, and truth.

5 Extending Controllability to Video, 3D, and Multi-View Synthesis

5.1 Dynamic Neural Radiance Fields and Spatio-Temporal Representations

5.1 Dynamic Neural Radiance Fields and Spatio-Temporal Representations

The extension of controllable image generation into the temporal domain necessitates robust frameworks capable of modeling dynamic scenes with high fidelity, geometric consistency, and computational efficiency. Foundational approaches in this realm have evolved from implicit neural representations to explicit, point-based primitives, fundamentally altering how spatio-temporal data is synthesized and rendered. Early methodologies primarily relied on Neural Radiance Fields (NeRFs), which model scenes as continuous functions of space and view direction. To accommodate dynamics, these models were extended to incorporate time as an additional input dimension, often utilizing deformation fields that map coordinates from observed frames back to a canonical, time-invariant space. However, these implicit approaches frequently struggle with the computational bottlenecks associated with volumetric rendering, resulting in slow training and inference speeds that hinder real-time applications. Furthermore, standard deformation-based NeRFs often fail to capture topological changes or complex non-rigid motions due to the continuity constraints of the underlying neural networks [269]. To address topological variations, some frameworks have lifted the representation into higher-dimensional hyper-spaces, allowing for smooth interpolation between distinct scene configurations that would otherwise require discontinuous deformation fields [269].

Despite these advancements, the reliance on dense sampling and volumetric integration in NeRF-based methods limits their scalability for long video sequences and large-scale environments. Recent innovations have shifted towards factorized representations that decompose the spatio-temporal volume into more manageable components. For instance, tensor decomposition techniques have been employed to represent dynamic scenes as 4D or 5D tensors, breaking them down into low-rank components to significantly reduce memory footprints while maintaining rendering quality [270] [271]. These methods effectively capture both structural motions and fine-grained dynamic details by hierarchically factoring the 4D fields, enabling reconstruction from sparse-view inputs or even monocular cameras [271]. Additionally, approaches like BLiRF have sought to bridge the gap between classic non-rigid structure-from-motion and neural fields by formulating scenes

as compositions of bandlimited signals, thereby leveraging well-established geometric priors to regularize the under-constrained problem of monocular dynamic reconstruction [272].

A paradigm shift occurred with the introduction of 3D Gaussian Splatting (3DGS), which replaced implicit volumetric rendering with an explicit representation based on anisotropic Gaussian ellipsoids. This transition enabled real-time rendering speeds through efficient rasterization pipelines, sparking a surge in research dedicated to extending Gaussian Splatting to dynamic scenarios. The core challenge in this domain lies in modeling the temporal evolution of Gaussian parameters—position, rotation, scale, and opacity—without incurring prohibitive memory costs or losing temporal coherence. One prominent strategy involves defining a canonical set of 3D Gaussians and learning a time-dependent deformation field, typically parameterized by a Multi-Layer Perceptron (MLP), to predict the state of these Gaussians at any given timestamp [273] [274]. To enhance the geometric consistency of these deformations, recent works have integrated explicit 3D geometry features into the learning process, ensuring that the motion of individual Gaussians remains physically plausible and coherent with the underlying scene structure [275].

To further improve efficiency and handle complex motions, researchers have proposed representing scenes directly in 4D space using 4D Gaussians. Unlike methods that slice 3D Gaussians per frame, 4D Gaussian Splatting (4DGS) models dynamics using anisotropic 4D XYZT Gaussians, which are temporally sliced during rendering to compose dynamic 3D structures [276]. This holistic representation allows for the seamless projection of spatio-temporal primitives into images, demonstrating superior capabilities in capturing abrupt motions and fine details while achieving real-time inference speeds exceeding 200 FPS on modern GPUs [276]. Alternative formulations, such as Spacetime Gaussian Feature Splatting, enhance this by incorporating temporal opacity and parametric motion into the Gaussian definition, alongside splatted neural features that replace traditional spherical harmonics to model view- and time-dependent appearance compactly [277].

Efficiency remains a critical concern, particularly for long video sequences where storing per-frame parameters is infeasible. Solutions like Gaussian-Flow introduce dual-domain deformation models that capture time-dependent residuals using polynomial fitting in the time domain and Fourier series in the frequency domain, eliminating the need for separate models per frame or heavy implicit fields [278]. Similarly, motion-aware frameworks leverage optical flow cues from 2D observations to guide the optimization of 3D Gaussian movements, reducing model redundancy and improving reconstruction quality in both monocular and multi-view settings [279]. For scenarios involving articulated objects or specific semantic controls, methods like CoGS and SC-GS decompose motion into sparse control points or utilize mesh-based anchoring to enable user-controllable editing of dynamic scenes while preserving high-fidelity appearance [280] [281].

Furthermore, the integration of semantic information and physical properties into these spatio-temporal representations has opened new avenues for controllable synthesis. Feature Splatting distills vision-language features into 3D Gaussians, enabling semi-automatic scene decomposition and physics-based dynamics synthesis grounded in natural language [282]. In specialized domains such as surgical robotics, Endo-4DGS adapts these principles by utilizing pseudo-depth priors from foundation models to reconstruct endoscopic scenes in real-time, addressing the unique challenges of monocular depth estimation in medical settings [283]. Collectively, these advancements signify a maturation of dynamic scene representation, moving from slow, implicit volumetric models to fast, explicit, and highly controllable spatio-temporal frameworks. These robust geometric backbones provide the essential foundation for the subsequent integration of diffusion priors, enabling the leap from reconstructing observed dynamics to synthesizing novel 4D content from text and image prompts.

5.2 Score Distillation and Diffusion Priors for Text-to-4D Synthesis

5.2 Score Distillation and Diffusion Priors for Text-to-4D Synthesis

Building upon the robust geometric backbones of dynamic spatio-temporal representations established in Section 5.1, the extension of controllable generative modeling from static 3D assets to dynamic 4D content represents a significant leap in complexity. This transition necessitates the simultaneous enforcement of geometric consistency across spatial views and temporal smoothness across time steps. The primary methodology driving this advancement is Score Distillation Sampling (SDS), which leverages the rich priors embedded in pre-trained 2D image and video diffusion models to optimize dynamic 3D representations without requiring large-scale paired 4D training datasets. However, directly applying standard SDS to 4D generation introduces a critical three-way trade-off between appearance quality, 3D structural fidelity, and motion realism. Text-to-image models provide high-fidelity appearance and structure but lack motion priors, while text-to-video models capture temporal dynamics but often suffer from compromised geometry and lower resolution. To address these challenges, recent research has focused on hybrid distillation strategies, deterministic sampling priors, and compositional frameworks that effectively synthesize these complementary strengths, laying the groundwork for the precise motion control mechanisms discussed in Section 5.3.

A fundamental limitation of traditional SDS in the 4D domain is the stochastic nature of the sampling trajectory, which can lead to inconsistent guidance signals during optimization. The randomness inherent in Stochastic Differential Equation (SDE) sampling often results in geometry collapse or flickering textures because the guidance at each step is not guaranteed to be less noisy or consistent with previous frames. To mitigate this, [284] proposes shifting from stochastic SDE trajectories to Ordinary Differential Equation (ODE) based deterministic sampling. By constructing an ODE for trajectory sampling and utilizing a consistency distillation loss, this approach ensures that the guidance signal converges deterministically to the target, significantly enhancing the stability and fidelity of the generated dynamic assets. This deterministic prior is crucial for 4D synthesis, where temporal coherence relies on the predictability of the generative process across consecutive time steps.

Beyond stabilizing the sampling trajectory, another major frontier involves the strategic composition of scores from orthogonal diffusion models. Recognizing that no single foundation model possesses all necessary priors, [285] introduces a framework that composes score functions from separate video and multi-view diffusion models. This method leverages the geometric consistency priors from multi-view models and the temporal smoothness priors from video models to directly sample dense multi-view, multi-frame images. By designing a denoising strategy based on the probability structure of the target images, this approach circumvents the scarcity of synchronized 4D training data and enables the generation of dynamic content within minutes, demonstrating the efficacy of decoupling and recombining distinct generative priors. Similarly, [286] addresses the aforementioned three-way trade-off by introducing a hybrid SDS procedure. This method alternates optimization steps, blending supervision signals from text-to-image and text-to-video models to synthesize 4D scenes that exhibit compelling appearance, robust 3D structure, and natural motion, effectively alleviating the weaknesses inherent in using either model type in isolation.

Further advancements have been made by incorporating explicit structural representations and grounding mechanisms to enhance control and consistency, bridging the gap between abstract generative priors and concrete scene instantiation. [287] combines pre-trained diffusion models with dynamic 3D Gaussian splatting, utilizing a multi-view diffusion model to initialize images anchored

on input video frames. A key innovation in this work is the introduction of a temporal anchor in the self-attention computation, using the first frame to ensure temporal consistency across the multi-view sequence before applying score distillation. This spatial-temporal anchoring strategy, coupled with an adaptive densification mechanism for the Gaussian point cloud, achieves state-of-the-art robustness and rendering quality. In a similar vein, [288] decomposes the generation task by identifying static 3D assets and monocular video sequences as key components. By employing spatial-temporal pseudo-labels and seamless consistency priors through 3D-aware score distillation, this framework supports grounded generation, allowing users to specify geometry and motion separately, thereby offering superior control compared to end-to-end text-only approaches.

The challenge of maintaining consistency is also addressed through collaborative and canonical distillation techniques. [289] proposes treating multiple samples as particles in a Stein Variational Gradient Descent (SVGD) update, combining their score functions to distill generative priors synchronously across a set of images. This method facilitates seamless information integration, which is essential for ensuring inter-frame consistency in video and 4D scene synthesis. For non-rigid objects, [290] introduces Canonical Score Distillation (CSD). This technique lifts 2D diffusion priors to generate a canonical model that remains robust across different articulations, integrating inductive priors to enhance the authenticity of bones and skinning while inferring invisible regions through multi-view distillation.

Finally, efficient optimization paradigms continue to evolve to make 4D generation more accessible and scalable. [291] presents an efficient framework that generates spacetime-consistent images under different camera views and uses them as labeled data to train a 4D Gaussian splatting model directly, achieving a remarkable speedup over prior optimization-heavy methods. Additionally, [292] leverages a compositional approach, combining text-to-image, text-to-video, and 3D-aware multiview models to provide feedback during optimization. This method includes novel regularization techniques for moving 3D Gaussians and motion amplification mechanisms, enabling the synthesis of vivid, long-duration dynamic scenes. Collectively, these methodologies illustrate a rapid maturation of score distillation techniques, moving from the naive application of 2D priors to sophisticated, hybrid, and deterministic frameworks capable of generating high-fidelity 4D content. While these approaches successfully establish the existence and basic coherence of dynamic scenes, they primarily rely on implicit consistency objectives. The subsequent section addresses the critical need to transcend these general consistency goals by introducing explicit constraints—such as trajectories and skeletons—to dictate precise object behaviors within these generated 4D worlds.

5.3 Controllable Animation via Trajectory, Skeleton, and Motion Guidance

5.3 Controllable Animation via Trajectory, Skeleton, and Motion Guidance

While Section 5.2 established the foundational role of score distillation and diffusion priors in synthesizing dynamic 4D content from text, a critical remaining challenge is the precise enforcement of user-defined motion constraints within these generated scenes. Moving beyond the general consistency objectives discussed previously, this subsection focuses on explicit control mechanisms that dictate *how* objects move, rather than merely ensuring they move coherently. The extension of controllable image generation into the fourth dimension introduces the formidable task of maintaining spatio-temporal fidelity while adhering to specific kinematic or dynamic constraints. Unlike static 3D synthesis, 4D generation requires models to disentangle geometry and appearance from

the dynamics of deformation and translation over time. Recent advancements have categorized these motion control strategies into three primary paradigms: trajectory-conditioned synthesis for global path planning, skeleton-driven frameworks for articulated non-rigid deformation, and sparse-controlled mechanisms that balance efficiency with fidelity. These approaches collectively enable the creation of dynamic scenes where object paths, poses, and interactions can be dictated with high precision, bridging the gap between generative priors and deterministic animation requirements.

Trajectory-conditioned synthesis has emerged as a critical method for controlling the global movement of objects within a scene, addressing the limitations of methods that struggle to generate motion extending beyond fixed bounding boxes. A notable advancement in this domain is the factorization of motion into global and local components. For instance, [293] proposes a framework where the global motion of a scene’s bounding box is represented by rigid transformations along a spline-parameterized trajectory, while local deformations are learned to conform to this global path. This decoupling allows for the synthesis of scenes animated along arbitrary trajectories, significantly enhancing the realism and range of generated motion compared to methods limited by static volume rendering. Similarly, [294] addresses the complexity of managing multiple interacting objects by introducing fine-grained trajectory control via diffusion models. By utilizing tracklet conditions, this framework ensures inter-frame consistency and manages appearance changes, appearance disappearances, and scale variations, which are often overlooked in standard video generation literature. Furthermore, [295] distinguishes between camera motion and object motion, enabling independent control over both through appearance-free conditions derived from camera poses and trajectories. This separation facilitates flexible combinations of movements, allowing for diverse and complex scene dynamics that were previously difficult to achieve with unified motion representations.

For articulated subjects such as humans and robots, skeleton-driven non-rigid deformation remains the gold standard for achieving physically plausible and semantically meaningful animation. Traditional kinematics-based models often fail to account for physical constraints, leading to artifacts like foot sliding or penetration. To mitigate these issues, [296] introduces a framework that interfaces a generative motion prior with projective dynamics, effectively acting as a minimalist physics simulator. This approach seamlessly incorporates control from the motion prior while ensuring adherence to Newtonian dynamics, resulting in responses that are both diverse and physically valid. In the realm of real-time synthesis, [297] proposes a forward dynamics-guided imitation method. This technique generates human-like reactions in real-time by leveraging a policy trained to mimic reference states while strictly respecting physical laws, thereby overcoming the latency and pliability issues inherent in purely diffusion-based kinematic approaches. Additionally, [298] presents a system that performs physics-oriented motion correction at test time. By regularizing synthesized skeleton motions within a physics simulator and utilizing reinforcement learning for imitation, this method ensures that the resulting animations are not only accurate but also compliant with physical constraints, significantly outperforming prior methods in terms of plausibility.

Beyond rigid trajectories and skeletal structures, sparse-controlled frameworks offer a versatile middle ground for dictating object paths and poses, particularly when dense supervision is unavailable or computationally prohibitive. [299] exemplifies this approach by decoupling motion and appearance to achieve superior video-to-4D generation. By employing sparse control points and introducing adaptive Gaussian initialization, SC4D mitigates shape degeneration and ensures the fidelity of learned motion, while also enabling novel applications such as transferring learned motion to diverse 4D entities based on textual descriptions. Complementing this, [281] utilizes sparse control points to learn compact 6-DoF transformation bases. These bases are interpolated to yield the motion field of 3D Gaussians, reducing learning complexity and facilitating user-controlled motion

editing while preserving high-fidelity appearances. The efficacy of sparse controls is further demonstrated in [300], which synthesizes animator-guided human motion using only a few joint locations. By decomposing motion synthesis into path walking and action transitions within a goal-centric canonical coordinate frame, this method generates long, plausible motion sequences that satisfy scene constraints without requiring explicit scene geometry during training.

Moreover, the integration of high-level semantic guidance with low-level trajectory control is becoming increasingly sophisticated, creating a bridge between abstract intent and concrete motion execution. [301] combines low-level trajectory constraints with high-level language semantics to synthesize realistic human motion. By employing a masked trajectories transformer and test-time optimization, TLControl achieves precise alignment with user-specified trajectories, addressing the precision gaps found in methods that rely solely on language or full trajectory inputs. Similarly, [302] leverages Large Language Models to decompose input prompts into distinct entities and map their trajectories, constructing compositional 4D scenes where objects are positioned accurately along designated paths. This LLM-guided approach enhances object interactions and motion fidelity, showcasing the potential of combining semantic reasoning with geometric control. Finally, [303] offers a skeleton-free alternative that uses integrative latent mapping to control 4D sequences. This framework encodes shape and color information into a unified latent representation, allowing diffusion models to generate coherent animations with dynamic evolution of shape and color, thus expanding the toolkit available for controllable 4D synthesis beyond traditional skeletal rigs.

Collectively, these methodologies illustrate a rapid evolution in controllable 4D generation, moving from the implicit consistency enforced by score distillation (Section 5.2) to complex, multi-modal control systems that integrate trajectories, skeletons, and sparse constraints. The ability to precisely dictate motion paths, enforce physical plausibility, and edit dynamic scenes with sparse inputs marks a significant frontier in generative modeling. These advances lay the groundwork for the subsequent discussion in Section 5.4, which addresses the inverse problem: reconstructing coherent 4D scenes and recovering implicit motion trajectories from limited observational data, such as single images or uncalibrated monocular videos, where explicit control signals are absent.

5.4 Single-View and Monocular Video Reconstruction with Temporal Consistency

5.4 Single-View and Monocular Video Reconstruction with Temporal Consistency

While Section 5.3 detailed methods for imposing explicit user-defined motion constraints—such as trajectories and skeletons—onto generative models, this subsection addresses the inverse and equally critical challenge: recovering coherent 4D scenes and implicit motion trajectories from limited observational data where such control signals are absent. Moving from deterministic animation to ill-posed reconstruction, the focus shifts to synthesizing dense, animatable 4D representations from sparse inputs like single images or uncalibrated monocular videos. Unlike multi-view setups endowed with abundant geometric constraints, monocular settings suffer from severe ambiguity due to the lack of explicit depth cues and camera parameters. To bridge the gap between these sparse 2D observations and rich 4D outputs, recent research has pivoted toward leveraging powerful generative priors, interpolation-driven consistency mechanisms, and canonical space mappings. These strategies aim to enforce temporal coherence and spatial fidelity without relying on cumbersome multi-camera arrays or precise calibration, laying the foundational recovery capabilities necessary for the complex, multi-entity scene synthesis discussed in Section 5.5.

A primary strategy for achieving temporal consistency in these under-constrained monocular settings involves the utilization of interpolation-driven consistency losses. By treating the video se-

quence as a continuous spatiotemporal volume, methods can enforce smoothness and coherence by minimizing discrepancies between rendered frames and their interpolated counterparts. For instance, [304] introduces a novel framework that casts 360-degree dynamic object reconstruction as a 4D generation problem using uncalibrated monocular videos. This approach leverages a Cascade Dynamic Neural Radiance Field (DyNeRF) to facilitate stable convergence and explicitly employs an interpolation-driven consistency loss. This loss function optimizes the model by minimizing the difference between frames rendered from the DyNeRF and those interpolated by a pre-trained video interpolation model, effectively ensuring both spatial and temporal continuity even when the supervision signal is discrete along the time axis. Similarly, [305] highlights the limitations of linear motion models in traditional interpolation, proposing a quadratic method that exploits acceleration information to predict curvilinear trajectories. Integrating such advanced motion modeling into 4D reconstruction pipelines allows for more accurate handling of complex non-rigid deformations that linear assumptions fail to capture, providing a smoother temporal baseline for subsequent editing or composition.

Another critical paradigm involves mapping dynamic observations into a time-invariant canonical space. This decoupling of appearance and geometry from motion allows models to learn a static representation that can be deformed to any time step, thereby ensuring long-term consistency across frames. [306] exemplifies this by reconstructing general non-rigid scenes through a canonical model that remains invariant over time. By decomposing deformations into strongly regularized coarse components and weakly regularized fine components, this method enables the tracking of studio-scale motions and establishes correspondences over long durations, overcoming the limitations of prior works restricted to small movements. In a similar vein, [307] proposes an implicit neural representation for non-rigid scenes in a canonical space, utilizing a novel dynamic scene flow loss to handle large deformations and perform shape completion in occluded regions directly from monocular RGB videos. The use of canonical spaces is further refined in [308], which employs a two-stage approach where a low-rank neural deformation model is first fitted to preserve consistency in novel views, subsequently regularizing the optimization of 3D Gaussians anchored in local oriented volumes.

Beyond canonical mappings, the effective decomposition of static and dynamic elements is essential for resolving the ambiguities inherent in single-view inputs, particularly when distinguishing between camera motion and object deformation. [309] addresses the under-constrained nature of stationary monocular reconstruction by utilizing decomposed static and dynamic feature planes. This framework emphasizes learning dynamic regions through dense ray casting and employs deep supervised optimization to overcome challenges posed by occlusions and inadequate 3D clues. Furthermore, [310] introduces a decomposition strategy based on temporal characteristics, categorizing points in the 4D space as static, deforming, or new areas, each represented by separate neural fields. This decomposition not only aids in reconstruction accuracy but also facilitates compact modeling and streamable rendering, which is crucial for interactive applications and serves as a precursor to the object-level decomposition required for compositional scene generation.

The integration of generative priors has also emerged as a potent tool for filling in missing information and ensuring consistency in scenarios with extreme sparsity, effectively “hallucinating” plausible content where observations are missing. [311] demonstrates the ability to generate long-term dynamic videos from a single static image by utilizing layered depth images unprojected into a feature point cloud. To maintain global consistency under large camera motions, the method displaces this cloud based on estimated scene flow and leverages a pretrained diffusion model to inpaint and outpaint occluded regions, effectively hallucinating consistent content where no observation ex-

ists. Additionally, [312] tackles the reconstruction of multiple dynamic humans from monocular video with severely limited observations. By optimizing a 3D Gaussian Splatting representation in a canonical space and fusing sparse cues, the method leverages a pre-trained 2D diffusion model to synthesize unseen views while maintaining consistency with observed appearances, enabling high-quality animation even in the presence of severe occlusion and cropping.

Finally, ensuring geometric and scale consistency remains a persistent hurdle in monocular reconstruction, as scale ambiguity often leads to drifting or distorted geometries over time. [313] proposes a test-time optimization approach that transfers the robustness of affine-invariant depth models to diverse scenes by optimizing unknown scale-shift values to ensure inter-frame consistency. This allows for robust camera pose estimation and dense reconstruction even in low-texture regions without requiring large-scale 3D training data. Complementing this, [314] introduces a learning framework that promotes depth estimation consistency across different views by rendering novel scenes and designing specific loss functions, enabling the autonomous recovery of domain-specific scale-and-shift coefficients using solely unlabeled images. Together, these advancements in interpolation, canonical mapping, decomposition, and generative priors are rapidly closing the gap between limited monocular inputs and high-fidelity, temporally coherent 4D reconstructions. By establishing robust methods to recover dynamic geometry from sparse data, this section provides the essential building blocks for the next frontier: scaling from single-object recovery to holistic, multi-entity scene synthesis where interactions and compositionality take center stage.

5.5 Compositional Scene Generation and Object Interaction Modeling in 4D

5.5 Compositional Scene Generation and Object Interaction Modeling in 4D

Building upon the foundations of monocular reconstruction and temporal consistency established in the previous section, the frontier of 4D generation now faces the challenge of scaling from single-object recovery to holistic, multi-entity scene synthesis. While Section 5.4 addressed the ill-posedness of recovering dynamic geometry from sparse views, generating complex environments where multiple dynamic entities coexist, interact, and adhere to physical laws introduces a distinct “compositionality” problem. Traditional monolithic generation approaches often suffer from semantic collapse, inconsistent object counts, and a lack of coherent interaction between distinct elements. To overcome these limitations, recent research has pivoted towards compositional frameworks that decompose complex 4D scenes into individual dynamic entities, leveraging Large Language Models (LLMs) for high-level planning and advanced score distillation techniques for low-level geometric and temporal consistency. This paradigm shift not only resolves the ambiguities of single-entity generation but also lays the necessary groundwork for the advanced editing and stylization tasks discussed in the subsequent section.

A cornerstone of this compositional paradigm is the integration of LLMs to act as cognitive planners that decompose abstract textual prompts into structured scene descriptions. Instead of treating a prompt as a single holistic signal, these frameworks utilize the reasoning capabilities of LLMs to identify distinct entities, define their spatial relationships, and map out their individual trajectories within the 4D volume. For instance, the [302] framework exemplifies this approach by first employing an LLM to parse an input prompt into separate entities and their corresponding motion paths. This decomposition allows the system to construct each 4D object separately before assembling them into a unified scene. By decoupling the generation of individual objects, the method effectively mitigates the interference often seen in monolithic generation pipelines, ensuring that each entity maintains its structural integrity and motion fidelity. Furthermore, [302] employs a compo-

sitional score distillation technique that utilizes pre-trained diffusion models across text-to-image, text-to-video, and text-to-3D domains to refine the scene, guided by the LLM-defined trajectories. This multi-modal guidance ensures that the resulting 4D environment not only looks realistic from any viewpoint but also exhibits coherent temporal evolution.

Beyond simple decomposition, achieving realistic object interaction requires sophisticated modeling of physical constraints and collision avoidance. In dynamic environments, objects must not only move independently but also react to one another in physically plausible ways. The [315] framework addresses this by leveraging LLMs to derive spatial transformations and relationships directly from text prompts, creating a compositional representation consisting of semantic primitives. Crucially, [315] introduces a mechanism to model collision relationships between objects at a global level during the optimization process. This step is vital for enhancing physical correctness, as it prevents the interpenetration of objects—a common artifact in non-compositional methods—and ensures that the generated scene adheres to basic laws of physics. Similarly, the concept of modeling interactions is further explored in robotic and agent-based contexts, where understanding the dynamics between multiple agents is essential. While primarily focused on planning, the insights from [316] highlight the complexity of multi-agent interactions, underscoring the necessity for generative models to capture not just individual motion but also strategic interdependencies and team collaboration dynamics within a shared 4D space.

To facilitate these complex interactions, the underlying representation of the 4D scene must be both flexible and efficient. Dynamic 3D Gaussian Splatting has emerged as a preferred representation due to its ability to handle explicit geometry and appearance modeling while supporting real-time rendering, bridging the gap between the implicit fields discussed earlier and the editable assets required for downstream tasks. Methods like [292] demonstrate how dynamic 3D Gaussians can be leveraged to synthesize animated objects with high fidelity. By combining text-to-image, text-to-video, and 3D-aware multiview diffusion models, [292] enforces temporal consistency and realistic geometry simultaneously. The compositional nature of this approach allows different 4D animations to be seamlessly combined, enabling the creation of complex scenes where multiple characters or objects interact without merging into a singular, indistinct mass. Moreover, the ability to regularize the distribution of moving Gaussians, as proposed in [292], stabilizes the optimization process, which is critical when multiple moving parts are optimized jointly.

The challenge of interaction modeling also extends to the domain of trajectory control and motion transfer, ensuring that objects follow specific paths while maintaining their shape and appearance. The [293] framework addresses this by factoring motion into global and local components, representing global motion via rigid transformations along a spline-parameterized trajectory. This factorization enables the synthesis of scenes where objects move along arbitrary, user-defined paths, a capability essential for compositional scene generation where specific interaction scripts need to be enacted. By learning local deformations that conform to these global trajectories, [293] achieves a level of realism and motion magnitude that surpasses previous methods, allowing for the creation of dynamic scenes with significant spatial displacement. Similarly, [299] decouples motion and appearance to achieve superior video-to-4D generation, introducing mechanisms to mitigate shape degeneration and ensure the fidelity of learned motion. This disentanglement is particularly powerful for compositional scenarios, as it facilitates the transfer of learned motion patterns onto diverse 4D entities, enabling the rapid prototyping of interactive scenes with varied characters and objects.

Furthermore, the integration of physical priors and interaction reasoning is becoming increasingly important for generating truly interactive 4D environments. While many generative models focus

on visual plausibility, few inherently understand the physics of collision and contact. Research in related fields, such as [317], demonstrates the value of decomposing interactions into fundamental processes like “Interacting” and “Navigating” to synthesize long-term, physically plausible motions in cluttered scenes. Although focused on human motion, the principles of decomposing complex interactions into reusable controllers can be adapted for 4D scene generation to ensure that generated objects do not merely visually overlap but interact with physical validity. Additionally, the work on [318] proposes an optimization-based approach that infers hidden shape information from intersection constraints, incorporating physical plausibility constraints on object intersections. Adapting such energy minimization frameworks to the 4D generative setting could significantly enhance the realism of object collisions and occlusions in compositional scenes.

In summary, the evolution of 4D scene generation is rapidly advancing towards compositional frameworks that marry the semantic reasoning of LLMs with the geometric precision of score distillation and dynamic representations like 4D Gaussian Splatting. By decomposing scenes into individual entities, explicitly modeling their trajectories, and enforcing physical interaction constraints, these methods are paving the way for the creation of rich, interactive 4D environments. These advancements provide the structured, multi-object foundations necessary for the subsequent exploration of fine-grained manipulation, where the ability to edit, inpaint, and stylize specific components of a dynamic scene becomes paramount. Future research will likely focus on tightening the coupling between high-level semantic planning and low-level physical simulation, enabling the autonomous generation of complex narratives involving multiple interacting agents in dynamic worlds.

5.6 Advanced Editing, Inpainting, and Stylization in 4D Spaces

5.6 Advanced Editing, Inpainting, and Stylization in 4D Spaces

Building upon the compositional foundations and interaction modeling capabilities established in Section 5.5, the frontier of 4D generation now shifts toward manipulation: the ability to edit, inpaint, and stylize specific components of a dynamic scene without compromising spatio-temporal fidelity. While the previous section addressed how to construct multi-entity scenes with coherent physical interactions, this section focuses on the post-generation refinement of these assets. The challenge here is distinct; unlike static 3D scenes or 2D videos, 4D representations demand manipulation strategies that simultaneously respect complex geometric constraints, appearance consistency across novel viewpoints, and continuous motion dynamics. Recent advancements have begun to tackle these hurdles through a synergy of training-free adaptations of pre-trained diffusion models and specialized architectures fine-tuned for dynamic neural representations, bridging the gap between structured scene assembly and the flexible editing required for practical applications.

A primary focus in 4D editing is the development of unified pipelines capable of diverse manipulation tasks while preserving the underlying spatio-temporal structure. Stylization, in particular, has seen significant progress with methods that decouple content geometry from artistic style, ensuring consistency across both space and time. For instance, [181] introduces a zero-shot framework for 4D style transfer on dynamic Neural Radiance Fields (NeRFs). By representing the 4D feature space through a deformable canonical volume and learning a linear style transformation matrix based on feature covariance, this approach successfully transfers arbitrary visual styles to dynamic scenes. Crucially, it maintains consistency on moving objects by deforming rays according to geometric priors, thereby overcoming the static scene assumptions that limit traditional 3D style transfer methods. Similarly, [319] addresses the specific complexities of human video by employing a dual-NeRF representation for the subject and the scene. This framework utilizes a geometry-guided

tri-plane representation to enhance feature robustness, allowing for stylization that supports novel poses and viewpoints simultaneously, thus striking a superior balance between texture fidelity and temporal coherence.

Parallel to NeRF-based approaches, the emergence of 3D Gaussian Splatting (3DGS) as an efficient rendering primitive has catalyzed new paradigms for controllable 4D stylization and editing. The explicit nature of 3DGS allows for direct manipulation of geometric and appearance attributes with unprecedented efficiency. [320] proposes a framework that leverages these explicit properties to achieve high-efficiency stylization with adaptable control over perceptual factors. By introducing a GS filter to eliminate reconstruction artifacts and employing a nearest neighbor-based style loss, the method fine-tunes geometry and color parameters while preserving depth structures. This allows users to exert flexible control over color, scale, and specific regions, fostering an environment conducive to creative exploration in XR applications. Furthermore, [321] extends scene stylization to multi-view settings by processing pretrained Gaussians through a multi-resolution hash grid and a tiny MLP. This architecture ensures uniform stylized appearance across views, offering inherent geometric consistency and fast rendering speeds that are critical for real-time augmented reality applications.

In the domain of inpainting and structural editing, the core challenge lies in filling spatio-temporal voids or modifying object trajectories without introducing flickering artifacts or geometric discontinuities. While many video inpainting methods operate strictly in 2D, recent works are adapting these principles for 3D-aware dynamic scenes. [322] presents a framework that learns spatial and temporal residuals on downsampled videos and aggregates them to refine high-resolution outputs, ensuring temporal coherence in video completion tasks. Although primarily 2D-focused, the principles of decoupling spatial and temporal attention found in [323] are increasingly being adapted for 4D neural representations to attend precisely to moving objects and background textures separately. For 3D-specific editing, [324] demonstrates how pre-trained diffusion models can serve as geometry critics to correct misalignments during single-image 3D editing. By iteratively synthesizing novel views and aligning geometry, this method achieves high-fidelity edits with large viewpoint transformations, a capability that is foundational for extending editing to dynamic 4D sequences where object motion must be preserved.

To reduce the computational burden associated with per-scene optimization, training-free techniques have gained significant traction for manipulating 4D content. [325] proposes a zero-shot text-driven editing technique for 3D scenes that avoids multi-view inconsistency by modifying only a single view and enforcing consistency through an Edit Transformer. This transformer utilizes self- and cross-attention to propagate edits across views, significantly reducing runtime and memory resources. While currently focused on static scenes, the underlying mechanism of enforcing intra-view consistency and inter-view style transfer is directly applicable to 4D scenarios, suggesting a pathway for efficient dynamic scene editing. Additionally, [326] introduces a tuning-free framework leveraging Large Language Models (LLMs) for attention control to generate smooth animations from text prompts. By manipulating spatial features and self-attention mechanisms, LASER ensures structural consistency during image morphing, effectively bridging the gap between static image editing and dynamic animation generation without fine-tuning.

The precision of these 4D manipulations is further enhanced by the integration of depth awareness and promptability. [327] offers a unified depth-aware inpainting model for interactive text-to-texture synthesis, integrating depth cues to mitigate 3D inconsistencies. This approach allows for region-specific repainting and precise texture editing, which is essential for maintaining the physical plausibility of edited 4D assets. Moreover, [328] addresses the generation of lighting-less texture

maps for 3D meshes through a coarse-to-fine diffusion process, resolving illumination artifacts that often plague 4D texturing tasks. By specialized UV inpainting and high-resolution refinement models, it ensures semantic consistency across the generated textures, a prerequisite for realistic 4D stylization and editing.

Collectively, these works illustrate a rapid evolution in the capabilities for manipulating 4D content, moving from rigid generative pipelines to flexible, user-controllable frameworks. The convergence of training-free attention manipulation, depth-aware conditioning, and efficient explicit representations like 3DGS is enabling increasingly sophisticated control over dynamic scenes. These advancements provide the necessary toolkit for immersive applications in virtual production, gaming, and interactive media, where the ability to refine and restyle generated content is paramount. However, the computational cost of these editing operations, particularly when involving complex dynamic representations, remains a bottleneck for real-time deployment. Addressing these efficiency challenges is the critical next step, motivating the transition to the optimization and rendering paradigms discussed in the following section, which aim to make high-fidelity 4D interaction feasible on commodity hardware.

5.7 Efficient Optimization and Real-Time Rendering Paradigms for Higher Dimensions

5.7 Efficient Optimization and Real-Time Rendering Paradigms for Higher Dimensions

Building upon the advanced editing and stylization capabilities in 4D spaces discussed in the previous section, the practical deployment of these dynamic generative models hinges on overcoming profound computational barriers. The transition from static 3D content to dynamic 4D synthesis introduces an exponential increase in data dimensionality, imposing stringent requirements for temporal consistency alongside real-time performance. As the field evolves from proof-of-concept demonstrations to applications in virtual reality, gaming, and digital twins, research has bifurcated into two complementary streams: optimization-based implicit methods that prioritize geometric fidelity through continuous representations, and explicit point-based renderers that leverage hardware rasterization for unprecedented inference speeds. The synthesis of these approaches, mediated by advanced distillation techniques and adaptive spatial management, defines the current state-of-the-art in high-dimensional generative modeling.

Implicit neural representations, particularly Neural Radiance Fields (NeRFs) and their dynamic extensions, have historically provided a robust framework for modeling complex spatio-temporal volumes. However, their reliance on dense ray marching and multi-layer perceptron (MLP) evaluations has traditionally resulted in prohibitive rendering latencies. To address this, recent innovations have focused on accelerating the sampling process and optimizing underlying network architectures. For instance, [329] introduces a projection-aware sampling network that adaptively selects critical points along rays, achieving rendering speeds 15 to 23 times faster than vanilla NeRF while maintaining superior image quality. Similarly, [330] employs vector quantization to reduce the sampling space to a lower resolution, subsequently reconstructing fine details via a pre-trained decoder to mitigate high-resolution computational bottlenecks. In the domain of large-scale scenes, [331] proposes a hierarchical subdivision strategy that assigns smaller, specialized NeRFs to scene sub-regions based on complexity, enabling highly parallelized GPU inference. Furthermore, [332] demonstrates that through hierarchical model partitioning and distillation, real-time navigation on commodity hardware, including smartphones, is achievable for scenes with footprints up to 300 square meters. Additionally, [333] integrates partitioned NeRFs with the rasterization pipeline in Unreal Engine 4, achieving real-time rendering of large-scale scenes at 4K resolution.

Despite these advances in implicit modeling, a paradigm shift toward explicit representations—specifically 3D and 4D Gaussian Splatting—has emerged as the dominant solution for achieving real-time performance in dynamic settings. Unlike implicit fields requiring iterative volume rendering, Gaussian Splatting utilizes an explicit collection of anisotropic primitives that can be directly rasterized using standard graphics pipelines. The foundational work [334] established that representing scenes with 3D Gaussians allows for high-quality novel view synthesis at over 30 frames per second (FPS) for unbounded scenes. Extending this to the temporal domain, [276] introduces anisotropic 4D XYZT Gaussians that model dynamics by temporally slicing 4D primitives, achieving real-time rendering speeds of up to 583 FPS on modern GPUs. This explicit formulation is particularly advantageous for capturing abrupt motions and fine details that often challenge deformation-based implicit models. The efficiency of this approach is further highlighted in [335], which reduces 4D generation optimization time from several hours to mere minutes while enabling flexible motion control and export to standard 3D engines.

To push the boundaries of resolution and fidelity, researchers have developed specialized architectures tailored for extreme performance metrics. [336] proposes a 4D point cloud representation built on a 4D feature grid, combined with a differentiable depth peeling algorithm, to achieve 80 FPS rendering at 4K resolution—a 30x speedup compared to previous methods. For human performance capture, [337] marries 3D Gaussian representations with non-rigid tracking, utilizing a dual-graph mechanism to achieve a compression rate of approximately 25 times with less than 2MB of storage per frame, facilitating immersive experiences on diverse platforms. The efficiency of these explicit methods relies not only on the representation itself but also on sophisticated optimization strategies. [338] addresses spatial distribution inefficiencies in standard Gaussian representations by introducing densification strategies like blur split and depth reinitialization, alongside simplification techniques such as Gaussian binarization, to maximize rendering quality under strict resource constraints. Similarly, [339] leverages voxel-guided efficient sampling to substantially reduce optimization time for indoor scenes.

The generation of 4D content from text or single-view videos further necessitates efficient optimization loops to make the technology accessible. While Score Distillation Sampling (SDS) has been the cornerstone of this effort, it often suffers from slow convergence. [291] circumvents the need for back-propagating through large pretrained models at every step by generating spacetime-consistent images first and then directly training a 4D Gaussian splatting model, resulting in a 20-fold speedup. Similarly, [285] leverages the parallelism of image generation by composing scores from video and multi-view diffusion models to directly sample dense multi-view, multi-frame images, significantly accelerating the creation of continuous 4D representations. For scenarios requiring precise motion control, [293] factors motion into global rigid transformations and local deformations, enabling the synthesis of scenes with extensive motion ranges previously unattainable with bounded volume representations. Moreover, [299] decouples motion and appearance using sparse control signals and adaptive Gaussian initialization, ensuring high fidelity while maintaining computational efficiency. Other works like [288] and [287] also employ dynamic 3D Gaussians to permit efficient, high-resolution supervision through rendering during training.

Looking forward, the convergence of these optimization and rendering paradigms suggests a future where high-fidelity 4D content can be generated and interacted with in real-time on edge devices. Strategies such as adaptive densification, as seen in [287], which employs adaptive strategies to mitigate unstable Gaussian gradients, are critical for robust optimization. Additionally, the integration of learning-based surrogates for non-differentiable components, as proposed in [340], opens new avenues for optimizing complex graphics pipelines that were previously inaccessible to gradient-based

methods. As these techniques mature, the distinction between offline generation and real-time interaction will continue to blur, enabling autonomous agents and unified world models to operate within dynamically rich, photorealistic environments.

6 Domain-Specific Adaptations and Scientific Applications

6.1 Anatomically Constrained Synthesis and Morphological Fidelity

6.1 Anatomically Constrained Synthesis and Morphological Fidelity

The generation of high-fidelity medical imagery, particularly in three dimensions, presents unique challenges that distinguish it fundamentally from natural image synthesis. While standard generative models excel at capturing texture and global statistics, they frequently fail to adhere to the stringent topological and geometric constraints inherent in biological systems. In clinical and research contexts, synthesized data must not only appear realistic but also maintain valid structural morphology to ensure utility in downstream tasks such as surgical planning, radiation therapy, and in-silico clinical trials. Consequently, a significant body of research has emerged focusing on embedding explicit anatomical priors—ranging from probabilistic atlases to geometric hard constraints—directly into the architecture and loss functions of diffusion and flow-based models. This focus on intrinsic structural validity serves as a critical prerequisite for the robust domain adaptation strategies discussed in the subsequent section, as models must first capture correct anatomy before they can be successfully adapted across varying imaging modalities.

A foundational approach to ensuring morphological fidelity involves the integration of probabilistic atlases, which serve as spatial priors to guide the generative process. Traditional atlases are often constructed from grouped demographic data but may lack the specificity required for individual variation. Recent advancements have shifted toward data-driven frameworks that construct subject-specific probabilistic atlases by clustering regions with similar features, thereby adapting whole-brain atlases to new subjects with high precision [341]. These atlases provide essential contextual information that regularizes the generation process, reducing the likelihood of anatomical hallucinations. Furthermore, the concept of conditional atlases has been extended through diffusion models that generate deformation fields rather than images directly. By transforming a general population atlas into one representing a specific sub-population via learned deformation fields, methods like Diff-Def ensure structural plausibility and smooth transformations, effectively mitigating the training instabilities and artifacts common in direct image synthesis [342]. This deformation-based strategy is complemented by unsupervised learning frameworks that derive anatomical priors from unpaired segmentation data, enabling robust segmentation and synthesis even when paired training data is unavailable [343].

Beyond statistical priors, enforcing strict geometric constraints is critical for preserving the topology of complex organs. Generative models that operate solely in pixel or voxel spaces often struggle to maintain connectivity and surface smoothness, leading to fragmented or non-manifold structures. To address this, researchers have developed hybrid representations that combine implicit neural fields with explicit geometric scaffolds. For instance, the GEM3D framework utilizes a neural skeleton-based representation derived from the Medial Axis Transform to encode both topology and geometry. By first generating skeleton-based representations via a denoising diffusion process and subsequently deriving surfaces through a skeleton-driven implicit formulation, this approach ensures that synthesized shapes possess accurate topological properties, even for high-genus struc-

tures [344]. Similarly, the Neural Volumetric Mesh Generator (NVMG) employs a diffusion model to create voxelized shapes with realistic outlines, which are then converted into high-quality tetrahedral meshes through a voxel-conditional network. This pipeline incorporates regularization terms specifically designed to eliminate defects such as flipping and high distortion, ensuring the regularity of both interior and surface structures [345].

The challenge of maintaining morphological fidelity is further compounded by the need to model shape variability across diverse populations. Deep implicit statistical shape models (DISSMs) address this by marrying the representation power of convolutional neural networks with the robustness of traditional statistical shape models. By utilizing a deep implicit surface representation, DISSMs create a compact latent space that captures anatomical variance, allowing for the reliable fitting of anatomically plausible shapes to images through a Markov decision process-based pose estimation pipeline [346]. Complementing this, multiscale metamorphic variational autoencoders apply composable morphological transformations to reference images, embedding strong anatomical inductive biases directly into the latent space to ensure that generated volumes cover the data distribution while maintaining visual fidelity [347].

In scenarios where data is sparse or incomplete, the role of anatomical priors becomes even more pronounced. Methods leveraging “mental maps” or environment mapping utilize Bayesian deep learning to extrapolate full volumetric anatomy from sparse 2D slices, effectively mimicking the reasoning process of radiologists who rely on normative anatomical knowledge [348]. Similarly, approaches that model 3D brain MRI distributions by combining 2D slice variational autoencoders with Gaussian models capturing inter-slice relationships demonstrate how lower-dimensional priors can be aggregated to synthesize coherent high-resolution 3D volumes [349]. For specific applications like computational fluid dynamics, statistical generative models are employed to synthesize patient-specific 3D geometries from baseline shapes, enabling rapid hemodynamic predictions without the computational cost of traditional simulation-based parameterization [350].

Furthermore, the synthesis of pathological data requires models that can disentangle healthy anatomy from disease-specific morphological changes. Frameworks like brainSPADE3D enable the joint generation of multi-modal MR images and segmentations conditioned on specific pathological phenotypes, ensuring that the synthesized data reflects realistic combinations of pathologies and contrasts [351]. This capability is crucial for training robust diagnostic models that can handle rare or unexpected conditions. Additionally, generative enhancement techniques such as GEM-3D facilitate dataset expansion by generating versatile 3D images from existing datasets using informed slices and editable mask volumes, thereby preserving patient-wise information while introducing controlled variations [352].

Ultimately, the integration of anatomical constraints into generative models represents a paradigm shift from purely data-driven synthesis to physics-and-biology-informed generation. Whether through probabilistic atlases [353], geometric skeletons [344], or deformation fields [342], these methods ensure that synthesized 3D volumes possess the structural integrity required for clinical deployment. By establishing a foundation of morphologically accurate synthetic data, these approaches not only support personalized medicine and education [354] but also provide the reliable, structure-preserving source data necessary for the cross-modality adaptation and generalization challenges addressed in the following section.

6.2 Domain Adaptation and Generalization Across Imaging Modalities

6.2 Domain Adaptation and Generalization Across Imaging Modalities

Building upon the foundation of anatomically constrained synthesis established in the previous section, where models are regularized to preserve intrinsic biological structures, the next critical frontier involves ensuring these models remain robust when deployed across diverse imaging environments. The utility of morphologically faithful synthetic data is frequently compromised by the pervasive issue of domain shift, where models trained on data from specific scanners, protocols, or patient populations fail to generalize to unseen target domains. This distributional discrepancy is particularly acute in cross-modality scenarios, such as adapting between Magnetic Resonance Imaging (MRI) and Computed Tomography (CT), or between different MRI weightings like T1 and T2. To address these challenges without the prohibitive cost of acquiring new annotated datasets, the research community has developed sophisticated strategies ranging from unsupervised domain adaptation (UDA) leveraging cycle-consistency to few-shot meta-learning frameworks. These approaches aim to align feature distributions while strictly preserving the semantic and structural integrity ensured by the anatomical priors discussed previously, thereby learning invariant representations that are robust to the heterogeneity inherent in medical imaging devices.

A dominant paradigm in addressing cross-modality shifts involves image-to-image translation techniques grounded in cycle-consistency. These methods seek to translate images from a source domain to a target domain style while preserving the underlying anatomical structure necessary for accurate segmentation. However, standard generative adversarial networks often fail to maintain perfect semantic consistency during style transfer, leading to structural distortions that degrade downstream task performance—a failure mode that directly contradicts the morphological fidelity goals of Section 6.1. To mitigate this, researchers have introduced semantic consistency constraints directly into the translation process. For instance, the [355] framework enforces that synthesized images, regardless of their modality style, yield identical segmentation maps, thereby ensuring that the translation module does not alter critical diagnostic features. Similarly, [356] integrates a segmentation loss term into the Cycle-GAN training objective, explicitly promoting the preservation of medical structures such as blood vessels and cardiac chambers during the domain translation process. Further refining this concept, [357] introduces an edge-based loss to ensure that fine details, which are often lost in standard adaptation processes, are retained during the harmonization of fundus images across different domains.

Beyond pixel-level alignment, advanced UDA strategies focus on feature-space alignment and self-training mechanisms to bridge the domain gap more effectively. The challenge of aligning distributions is complicated by the fact that global alignment may ignore local discrepancies where specific anatomical regions exhibit larger domain gaps. Addressing this, [358] proposes a Global-Local Union Alignment strategy that synthesizes target-like source content to reduce global shifts while prioritizing the alignment of discriminative local features. In scenarios where source data cannot be shared due to privacy regulations, Source-Free UDA (SFUDA) has emerged as a critical area of study. Methods such as [359] and [360] demonstrate how to adapt pre-trained models to target domains using only batch normalization statistic adaptation and memory-consistent self-training, eliminating the need for source data access during the adaptation phase. Furthermore, [361] enhances this privacy-preserving approach by introducing uncertainty estimation and prototype-based denoising schemes to filter noisy pseudo-labels generated by the source model, thereby stabilizing the self-training process on unlabeled target data.

When labeled data in the target domain is scarce but not entirely absent, few-shot learning and

meta-learning paradigms offer robust solutions for rapid adaptation, complementing the data-efficient synthesis methods mentioned earlier. These methods are designed to leverage a small number of annotated target samples to calibrate models effectively. The [362] framework utilizes a gradient-based meta-learning strategy to jointly train hallucination and segmentation models, generating diverse synthetic examples that enhance cross-modality performance in few-shot settings. Complementing this, [363] addresses the instability of adapting object detectors with limited target data by employing a bi-level module that aligns paired local patch features and regularizes source model features to prevent over-adaptation. The integration of causal reasoning into few-shot adaptation is also gaining traction; [364] posits that while data distributions may shift drastically, the underlying causal mechanisms generating the data often remain invariant, providing a solid statistical basis for adaptation even when source and target domains appear vastly different. Additionally, [365] incorporates shape compactness and smoothness constraints into a meta-learning scheme, simulating domain shifts during training to improve generalization to unseen scanners and protocols, thus reinforcing the geometric constraints vital for clinical reliability.

The complexity of domain shift is further compounded by multi-source scenarios and heterogeneous modalities with differing dimensionalities. Multi-source domain adaptation (MSDA) frameworks like [366] aggregate knowledge from multiple labeled sources by generating adapted domains for each source and enforcing consistency across them before aligning with the target. In cases where modalities differ fundamentally, such as adapting between 2D images and 3D point clouds or between MRI sequences with different channel counts, specialized approaches are required. [367] enforces consistency between predictions across different modalities via mutual mimicking, while [368] employs stochastic generative modeling to translate between heterogeneous domains, introducing semantic cycle-consistency losses to handle the mismatch in imaging channels. Finally, addressing the issue of sub-population shifts and biases within modalities, [369] leverages vision-language models to reshape feature representations, using natural language inputs to debias image features and improve robustness across diverse patient sub-groups. Collectively, these advancements illustrate a maturing field moving from simple global alignment to nuanced, structure-preserving, and privacy-aware adaptation strategies. By securing the generalizability of models across varying acquisition conditions, these methods provide the stable, domain-robust foundation necessary for the subsequent challenge: solving ill-posed inverse problems where physical laws must guide the reconstruction of high-fidelity images from sparse measurements, as detailed in the following section.

6.3 Solving Inverse Problems with Physics-Informed and Geometry-Aware Priors

6.3 Solving Inverse Problems with Physics-Informed and Geometry-Aware Priors

While the preceding discussion on domain adaptation focuses on aligning statistical distributions across varying imaging protocols and scanners, a distinct yet complementary challenge arises in solving inverse problems where the goal is to recover high-fidelity images from severely undersampled or noisy measurements. In scientific imaging domains such as medical diagnostics, non-destructive testing, and geophysical sensing, these inverse problems are notoriously ill-posed, characterized by nonlinear forward operators and extreme sparsity. Traditional deep learning approaches, which rely heavily on supervised pairs of corrupted and ground-truth data, frequently struggle with generalization and hallucination when faced with distribution shifts or measurement conditions unseen during training. To address these limitations, recent research has pivoted toward integrating physics-informed priors and geometry-aware mechanisms directly into the architecture and sampling trajectories of diffusion and flow-based models. This integration ensures that synthesized solutions not only adhere to the statistical manifold of natural images but also strictly satisfy the under-

lying physical governing equations and system geometries, bridging the gap between data-driven flexibility and model-based rigor.

A primary avenue for this integration involves modifying the diffusion process itself to align with the physical acquisition model, rather than treating physics as an external post-processing step. In Magnetic Resonance Imaging (MRI), where the forward model is defined by the Fourier transform of undersampled k-space data, standard diffusion models operating solely in the image domain can introduce inconsistencies. To overcome this, researchers have proposed formulating the diffusion process directly in the measurement domain or adapting the Stochastic Differential Equation (SDE) to respect the specific properties of k-space. For instance, the [370] paper introduces a framework that models high-frequency attenuation in k-space as a heat diffusion equation, subsequently employing score-based methods to solve the modified reverse heat equation. This approach ensures that the generative prior is physically interpretable and specifically tailored to the interpolation challenges of parallel imaging. Similarly, [371] leverages the iterative self-consistent term from the traditional SPIRiT algorithm to formulate a novel SDE. By driving the diffusion process with k-space physical priors, this method achieves high-quality reconstruction even at substantial acceleration rates, effectively bridging the gap between model-based optimization and deep generative modeling.

Beyond modifying the diffusion trajectory, another critical strategy involves the explicit incorporation of geometric and structural priors to regularize the solution space, particularly in ultra-sparse tomographic scenarios. In Computed Tomography (CT) and other volumetric imaging tasks, the lack of projection data can lead to severe artifacts that purely data-driven models may fail to correct. The [372] study demonstrates that seamlessly integrating known geometric priors of the imaging system is essential for enhancing performance in ultra-sparse sampling regimes. By embedding these constraints, the model avoids the pitfalls of learning entirely from data, thereby improving generalizability and reducing the complexity required for the neural network to infer structural validity. Furthermore, the challenge of three-dimensional consistency in volumetric reconstruction has been addressed by frameworks that augment 2D diffusion priors with model-based constraints in the remaining dimensions. The work [373] illustrates how combining a pre-trained 2D diffusion prior with a model-based prior allows for coherent 3D reconstructions from sparse-view tomography, achieving state-of-the-art results on commodity hardware while maintaining fidelity across slices.

The integration of physics also extends to handling nonlinear forward models and complex noise structures that deviate from standard Gaussian assumptions. In many scientific applications, the relationship between the latent image and the observed measurements is governed by nonlinear partial differential equations (PDEs) or involves structured noise. The [374] paper proposes a method that combines an unconditional diffusion prior with a likelihood score function derived from a nonlinear physical model. This plug-and-play approach enables the solution of inverse problems for generalized CT system designs without retraining, effectively handling the nonlinearity of x-ray transmission physics. Additionally, when measurements are corrupted by structured noise rather than simple additive white noise, joint conditional reverse diffusion processes have been developed. As shown in [375], learning scores for both the noise and signal-generating distributions allows the model to disentangle the true signal from complex interference patterns, outperforming baselines that assume simplified noise models.

Advanced optimization strategies further enhance the synergy between physical constraints and generative priors. Bi-level guidance frameworks have emerged as a powerful tool for steering unconditional predictions toward measurement consistency. The [376] introduces a strategy that first approximates a conditional posterior mean as a reference point and then solves an outer-level prox-

imal optimization objective. This dual-layered approach effectively reduces hallucinatory artifacts in severely degraded MRI and CT scans. Moreover, the concept of “hard data consistency” has been formalized to ensure that every step of the reverse sampling process adheres strictly to the observed measurements. The [377] study proposes an algorithm that solves an optimization problem during reverse sampling to map samples back onto the measurement-consistent manifold, demonstrating superior performance in both linear and nonlinear inverse problems compared to soft-constraint methods.

Finally, the fusion of physics-informed learning with sparse representation theories offers a robust pathway for solving underdetermined systems where data is extremely scarce. Approaches that combine compressed sensing principles with deep generative models leverage the sparsity of signals in specific domains to regularize the inversion. For example, [378] jointly optimizes sampling patterns and reconstruction using continuous spherical harmonic representations, integrating unique dMRI properties to achieve robustness against noise. Similarly, methods like [379] generalize iterative recovery models from the image domain to high-dimensional feature domains, incorporating real physics into the unrolled network architecture to ensure real-time inference speeds without sacrificing accuracy. Collectively, these advancements establish diffusion and flow models not merely as black-box generators, but as rigorous, physics-compliant solvers. This evolution from statistical restoration to physically grounded reconstruction provides the high-fidelity, structurally accurate imagery necessary for the subsequent stage of analysis: causal reasoning and biomarker discovery, where the integrity of generated counterfactuals is paramount.

6.4 Causal Reasoning and Counterfactual Generation for Biomarker Discovery

6.4 Causal Reasoning and Counterfactual Generation for Biomarker Discovery

While the preceding section established how physics-informed priors ensure the structural fidelity and measurement consistency of reconstructed images, a distinct yet complementary challenge arises in moving beyond static restoration to dynamic, interventional reasoning. In clinical contexts, understanding “what-if” scenarios—such as how a patient’s anatomy would evolve under a specific treatment or how disease progression might differ given altered demographic factors—is critical for personalized medicine and biomarker discovery. Traditional deep learning approaches, while powerful in pattern recognition, often fail to distinguish between spurious correlations and genuine causal mechanisms, leading to models that may not generalize well across different populations or clinical settings. To address this, recent research has focused on developing Structural Causal Models (SCMs) that explicitly encode the causal relationships between demographic variables, clinical indices, and high-dimensional medical imagery. By leveraging these models, researchers can generate counterfactual images that serve as a rigorous tool for isolating causal effects and discovering patient-specific biomarkers, thereby bridging the gap between the physically accurate reconstructions discussed previously and the actionable clinical insights required for the molecular design applications explored next.

A foundational aspect of this domain involves modeling the causality between non-image variables, such as age or cognitive scores, and medical images like brain MRIs. For instance, in the context of Alzheimer’s Disease, novel methods have been proposed to depict the causality between demographic variables, clinical indices, and brain MR images using structural causal models combined with styled generators [380]. These approaches utilize low-dimensional latent feature representations to make the learning of causal relationships tractable, allowing for the synthesis of counterfactual brain MR images where specific attributes like age or brain volume are intervened

upon. Similarly, for Multiple Sclerosis, SCMs have been developed to model interactions between disease covariates and MR images, enabling the generation of counterfactuals that visualize how a patient’s brain would appear if specific disease parameters were altered [381]. Such capabilities are essential for modeling disease progression and performing image processing tasks where controlling for confounders is necessary.

Beyond volumetric imaging, causal reasoning has been extended to anatomical shape analysis through Deep Structural Causal Shape Models (CSMs). These models utilize geometric deep learning techniques within the framework of deep structural causal models to enable subject-specific prognoses [382]. Unlike traditional statistical shape models that operate at the population level, CSMs can answer counterfactual questions regarding morphological variations, such as predicting how a specific patient’s brain structure would change if they were ten years older. This level of granularity is vital for identifying biomarkers that are predictive of future disease outcomes on an individual basis. The discovery of patient-specific imaging markers is further advanced by counterfactual synthesis processes that perturb local imaging features pertinent to subject-specific future disease evolution [383]. By examining the features that must be perturbed to alter a predicted outcome, researchers can identify novel, subject-specific predictive markers that reflect established clinical indicators.

To realize these complex interventional capabilities, generative architectures have evolved to specifically support counterfactual estimation in high-dimensional spaces. Diffusion-based approaches, such as Diff-SCM, have been introduced to perform counterfactual estimation by inferring latent variables via deterministic forward diffusion and intervening on the reverse process using gradients of an anti-causal predictor [384]. These models have demonstrated the ability to produce more realistic and minimal counterfactuals compared to baselines, ensuring that the generated changes are strictly attributable to the intervention rather than artifacts. Furthermore, instruction-learning frameworks like BiomedJourney have emerged to leverage multimodal patient journeys, using natural language descriptions of disease progression to train latent diffusion models for counterfactual biomedical image generation [385]. This approach effectively bridges the gap between scarce time-series imaging data and abundant single-image report pairs, enhancing the robustness of disease progression modeling.

However, the deployment of causal generative models in healthcare is not without pitfalls. A significant issue identified in recent literature is “attribute amplification,” where unrelated attributes are spuriously affected during interventions due to the use of hard labels in training, leading to biases across protected characteristics [386]. To mitigate this, soft counterfactual fine-tuning strategies have been proposed to ensure that generated counterfactuals remain faithful and unbiased. Additionally, the handling of missing data and incomplete clinical records is addressed through semi-supervised deep causal generative models that exploit causal relationships to infer missing values and generate realistic counterfactuals even when labels are partially absent [387].

The evaluation of these counterfactual generations remains a complex task due to the lack of observable ground truth for hypothetical scenarios. Comprehensive benchmarking frameworks have been developed to assess diverse aspects of counterfactuals, including composition, effectiveness, minimality of interventions, and image realism [388]. These frameworks are crucial for validating that the discovered biomarkers are indeed causally linked to the disease outcome rather than being artifacts of the generative process. Moreover, the concept of backtracking counterfactuals, where causal laws are kept intact while altering specific circumstances, offers a causally compliant alternative for explaining model decisions and identifying root causes of disease [389]. By assigning root causal contribution scores using Shapley values, these methods provide a rigorous mathematical

formulation for automatically detecting root causes from data, adapting to disease prevalence and admitting fast computation [390].

In summary, the convergence of causal inference and generative modeling is unlocking new frontiers in biomarker discovery. By moving beyond association to intervention and counterfactual reasoning, these models offer a powerful toolkit for personalized disease modeling, equitable healthcare, and the identification of robust, causally validated imaging markers. The continued development of scalable algorithms, such as those leveraging normalizing flows for personalization in public policy analysis [391], and the integration of unstructured data proxies for confounder adjustment [392], promise to further refine our ability to understand and intervene in complex biological systems. Having established the capacity to generate causally sound counterfactuals at the organ and tissue level, the discourse now naturally extends to the atomic scale, where similar principles of physical adherence and generative control are applied to the design of novel molecules and materials.

6.5 Molecular and Material Design via Equivariant Generative Flows

6.5 Molecular and Material Design via Equivariant Generative Flows

Building upon the causal frameworks discussed in the previous section, which enable counterfactual reasoning in biomedical imaging, the application of generative AI extends critically to the atomic scale for the design of novel molecules and advanced materials. This domain represents a frontier where generative models must not only learn data distributions but also strictly adhere to the fundamental laws of physics. Unlike medical images where causal interventions might alter demographic attributes, molecular and material generation is governed by the rigid symmetries of Euclidean space. Any valid generative model must inherently respect rotation, translation, and permutation equivariance to ensure that predicted properties and structures remain physically consistent regardless of the coordinate system. Traditional generative approaches often struggle to enforce these constraints, leading to unstable sampling dynamics or geometrically invalid outputs. Consequently, the field has increasingly pivoted toward group-equivariant stochastic differential equations (SDEs) and flow matching paradigms, embedding physical symmetries directly into model architectures and training objectives to bridge the gap between data-driven generation and physical reality.

A foundational strategy in this domain involves adapting Normalizing Flows to respect Euclidean symmetries. Early breakthroughs demonstrated the efficacy of $E(n)$ Equivariant Normalizing Flows, which integrate discriminative equivariant graph neural networks into continuous-time differential equations to create invertible functions capable of jointly generating molecule features and 3D positions [393]. These models significantly outperform non-equivariant baselines on particle systems and small molecules by ensuring that the generated probability density transforms correctly under symmetry operations. Advancing this paradigm, researchers have introduced equivariant continuous normalizing flows (CNFs) trained via flow matching objectives. Leveraging optimal transport theory, these models learn direct vector fields between distributions, a approach known as Equivariant Flow Matching, which allows for simulation-free training and yields flows with shorter integration paths, thereby improving sampling efficiency for many-body systems such as proteins and small molecules [394]. Furthermore, the development of $SE(3)$ Equivariant Augmented Coupling Flows has enabled fast sampling and exact density evaluation while preserving both $SE(3)$ and permutation equivariance, marking a significant step toward practical Boltzmann generators that can sample from complex energy landscapes without relying on tailored internal coordinate featurizations [395].

Beyond continuous flows, the discrete and hybrid nature of molecular data—comprising both cat-

egorical atom types and continuous coordinates—has spurred the development of specialized architectures. For instance, Equivariant Finite Normalizing Flows utilize discrete layers, such as G-Residual and G-Coupling Flows, to elevate classical flow architectures with equivariant maps, proving that equivariant diffeomorphisms can be exactly mapped even in discrete settings [396]. In parallel, diffusion models have emerged as powerful alternatives for geometry generation. Models like GeoDiff treat atoms as particles and reverse a diffusion process using equivariant Markov kernels, theoretically guaranteeing that the resulting distribution is roto-translational invariant [397]. To further enhance the quality and diversity of generated molecules, recent innovations such as MDM incorporate dual equivariant encoders to capture interatomic forces and introduce distributional controlling variables to enforce thorough exploration during the reverse diffusion process [398].

The necessity of equivariant principles extends critically to periodic materials, which possess distinct symmetry groups compared to isolated molecules. The SyMat framework addresses this by employing a score-based diffusion model with a novel symmetry-aware probabilistic model specifically designed to capture the physical symmetries of periodic material structures, including lattice lengths and angles [399]. Similarly, the generation of crystallographic packings has been approached through information-geometric frameworks that define trust regions on statistical manifolds, enabling the search for densest packings restricted to crystallographic symmetry groups [400]. For more complex material tensors, such as thermal conductivity in anisotropic bones, stochastic modeling frameworks have been formulated to respect positive definiteness and spatial invariances simultaneously, utilizing Lie algebra representations to control orientation and strength independently [401].

Efficiency remains a paramount concern in scientific applications where high-throughput screening is required. To this end, lightweight equivariant interaction graph neural networks have been proposed, utilizing scalar-vector dual representations to achieve accuracy comparable to *ab initio* molecular dynamics while retaining the computational efficiency of classical simulations [402]. Additionally, acceleration schemes specifically designed for SE(3)-invariant spaces have been developed, utilizing projected differential equations to reduce approximation errors and achieve speedups of 50 to 100 times compared to standard solvers [403]. The integration of hybrid probability paths, where coordinate paths are regularized by equivariant optimal transport, has also shown to stabilize probability dynamics and significantly accelerate sampling in 3D molecule generation [33].

Moreover, the unification of 2D topological graphs and 3D geometric point clouds has been explored through models like MUDiff, which combines discrete and continuous diffusion processes within a unified transformer architecture that adheres to 3D roto-translation equivariance constraints [404]. This holistic approach ensures that both the connectivity and spatial arrangement of atoms are generated coherently. In the realm of dynamics, DiffMD proposes a geometric diffusion model that directly estimates the gradient of the log density of molecular conformations, incorporating atomic velocities and directions via 3D spherical Fourier-Bessel representations to simulate kinetic molecular processes [405]. Collectively, these advancements illustrate a paradigm shift where strict adherence to physical symmetries via equivariant flows and diffusion processes is no longer an optional enhancement but a fundamental requirement for generating reliable, high-fidelity molecular and material structures. While these physics-informed generative models excel when the underlying laws are known, they often rely on specific architectural inductive biases. In contrast, the following section explores how foundation models and self-supervised learning address scenarios where data is scarce and explicit physical rules may be less defined, leveraging vast unlabeled corpora to learn universal representations across diverse scientific domains.

6.6 Foundation Models and Self-Supervised Learning for Data-Scarce Environments

6.6 Foundation Models and Self-Supervised Learning for Data-Scarce Environments

While the preceding section demonstrated how strict adherence to physical symmetries via equivariant flows enables reliable generation in domains with well-defined laws, many scientific frontiers face a different bottleneck: the scarcity of labeled data and the absence of explicit governing rules. In these contexts, the proliferation of foundation models has fundamentally altered the landscape of machine learning, offering a paradigm shift from training task-specific models from scratch to adapting large-scale pre-trained systems. This transition is particularly critical where high annotation costs, privacy constraints, and data heterogeneity render traditional supervised learning impractical. By leveraging vast quantities of unlabeled scientific data, foundation models and self-supervised learning (SSL) strategies learn universal representations that capture underlying physical, biological, or structural patterns. These representations can then be efficiently transferred to downstream tasks with minimal labeled examples, bridging the gap between data-poor regimes and high-performance inference.

The core challenge in scientific machine learning often lies in the heterogeneity of data sources and the scarcity of labeled instances required for robust generalization. Traditional transfer learning methods, which typically rely on fine-tuning models pre-trained on natural images like ImageNet, frequently suffer from significant domain shifts when applied to specialized fields. To address this, recent research advocates for the development of domain-specific foundation models trained via SSL on large-scale unlabeled datasets inherent to the target field. For instance, in computational pathology, the collection of annotated histology slides is prohibitively expensive; however, the availability of billions of unlabeled pathology images has enabled the training of visual foundation models that significantly outperform those pre-trained on natural images. Studies have demonstrated that pre-training on pathology-specific data using algorithms like DINO or masked autoencoders leads to superior generalization across diverse clinical tasks, ranging from cancer detection to mutation prediction, effectively establishing a new baseline for data-efficient learning in healthcare [406]. Similarly, in brain network analysis, the heterogeneity of functional patterns and the scarcity of clinical imaging data pose significant hurdles. The introduction of frameworks like BrainMass, which utilizes mask modeling and feature alignment on collated datasets comprising tens of thousands of participants, illustrates how self-supervised pre-training can yield models with exceptional few-shot and zero-shot learning capabilities for diagnosing neurological disorders [407].

The efficacy of these foundation models is not merely a function of scale but also of the strategic design of pre-training objectives that exploit the intrinsic structure of unlabeled data. In domains where labeled data is sparse, SSL serves as a powerful mechanism to learn informative representations without explicit supervision. For example, in electroencephalography (EEG) analysis, where labeling requires expert clinical interpretation, self-supervised strategies that predict temporal context or reconstruct masked signals have proven to outperform purely supervised approaches in low-data regimes. These methods capture essential physiological information, enabling the development of foundation models like EEGFormer that offer both transferable performance across various downstream tasks and interpretable insights into brain activity patterns [408] [409]. Furthermore, the concept of “big learning” has been proposed to describe a universal paradigm where a single foundation model exhaustively exploits information from diverse domains by modeling joint, conditional, and marginal data distributions, thereby unifying conventional machine learning paradigms and facilitating flexible cooperation in data-scarce environments [410].

Beyond simply pre-training on large datasets, advanced strategies such as meta-learning and efficient data selection have been integrated to enhance the transferability of self-supervised knowledge. In scenarios involving highly heterogeneous data, such as functional MRI for predicting post-traumatic epilepsy, combining meta-learning with self-supervised pre-training allows models to generalize effectively from abundant healthy control data to scarce clinical datasets. This bi-level optimization approach ensures that the learned features are robust to domain discrepancies, significantly improving performance on small-scale clinical tasks [411] [412]. Additionally, the efficiency of pre-training is paramount when computational resources are limited. Frameworks like SEPT (Scalable and Efficient visual Pre-Training) propose retrieving unlabeled samples with distributions similar to the target task, thereby reducing the volume of pre-training data required by an order of magnitude while maintaining competitive performance. This data selection perspective is crucial for scientific applications where curating massive datasets may be infeasible [413].

The application of foundation models extends to other scientific verticals, including molecular learning and particle physics, where data generation is costly and complex. In molecular learning, parameter-efficient foundation models like MiniMol are pre-trained on millions of molecules across thousands of tasks, demonstrating significant improvements in downstream ADMET predictions despite the inherent data scarcity in biological measurements [414]. Similarly, in particle physics, the OmniJet- α model represents a breakthrough as the first cross-task foundation model, successfully transferring knowledge from unsupervised jet generation to supervised jet tagging, thereby reducing the data and time requirements for physics simulations [415]. These examples underscore a broader trend: the move towards “foundation models for scientific machine learning,” where pre-training on a mixture of different physics problems enables adaptation to various downstream applications with orders of magnitude fewer examples than training from scratch [416].

Moreover, the robustness of self-supervised representations to dataset imbalance and distribution shifts makes them particularly suitable for real-world scientific data, which often exhibits long-tailed distributions. Research indicates that off-the-shelf self-supervised representations are inherently more robust to class imbalance than supervised counterparts, learning richer features from frequent data that aid in classifying rare classes [417]. This property is vital for applications like skin lesion analysis or rare disease detection, where positive cases are exceptionally rare. Finally, the integration of uncertainty-aware mechanisms and Bayesian approaches into self-training frameworks further enhances reliability in low-label scenarios, allowing models to leverage unlabeled data confidently while mitigating the risks of pseudo-label noise [418] [419]. As the field progresses, the synergy between large-scale self-supervised pre-training and efficient adaptation techniques promises to unlock new frontiers in scientific discovery. However, while these universal representations address data scarcity, specific domains like remote sensing introduce additional, stringent constraints regarding geospatial semantic consistency, necessitating specialized frameworks that preserve structural fidelity alongside generalization, a topic explored in the following section.

6.7 Remote Sensing and Geospatial Semantic Consistency

6.7 Remote Sensing and Geospatial Semantic Consistency

While Section 6.6 highlighted how foundation models and self-supervised learning address data scarcity through universal representations, the domain of remote sensing (RS) introduces a distinct and stringent constraint: the imperative of geospatial semantic consistency. Unlike natural images where aesthetic style transfer may prioritize visual appeal, RS applications demand that any generative transformation—whether for domain adaptation, data augmentation, or ecological

monitoring—strictly preserves the underlying physical and structural semantics. A failure to maintain these features can lead to catastrophic errors in critical downstream tasks such as land cover classification, change detection, and disaster assessment. Consequently, recent research has pivoted from merely leveraging large-scale pre-training to developing specialized frameworks that ensure semantic fidelity while bridging the gaps between diverse sensors, spectral bands, and geographical contexts.

A primary application of this semantic-consistent generation is unsupervised domain adaptation (UDA), which tackles distribution shifts arising from varying sensors, acquisition times, or locations. Traditional methods often struggle with the discrepancy between satellite datasets, necessitating models that can narrow these gaps without compromising per-pixel semantic integrity. Advanced image-to-image translation models have been proposed to preserve semantic consistency during style transfer, ensuring that segmentation models trained on adapted images achieve performance comparable to state-of-the-art supervised methods [420]. These approaches are particularly vital when transferring knowledge between sensors with distinct spectral responses, such as translating Landsat 8 imagery to match Sentinel-2 characteristics, thereby enabling the reuse of segmentation models across different satellite constellations [421]. The core challenge lies in performing this translation without corrupting semantic content, a problem exacerbated in heterogeneous change detection tasks involving optical and SAR imagery. To mitigate this, deep homogeneous feature fusion strategies have been introduced to segregate semantic content from style features, preventing the deterioration of change detection accuracy in regions affected by seasonal variations or sensor differences [422].

The complexity of maintaining geospatial semantic consistency is further compounded by the need to handle multi-scale objects and diverse environmental contexts across disparate regions. Models trained on urban scenes often fail to generalize to rural areas due to significant differences in landscape morphology and object distributions. To address this, datasets specifically designed for domain adaptive semantic segmentation, encompassing both urban and rural domains, have been constructed to benchmark methods capable of handling multi-scale objects and complex backgrounds [423]. Furthermore, the transferability of learning models across these regions remains a critical concern. Research has demonstrated that raw transferability is often insufficient, prompting the development of label-free assessment methods that utilize spectral indices to quantify model adaptability in target domains where ground truth is unavailable [424].

In the realm of ecological monitoring and climate change mitigation, preserving semantic fidelity is essential for accurate long-term analysis. Seasonal variations, such as changes in vegetation color between winter and summer, can be falsely interpreted as land cover changes if generative models do not effectively disentangle style from content. Innovative solutions include style-based recalibration modules and improved style discriminators that capture seasonal features effectively, allowing for consistent image translation that supports reliable change detection over time [425]. Moreover, the integration of geographical metadata has emerged as a powerful tool for enhancing semantic consistency. By leveraging geo-coordinates to align source and target domains, lightweight models like GeoMultiTaskNet have achieved state-of-the-art performance in land cover mapping, demonstrating that spatial context is a crucial prior for maintaining semantic alignment across diverse regions [426].

The advent of large vision-language models (VLMs) and multimodal learning has opened new frontiers for ensuring semantic consistency through textual guidance and metadata collaboration, echoing the foundation model trends discussed previously. Although the scarcity of large-scale, semantically diverse image-text pairs in remote sensing has historically hindered VLM development,

recent efforts have bridged this gap by constructing comprehensive datasets that connect open remote sensing images with rich semantics derived from OpenStreetMap, facilitating zero-shot transfer for fine-grained object attribute classification and cross-modal retrieval [427]. Building on this, metadata-collaborative frameworks have been developed to extract key characteristics, such as climate zones, from image metadata and convert them into text prompts. This approach enables vision-language representation learning that significantly improves generalization and reliability in semantic segmentation tasks by incorporating multimodal information often ignored by unimodal models [428]. Additionally, parameter-efficient transfer learning methods have been explored to adapt pre-trained vision-language models to the remote sensing domain with minimal computational cost, addressing the high intra-modal similarity inherent in RS data while achieving superior retrieval performance [429].

Finally, the generation of synthetic data for training robust models must adhere to strict semantic constraints to ensure utility in safety-critical scenarios. In change detection, where labeled bi-temporal samples are expensive to acquire, semantic latent diffusion models have been employed to generate large-scale, diverse datasets that respect the logical constraint that “change implies semantically different” while maintaining consistency in non-changing regions [430]. Similarly, hierarchy-oriented geo-aware datasets have been introduced to facilitate precise image-text retrieval, ensuring that generated or retrieved content aligns with detailed geospatial descriptions and hierarchical spatial sampling standards [431]. As the field progresses, the integration of these semantic-preserving mechanisms with emerging foundation models promises to enhance the reliability of remote sensing applications. However, just as in clinical settings where uncertainty quantification is paramount for trust, the deployment of these geospatial generative models necessitates rigorous validation to ensure that the preserved semantics translate into reliable decision-making, a theme that leads directly into the discussion of uncertainty and trustworthy deployment in the following section.

6.8 Uncertainty Quantification and Trustworthy Clinical Deployment

6.8 Uncertainty Quantification and Trustworthy Clinical Deployment

Building on the imperative for rigorous validation in geospatial applications discussed in Section 6.7, the deployment of deep learning models in clinical environments represents an equally high-stakes domain where prediction errors can lead to severe patient harm, misdiagnosis, or inappropriate treatment plans. While remote sensing demands geospatial semantic consistency to prevent catastrophic errors in land cover analysis, clinical AI necessitates robust mechanisms for uncertainty quantification (UQ) and explainability to ensure patient safety. The transition from research prototypes to trustworthy clinical tools relies on these mechanisms to provide a critical safety layer, enabling models to express confidence levels and facilitating human-in-the-loop decision-making when encountering ambiguous or out-of-distribution (OOD) data. Consequently, the integration of Bayesian inference and advanced probabilistic frameworks has emerged as a cornerstone for addressing these safety considerations, ensuring that AI systems can reliably identify their own limitations just as geospatial models must recognize domain shifts.

A primary challenge in clinical deployment is the detection of distribution shifts, where test data differs significantly from the training distribution due to variations in imaging protocols, scanner hardware, or unseen pathologies. Much like the sensor discrepancies in remote sensing, standard deterministic models in healthcare often fail to recognize these shifts, producing overconfident and potentially catastrophic predictions. To mitigate this, researchers have developed Bayesian approaches that quantify both epistemic (model) and aleatoric (data) uncertainties. For instance,

probabilistic Bayesian convolutional neural networks have been successfully applied to diagnose faults and detect OOD datasets in safety-critical applications, flagging samples with high predictive uncertainty for expert intervention [432]. This capability is equally vital in medical imaging, where models must distinguish between in-distribution cases and those exhibiting novel artifacts or diseases. Studies have demonstrated that methods capable of addressing multimodal characteristics in the posterior distribution offer more dependable uncertainty estimates under such distribution shifts, thereby enhancing the robustness of diagnostic predictions [433].

Several specific methodologies have been proposed to integrate uncertainty into clinical workflows effectively, mirroring the specialized frameworks developed for heterogeneous remote sensing data. Monte Carlo dropout and variational inference are widely adopted techniques that approximate posterior distributions to generate uncertainty metrics. In the context of computer-aided diagnosis for retinal conditions, both Bayesian ResNets using Monte Carlo sampling and Variational ResNets predicting posterior parameters have shown that incorrectly diagnosed cases correlate strongly with higher uncertainty, validating the use of variance as a reliability metric [434]. Similarly, in structural health monitoring, which shares safety-critical parallels with clinical settings, Bayesian inference via Monte Carlo dropout has proven effective in correlating uncertainty metrics like softmax variance and entropy with misclassifications, suggesting a pathway for triggering human review in medical inspections [435]. Furthermore, specialized architectures such as Laplacian Segmentation Networks have been introduced to jointly model spatially correlated aleatoric uncertainty and epistemic uncertainty, successfully assigning high uncertainty to OOD objects within medical images [436].

Beyond generic uncertainty estimation, there is a growing emphasis on tailoring these methods to specific clinical tasks and data modalities, akin to the domain-specific adaptations seen in ecological monitoring. For example, in proton therapy dose prediction, input reconstruction branches added to U-Net architectures have been utilized as surrogates for model uncertainty, offering faster inference times and superior OOD detection compared to traditional ensemble methods [437]. In seismic imaging and other inverse problems relevant to scientific diagnostics, deep priors combined with Bayesian model averaging have been employed to reduce overfitting and provide pointwise variance measures that highlight regions difficult to image, a technique adaptable to complex medical reconstruction tasks [438]. Additionally, the concept of “safe probability” has been formalized to restrict probabilistic inference to domains where predictions are valid, avoiding paradoxes and ensuring that models do not make unfounded claims in unfamiliar scenarios [439].

Explainability and visual attribution are inextricably linked to trustworthiness, particularly when uncertainty highlights a potential error, paralleling the need for interpretable metadata in geospatial analysis. Visual explanation methods help clinicians understand the basis of a model’s prediction, yet traditional gradient-based methods like Grad-CAM often lack robustness. Innovative approaches utilizing Information Bottleneck attribution have been developed to provide stable visual explanations for diagnosis and prognosis, even in the absence of dense segmentation labels, thereby overcoming the sensitivity issues of conventional attribution techniques [440]. Moreover, the integration of uncertainty into group analyses and quality control pipelines allows for automated assessment of segmentation fidelity. Bayesian QuickNAT, for instance, introduces structure-wise uncertainty metrics that correlate highly with manual annotation scores, enabling reliable quality control for large-scale population studies without exhaustive human review [441].

Despite these advances, challenges remain in ensuring that uncertainty estimates are well-calibrated and practically useful, echoing the validation hurdles faced in remote sensing deployment. Some studies indicate that popular uncertainty estimation methods may fail to reliably detect OOD data in classification settings due to the saturating nature of activation functions and the piece-

wise affine transformations of ReLU networks [442]. Consequently, there is a push towards more rigorous evaluation frameworks, such as the RETINA Benchmark, which assesses the reliability of predictive models in real-world, safety-critical scenarios rather than unrealistic test beds [443]. Furthermore, the distinction between visual quality and algorithmic quality in medical imaging necessitates uncertainty-aware frameworks that interpret predictive uncertainty as an estimation of achievable image recovery, ensuring that data exclusion criteria are aligned with downstream task requirements [444].

Looking forward, the convergence of conformal prediction, Bayesian deep learning, and foundation models offers promising avenues for trustworthy deployment, bridging the gap between the probabilistic rigor discussed here and the generative capabilities of modern diffusion and flow-based models. Conformal prediction has shown robust performance across diverse testing conditions, providing distribution-free guarantees that are particularly valuable for decision-making in high-risk medical applications [445]. Additionally, leveraging large language models for multi-modal OOD detection introduces new possibilities for incorporating world knowledge, provided that consistency-based calibration is employed to mitigate hallucinations [446]. Ultimately, achieving trustworthy clinical deployment requires a holistic approach that combines rigorous uncertainty quantification, interpretable visual attributions, and adaptive safety mechanisms to ensure that AI systems act as reliable partners in healthcare, setting the stage for future explorations into controllable generation where safety and fidelity are paramount.

7 Evaluation, Safety, Ethics, and Future Directions

7.1 Benchmarking Controllability, Fidelity, and Reproducibility

7.1 Benchmarking Controllability, Fidelity, and Reproducibility

The rapid advancement of diffusion and flow-based models for controllable image generation has necessitated a rigorous re-evaluation of performance benchmarking, specifically focusing on the critical triad of controllability, fidelity, and reproducibility. As these architectures transition from research prototypes to deployment in high-stakes domains, the alignment between conditional inputs—such as text prompts, layout masks, or semantic graphs—and the generated outputs has become the definitive metric of success. However, current evaluation ecosystems frequently suffer from a “Great Misalignment Problem,” where discrepancies between problem definitions, proposed methods, and human evaluation protocols lead to invalid or irreproducible conclusions [447]. This issue is further exacerbated by a reliance on static benchmarks that fail to capture the dynamic nature of real-world data distributions, creating a significant context shift between controlled test environments and production scenarios [448].

A primary challenge in benchmarking controllability lies in quantifying the degree to which a model adheres to specific constraints without compromising image quality. Traditional metrics often measure global similarity or perceptual quality but fail to assess local adherence to complex structural conditions. For instance, in tasks requiring precise spatial alignment, standard overlap metrics may not adequately reflect the clinical or practical value of a segmentation when reference annotations are uncertain, small, or empty [449]. Consequently, future benchmarks for controllable generation must incorporate dimensions that account for annotation uncertainty and the difficulty level of the conditioning signal, rather than assuming perfect ground truth. Furthermore, the evaluation of conditional generation must address the issue of spurious causality, where models inherit biases from training data, resulting in label-conditional distributions that are imbalanced with respect

to latent attributes [450]. Effective benchmarks must therefore include fairness interventions and corrective sampling metrics to ensure that enhanced controllability does not come at the cost of reinforcing societal biases.

Reproducibility remains a cornerstone of scientific progress, yet it is frequently compromised in the field of generative modeling. This crisis is often driven by undocumented preprocessing steps, missing code, and the reporting of baseline metrics that do not accurately reflect the comparison landscape [451]. In the context of diffusion and flow-based models, where stochastic sampling processes introduce inherent variability, establishing reproducible baselines is exceptionally difficult. Researchers must adopt consistency tests capable of identifying inconsistencies between reported performance scores and experimental setups, similar to tools developed for validating machine learning results in other domains [452]. Without such rigorous validation, claims regarding improved controllability or fidelity may be artifacts of experimental design rather than genuine algorithmic advancements. Moreover, ambiguity in rank-based evaluations can lead to misleading conclusions, as varying test set sizes automatically impact performance metrics, making cross-dataset comparisons unreliable [453]. To mitigate this, benchmarks must standardize test conditions and provide clear guidelines for statistical significance testing.

While the fidelity of generated images is often assessed through perceptual metrics, a more nuanced approach is required—one that considers the utility of the output for specific downstream tasks. The concept of “goal-oriented” evaluation suggests that metrics should directly quantify the impact of semantic mismatches on decision-making processes, rather than relying on proxy measures like Age of Information or Mean Square Error [454]. In controllable generation, this implies that fidelity should be measured by how well the generated image supports the intended application, whether it be medical diagnosis, material design, or creative composition. Additionally, the uncertainty inherent in generative models must be explicitly evaluated. Metrics that reward high confidence in correct assertions and penalize over-confidence in errors are essential for trustworthy deployment [455]. Visualizing the distributions of performance metrics under different models of uncertainty can reveal that the variance in empirical estimates often eclipses the differences between competing models, urging caution in claiming state-of-the-art status based on point estimates alone [456].

Furthermore, the evaluation of controllability must extend beyond simple alignment to include the robustness of the model against distribution shifts and adversarial perturbations. Benchmarks that rely on independent and identically distributed (IID) data fail to expose the brittleness of models when faced with out-of-distribution inputs [457]. Dynamic benchmarking strategies, which focus on capturing the underlying data generation process, are necessary to promote generalizability and reduce context shift errors [448]. Finally, the integration of causal reasoning into evaluation frameworks can help distinguish between genuine controllability and spurious correlations. By leveraging counterfactual data augmentation and contrastive learning, researchers can create more robust benchmarks that assess a model’s ability to generate consistent outputs under varying causal conditions [458]. In summary, the future of benchmarking in controllable image generation lies in developing comprehensive, reproducible, and context-aware evaluation protocols that prioritize alignment, fairness, and practical utility over superficial performance gains. These rigorous standards are not only vital for scientific integrity but also form the foundational baseline upon which mechanisms for attribution, fingerprinting, and copyright protection—discussed in the following section—must be built and verified.

7.2 Attribution, Fingerprinting, and Copyright Protection Mechanisms

7.2 Attribution, Fingerprinting, and Copyright Protection Mechanisms

Building upon the rigorous evaluation protocols established in the previous section, which emphasize reproducibility, fairness, and context-aware benchmarking, the deployment of high-fidelity diffusion and flow-based models necessitates equally robust mechanisms for attribution, intellectual property (IP) protection, and copyright resolution. As these architectures achieve a level of synthesis indistinguishable from natural data, the risks of malicious misuse, unauthorized redistribution, and legal ambiguity regarding ownership have escalated. Consequently, the research community has developed a diverse array of technical strategies ranging from latent space manipulation and adversarial training to data-centric provenance tracking. These approaches aim to embed unique identifiers within models or their outputs without compromising generation quality, thereby establishing a verifiable chain of custody for synthetic media that aligns with the trust standards required for high-stakes applications.

A primary avenue for IP protection involves the direct embedding of fingerprints into the generative model’s parameters or its latent semantic dimensions. Traditional watermarking methods often require invasive modifications to the training process, which can degrade model utility or introduce security vulnerabilities. In contrast, recent advancements propose leveraging the inherent capacity of deep neural networks to hide information within weight distributions or latent representations. For instance, frameworks have been developed to embed unique fingerprints into the probability density function of trainable weights, ensuring robustness against model compression and fine-tuning attacks [459]. Similarly, other approaches utilize latent semantic dimensions as fingerprints, analyzing design variables such as dimension choice and strength to optimize the trade-off between attribution accuracy and generation quality [460]. This non-invasive philosophy is further extended by methods that extract intrinsic features via explainable AI techniques, such as Deep Taylor Decomposition, to verify ownership based on the model’s decision logic rather than embedded triggers [461].

To address the specific challenges posed by user-specific attribution in distributed environments, novel frameworks have emerged that assign responsibility to individual users rather than just the model owner. The WOUAF framework, for example, modifies text-to-image diffusion models based on a user’s unique digital fingerprint, imprinting an identifier onto generated content that facilitates traceability with minimal impact on output quality [462]. Complementary to this, watermark-conditioned diffusion models manipulate the watermark as a conditioned input during the generation process, allowing for the recovery of user-specific information from black-box API queries [463]. These user-centric approaches are crucial for enforcing license compliance and mitigating the risks associated with the decentralized deployment of powerful generative tools, ensuring that accountability persists even when models are accessed via remote interfaces.

Beyond parameter and latent space modifications, a paradigm shift towards data-centric fingerprinting has gained traction. This strategy involves embedding artificial fingerprints directly into the training data, which are then inadvertently learned and reproduced by the generative model in its outputs. Research has demonstrated that such artificial fingerprints transfer effectively from training data to deepfakes generated by various architectures, including GANs and diffusion models, providing a proactive solution that remains robust against model evolution [464]. This method effectively closes the responsibility loop by linking generated content back to the specific data sources

used for training, thereby simplifying the attribution process. Furthermore, for large language models, watermarking frameworks like WASA enable the mapping of text from different data providers to unique watermarks, solving both source attribution and data provenance problems simultaneously [465].

The landscape of attribution also encompasses the detection and classification of synthetic content through the analysis of model-specific artifacts. Studies have shown that generative models, regardless of their architecture, leave distinct “fingerprints” in the form of residual artifacts or frequency patterns that can be learned by discriminative classifiers [466]. Advanced detection systems employ set-based artificial fingerprints and contrastive training to learn robust representations of these artifacts, enabling not only source attribution but also the unsupervised clustering of generative models into latent families [467]. Moreover, systematic studies on text-to-image models have confirmed that fake images can be effectively attributed to their source models, such as Stable Diffusion or DALL-E 2, based on these unique signatures [468].

However, the efficacy of these mechanisms is constantly challenged by adaptive adversaries capable of removing watermarks or obfuscating fingerprints. To counteract this, researchers have developed adversarially robust attribution methods, such as ARIA, which utilize robust contrastive learning to maintain accuracy even under imperceptible input perturbations [469]. Additionally, testing frameworks like DEEPJUDGE have been proposed to quantitatively assess similarities between victim and suspect models using diverse metrics, offering a non-invasive and flexible approach to copyright verification that is resilient to model extraction and adaptive attacks [470]. Despite these advancements, theoretical analyses reveal limitations in existing dataset inference techniques, highlighting the potential for false positives and negatives when adversaries employ regularization or adversarial training [471].

Finally, the intersection of fingerprinting and privacy presents a complex trade-off that serves as a critical bridge to the subsequent discussion on privacy-preserving synthesis. While generative models offer potential for privacy-preserving data sharing, there is a risk that identifiable training samples could be reproduced, creating a tension between the need for strong attribution and the imperative of data anonymity. Recent work introduces adversarial approaches to estimate the upper bound of reproducing identifiable images by searching for anatomic fingerprints, underscoring the necessity for careful model training to prevent privacy breaches [472]. As the field evolves, future research must continue to balance the robustness of attribution mechanisms with the preservation of data utility and individual privacy. This delicate equilibrium ensures that intellectual property rights are protected without stifling innovation, setting the stage for the exploration of differential privacy mechanisms and secure synthetic data generation in the following section.

7.3 Privacy-Preserving Synthesis and Differential Privacy Trade-offs

7.3 Privacy-Preserving Synthesis and Differential Privacy Mechanisms

While the preceding section addressed mechanisms for attributing ownership and protecting intellectual property, the deployment of generative models introduces an equally critical tension between data utility and individual privacy. The proliferation of diffusion and flow-based architectures has revolutionized synthetic data generation, offering a potent solution for sharing sensitive information in domains ranging from healthcare to finance. However, the assumption that synthetic data is inherently anonymous simply because it does not contain real records is dangerously flawed. Without rigorous mathematical guarantees, generative models can memorize training examples, leading to the reconstruction of sensitive attributes or the inference of individual membership in the training

set. Consequently, the integration of Differential Privacy (DP) into diffusion and flow-based models has become a focal point of research, aiming to provide provable privacy bounds while navigating the inevitable trade-offs in data quality.

The fundamental challenge lies in the fact that standard generative models are prone to memorizing details of individual training records, which can result in synthetic outputs that too closely resemble sensitive source data [473]. To mitigate this, researchers have adapted Differentially Private Stochastic Gradient Descent (DP-SGD) for training diffusion models, resulting in frameworks like Differentially Private Diffusion Models (DPDMs). These approaches enforce privacy by clipping gradients and injecting noise during the training process, ensuring that the model’s parameters do not reveal information about any single data point. Recent advancements have demonstrated that by carefully tuning the diffusion parameterization and sampling algorithms, it is possible to achieve state-of-the-art image generation quality even under strict privacy constraints [474]. Furthermore, studies have shown that classifiers trained on such privately generated synthetic data can perform on par with those trained using task-specific DP-SGD, a milestone that validates the utility of DP diffusion models for downstream machine learning tasks [475].

Despite these successes, the privacy-utility trade-off remains a complex, multi-dimensional problem governed by the “privacy budget,” denoted by ϵ . Lower values of ϵ offer stronger protection but typically degrade the statistical accuracy of the synthetic data. A significant body of work focuses on optimizing how this budget is allocated. For instance, naive allocation strategies often distribute the privacy budget evenly across all features, which can undervalue critical conditional probabilities and disproportionately harm the predictive quality for minority groups [476]. To address this, novel ensemble methods have been proposed that allocate the privacy budget “wisely” based on feature importance and “fairly” to bound disparities across demographic groups, thereby maintaining high predictive accuracy while adhering to fairness constraints [476]. Additionally, the dimensionality of the data plays a crucial role; deep generative models tend to spend their budget per iteration, making their behavior less predictable with varying dataset dimensions compared to graphical models which distribute the budget horizontally [477].

Beyond the training phase, the evaluation of privacy risks in synthetic data requires robust auditing frameworks that go beyond empirical similarity metrics. Relying solely on visual or statistical similarity between real and synthetic data is insufficient and can be misleading, as high similarity can sometimes indicate severe privacy leakage rather than fidelity. Research has demonstrated that reconstruction attacks can successfully recover attributes of training records even from models that appear to produce anonymous data, highlighting the inadequacy of similarity-based privacy metrics [478]. To counter this, comprehensive auditing frameworks like Anonymeter have been developed to quantify specific privacy risks such as singling out, linkability, and inference attacks, providing a legally aligned assessment of factual anonymization [479]. Moreover, visual privacy auditing using diffusion models themselves has emerged as a powerful tool, leveraging real-world image priors to assess reconstruction success and visualize privacy leakage under DP-SGD [480].

Theoretical explorations have also extended to discrete domains and specific data structures to broaden the applicability of these protections. For discrete denoising diffusion models, pioneering work has established mathematical characterizations of per-instance differential privacy, revealing how privacy leakage correlates with dataset distribution and diffusion coefficients [481]. In the realm of tabular data, which is prevalent in regulated industries, frameworks such as DP-TLDM (Differentially Private Tabular Latent Diffusion Model) have been introduced to balance utility and privacy by encoding data into a latent space before applying diffusion, significantly improving data resemblance and discriminability compared to other DP-protected generators [482]. Furthermore,

the challenge of missing data in privacy-preserving synthesis has been addressed through adaptive strategies that model missing mechanisms as sampling processes, ensuring tighter privacy bounds for the ground truth data [483].

Looking forward, the field is moving towards more nuanced definitions of privacy that account for temporal correlations and evolving datasets. Traditional DP assumes independent records, but real-world data often exhibits temporal dependencies that can amplify privacy leakage over time. Concepts such as “discounted differential privacy” have been proposed to handle infinite horizons by weighting past privacy losses less heavily, allowing for bounded noise magnitude in streaming scenarios [484]. Additionally, the integration of causal information into the training process has shown promise in strengthening differential privacy guarantees while improving resilience against membership inference attacks [485]. As the community advances, the focus must remain on developing holistic auditing methodologies that simultaneously evaluate bias, fidelity, utility, and robustness, ensuring that synthetic data solutions are not only mathematically private but also trustworthy and equitable for deployment in high-stakes regulated domains [267]. The ultimate goal is to transcend the zero-sum game of privacy versus utility, creating generative frameworks where strong privacy guarantees enhance, rather than hinder, the reliability and fairness of data-driven insights. This foundation of trusted, private generation sets the stage for the subsequent discussion on detecting malicious forgeries and ensuring adversarial robustness in the face of increasingly sophisticated synthetic media.

7.4 Deepfake Detection, Forgery Localization, and Adversarial Robustness

7.4 Deepfake Detection, Forgery Localization, and Adversarial Robustness

Building upon the foundation of trusted, private generation established in the preceding section, the deployment of diffusion and flow-based models introduces a parallel challenge: the potential for malicious exploitation. While differential privacy safeguards individual data points, the unprecedented realism of these models has eroded the reliability of “seeing is believing,” igniting an intense arms race between high-fidelity synthesis and forensic countermeasures. This section addresses the critical necessity for sophisticated detection frameworks capable of identifying subtle forgery clues while resisting adversarial evasion, serving as a bridge between the technical guarantees of privacy and the societal imperatives of fairness discussed subsequently.

A primary bottleneck in current deepfake detection is the tendency of models to overfit to specific artifact patterns or local regions present in training data, severely limiting their generalizability to unseen generation methods. To address this, recent research has shifted towards extracting broader, more robust forgery clues. For instance, novel frameworks have been proposed that capture multiple non-overlapping local representations and fuse them into global semantic-rich features, utilizing information bottleneck theory to ensure the orthogonality of these representations while preserving task-relevant information [486]. Furthermore, the disentanglement of image information into forgery-irrelevant, method-specific, and common forgery features has emerged as a powerful strategy to uncover universal traces that transcend specific generation algorithms [487].

Beyond binary classification, a significant frontier is forgery localization, which aims to pinpoint the exact spatial regions manipulated within an image or video. This granular analysis is crucial for providing explainable evidence of tampering, yet localization models remain particularly vulnerable to attacks designed to mislead pixel-level decisions. Traditional adversarial attacks often fail in this context because they merely induce decision flips without effectively concealing underlying tampering traces across diverse forensic models. In response, self-supervised anti-forensic techniques

have been developed to actively deceive forgery localization systems by reconstructing perturbations that erase tampering features, thereby exposing the fragility of current detectors against adaptive adversaries [488]. The challenge is compounded by the fact that many detectors rely heavily on a few dominant forgery regions, overlooking other equally important cues. Innovative two-stream networks have been introduced to enlarge the potential regions for evidence extraction, collaborating with semi-supervised learning strategies to estimate patch-level annotations even in the absence of exhaustive ground truth [489].

The robustness of these detection systems is further tested by an evolving landscape of adversarial perturbations. Adversaries are increasingly employing attribute-variation-based attacks that perturb the latent space of generative models, introducing inconspicuous semantic changes that bypass detection while remaining imperceptible to human observers [490]. Moreover, the emergence of user-customized generative models and vision foundation models has expanded the threat surface, allowing attackers to craft adversarial deepfakes through semantic manipulation rather than adding noticeable noise, a technique that renders many existing content-agnostic defenses ineffective [491]. These vulnerabilities extend beyond visual data; audio and multimodal systems face similar risks where heterogeneous spectro-temporal artifacts must be modeled efficiently to detect spoofing without relying on computationally demanding ensembles [492].

To counter these evolving threats, the community is exploring proactive defense mechanisms and robust training paradigms. One promising direction involves the integration of artificial fingerprints directly into training data, which transfer to the generative models and persist in the output, converting the complex problem of detection into a more tractable attribution task that is agnostic to the evolution of generation techniques [464]. Additionally, adversarial training strategies are being refined to enhance generalization. By dynamically synthesizing the most challenging forgeries during training, models can learn to be sensitive to a diverse spectrum of manipulation types, significantly improving performance against unseen methods [493]. Other approaches leverage latent space augmentation to simulate variations within and across forgery features, smoothing the decision boundaries and mitigating overfitting to method-specific artifacts [494].

Despite these advancements, significant gaps remain. Many state-of-the-art detectors still struggle when faced with decision-based black-box attacks that exploit frequency cues or cross-task correlations to generate high-quality adversarial examples that pass both detection and recognition systems [495]. Furthermore, the reliance on large labeled datasets for training limits the scalability of supervised approaches, prompting interest in few-shot detection methods that can adapt to novel forgery classes by interpolating artifacts between known and unknown distributions [496]. The integration of large language models (LLMs) and multimodal large language models (MLLMs) offers a new paradigm for auditing, where semantic reasoning can be applied to identify logical inconsistencies in forged content, although current evaluations suggest these models still struggle with highly sophisticated, realistic forgeries [497; 498]. Ultimately, achieving true adversarial robustness requires a holistic approach that combines advanced feature extraction, semantic specification auditing, and proactive defense mechanisms. While these technical measures secure the integrity of synthetic media, they must be complemented by the ethical frameworks and responsible curation practices discussed in the following section to ensure that generative technologies serve society equitably.

7.5 Mitigating Bias, Ensuring Fairness, and Responsible Dataset Curation

7.5 Mitigating Bias, Ensuring Fairness, and Responsible Dataset Curation

While the preceding section detailed the technical arms race against malicious forgeries and adversarial attacks, an equally critical dimension of trustworthy generative AI lies in addressing the systemic societal biases embedded within diffusion and flow-based models. The proliferation of these models for controllable image generation has precipitated a rigorous examination of how historical inequalities and underrepresentation in training data are not merely preserved but often amplified by generative algorithms, leading to discriminatory outcomes and the reinforcement of harmful stereotypes [499]. Consequently, the research focus has expanded from maximizing fidelity and robustness to developing comprehensive methodologies for ensuring fairness and responsible dataset curation throughout the entire generative lifecycle.

A primary strategy in this domain involves leveraging synthetic data not just for augmentation, but as a mechanism for “Fairness by Design.” Traditional curation of real-world datasets is often labor-intensive and insufficient to correct deep-seated demographic imbalances. In contrast, synthetic data allows for the controlled simulation of unlimited training samples tailored to specific fairness constraints. Recent frameworks incorporate fairness directly into the self-supervised learning process, enabling the generation of data where gender and racial biases are explicitly controlled while retaining faithful attribute relationships [500]. This paradigm empowers data consumers to adjust the trade-off between accuracy and fairness, ensuring that downstream models provide equitable predictions even when source data is heavily biased [501]. Furthermore, generative models are increasingly utilized to address class imbalance problems, with methods like SYNAuG exploiting synthetic generation to equalize unbalanced distributions, thereby reducing biased predictions in downstream tasks [502].

However, the reliance on generative models for bias mitigation introduces a paradox: these models can inadvertently learn and replicate the very biases present in their training distributions. Studies indicate that architectures such as Generative Adversarial Networks (GANs) can exhibit biased behaviors, such as over-representing specific demographic groups, which leads to disparate impacts when used for fine-tuning verification systems [503]. To counteract this, novel architectures like Bias-transforming Generative Adversarial Networks (Bt-GAN) have been introduced. These models employ information-constrained data generation processes and score-based weighted sampling to incentivize the generator to preserve sub-group densities and learn fair deterministic transformations, significantly improving fairness in sensitive applications like healthcare [504]. Similarly, architecture-agnostic pipelines have been proposed that utilize pre-processing algorithms to identify and remove bias-inducing samples before generation, ensuring the model focuses on informative rather than stereotypical data patterns [505].

Beyond architectural innovations, responsible dataset curation demands rigorous auditing and transparency frameworks. The complexity of bias necessitates holistic auditing mechanisms that evaluate synthetic datasets across multiple dimensions, including fidelity, utility, robustness, and privacy preservation. Frameworks proposing a “trustworthiness index” allow for the ranking of datasets based on their safeguard trade-offs, facilitating collaboration among data scientists, governance experts, and regulators [267]. Moreover, the concept of “dataset multiplicity” highlights how inaccuracies and social biases in training sets impact test-time predictions, urging researchers to consider counterfactual scenarios where unbiased versions of datasets are analyzed to understand

the stability of model outcomes [506]. To support these efforts, comprehensive benchmarks like the Chinese Bias Benchmark (CBBQ) provide standardized metrics for detecting biases across various social dimensions, enabling the community to measure progress and identify gaps in current models [507].

Responsible curation also extends to the documentation of the data development lifecycle, drawing from software engineering best practices to enhance transparency. Rigorous frameworks have been introduced to document deliberation processes, stakeholder perspectives, and bias measurement strategies employed during dataset creation, aiming to close the accountability gap by making visible the often-overlooked labor and decisions involved in dataset construction [508]. Additionally, methodologies for responsible data collection emphasize iterative analysis using granular metrics to assess internal reliability and external stability, ensuring that datasets remain representative over time and across diverse domains [509].

Finally, addressing bias requires a multi-faceted approach that includes correcting evaluation metrics and implementing post-processing interventions. Biases can permeate even the metrics used to evaluate generative models; for instance, model-based evaluation metrics for image captioning have been found to encode gender stereotypes, favoring captions that align with traditional roles. Mitigating these metric biases is essential to prevent the propagation of stereotypes through reinforcement learning loops [510]. Furthermore, modular approaches utilizing attribute-removal sub-networks allow for on-demand bias mitigation at inference time, offering flexibility to debias specific attributes without retraining the entire model [511]. By integrating these diverse strategies—from fair synthetic data generation and architectural innovations to rigorous auditing and transparent documentation—the field moves toward generative models that uphold ethical standards. This foundation of internal fairness and responsible curation is a prerequisite for the broader societal governance frameworks discussed in the following section, which address the regulatory and ethical structures necessary to manage these powerful technologies at scale.

7.6 Dual Governance Frameworks: Regulation, Ethics, and Decentralized Safety

7.6 Dual Governance Frameworks: Regulation, Ethics, and Decentralized Safety

Building upon the foundation of internal fairness and responsible dataset curation established in the preceding section, the scale and societal impact of generative AI necessitate a transition from technical mitigations to comprehensive external oversight. While Section 7.5 detailed strategies for correcting biases embedded within training data and model architectures, these internal adjustments alone are insufficient to address the broader landscape of ethical and safety concerns ranging from privacy violations to intellectual property theft. As these systems increasingly influence human creativity and livelihoods, the inadequacy of singular regulatory approaches has become apparent. Centralized government regulations, while necessary for establishing baseline legal standards, often suffer from a lack of clarity, uniformity, and interoperability across different jurisdictions, potentially stifling innovation and hindering free market competition. Conversely, purely decentralized protections reliant on crowdsourced safety tools frequently lack the necessary oversight and enforcement mechanisms to ensure consistent adherence to ethical standards. To address these systemic limitations and operationalize the fairness goals discussed previously, the concept of “Dual Governance” has emerged as a critical framework, proposing a synergistic marriage between centralized statutory regulations and community-driven safety mechanisms. This

approach posits that innovation and creativity can be robustly promoted only when paired with a cooperative structure that leverages the enforceability of state power and the agility of decentralized community oversight [512].

The theoretical underpinnings of such a framework require a nuanced understanding of the relationship between centralized authority and distributed autonomy. A purely top-down approach often fails to keep pace with the rapid evolution of AI technologies, whereas bottom-up mechanisms may lack the teeth to penalize bad actors effectively. The Dual Governance model seeks to bridge this gap by defining clear roles where centralized regulations set the macro-level boundaries and liability structures, while decentralized mechanisms provide real-time monitoring, auditing, and adaptive safety filters. This aligns with broader discussions on evolving data governance toward information governance, where the concept of “Autonomous Principals” capable of exercising transactional sovereignty is introduced to manage digital selves within a legitimate authority structure [513]. By treating diverse stakeholders as autonomous entities within a governed ecosystem, the framework facilitates a dynamic interplay between legal mandates and technical implementations, ensuring that the fair data practices from the previous section are upheld at a systemic level.

A significant challenge in implementing dual governance lies in the articulation of ethical charters, legal tools, and technical documentation. These three domains are often treated in isolation, yet their interdependence is crucial for creating a comprehensive accountability framework. Ethical charters provide the normative values, legal tools offer the enforcement mechanisms, and technical documentation ensures the verifiability of compliance. Stronger Together on the Articulation of Ethical Charters, Legal Tools, and Technical Documentation in ML emphasizes that leveraging complementary notions of compliance across these fields allows for a more robust governance structure that accounts for technical capabilities, societal impacts, and regulatory requirements. Furthermore, the integration of ordoliberal principles suggests that hard-law legislation must be merged with competition policy and existing AI guidelines to form a “Digital Bill of Rights,” ensuring that governance mechanisms are not merely voluntary codes of conduct but are embedded within the market structure itself [514].

In the context of autonomous agents, which are increasingly making moral and legal decisions in domains such as healthcare and transportation, the Dual Governance framework must extend to the agent level to ensure these systems act as responsible actors within the defined ecosystem. The concept of “Reasonable Machines” argues that future intelligent autonomous systems must be capable of justifying their actions through normative communication, thereby creating an ecosystem of trust [515]. This requires hybrid reasoning tools and ethico-legal ontologies that allow machines to engage in social reasoning, effectively internalizing the dual governance structure. Similarly, the development of “Ethical Layers” in robot controllers demonstrates how predetermined ethical rules can be computationally enforced to prevent harm, serving as a technical instantiation of decentralized safety within a regulated environment [516]. These agent-level controls serve as the bridge between the high-level policy discussions in this section and the technical alignment strategies required for future autonomous systems.

The practical deployment of dual governance also necessitates innovative mechanisms for verification and enforcement in decentralized settings. For instance, blockchain-based approaches offer potential solutions for validating the behavior of autonomous agents through smart contracts and reputation models, although challenges regarding scalability and the definition of sufficient decentralization remain [517; 518]. The risk of “Dark DAOs” and systemic bribery highlights the need for rigorous metrics, such as Voting-Bloc Entropy, to assess the true decentralization and integrity of these governance structures [519]. Moreover, the application of process mining can provide fact-based

visibility into compliance processes, surfacing bottlenecks and enabling automated remediation of regulatory gaps, thus strengthening the centralized oversight component of the dual framework [520].

Ultimately, the success of Dual Governance frameworks depends on the ability to balance the rigidity of legal compliance with the flexibility of community-driven adaptation. This involves creating actionable principles that can be operationalized through multi-stakeholder participation and cross-sectoral feedback [521]. It also requires a commitment to transparency, where voluntary safety commitments can serve as an escape from over-regulation while still ensuring socially beneficial outcomes [522]. By synthesizing centralized regulatory compliance with decentralized, crowdsourced safety mechanisms, the Dual Governance approach offers a promising pathway to navigate the ethical complexities of generative AI. This regulatory and ethical scaffolding provides the essential environment within which the next generation of technologies—specifically autonomous agents, unified world models, and explainable systems discussed in the following section—must operate to ensure they remain aligned with human values and societal well-being.

7.7 Future Frontiers: Autonomous Agents, Unified World Models, and Explainability

7.7 Future Frontiers: Autonomous Agents, Unified World Models, and Explainability

Building upon the Dual Governance frameworks established in the preceding section, the trajectory of controllable image generation is rapidly converging with the broader evolution of artificial intelligence toward autonomous agency and comprehensive world modeling. While Section 7.6 outlined the regulatory and ethical scaffolding necessary to govern these systems, this section explores the technical and cognitive frontiers that will define their operational reality. As generative models transition from static content creation tools to dynamic components of interactive systems, the next frontier of research must address the complex interplay between agent autonomy, environmental dynamics, and human alignment. The future of this field resides not merely in improving pixel-level fidelity but in achieving deep integration and embodiment, where AI systems can operate within ecologically valid conditions and engage in a cognitive arms race that bootstraps increasingly complex skills [523]. This shift necessitates a move from isolated generation tasks to unified frameworks where agents leverage foundation models to reason, make decisions, and interact with unpredictable real-world environments, effectively operationalizing the governance principles discussed previously.

A critical component of this future landscape is the development of unified alignment strategies for autonomous agents, which serve as the technical instantiation of the ethical charters mentioned earlier. Current research indicates that aligning agents solely with human intent is insufficient; instead, a holistic approach known as Unified Alignment for Agents (UA^2) is required. This paradigm advocates for the simultaneous alignment of agents with human intentions, environmental dynamics, and self-imposed constraints such as computational or monetary budgets [524]. In the context of controllable generation, this implies that future diffusion-based agents must not only adhere to user prompts but also dynamically adapt to changing scene physics and operational limits. Furthermore, the balance between autonomy and alignment remains a pivotal challenge, particularly in multi-agent architectures where specialized agents must collaborate to decompose and execute complex goals [525]. Future systems will likely employ interactive “alignment dialogues,” allowing users and agents to negotiate goals and maintain alignment through continuous feedback rather than static reward modeling [526], thereby creating a flexible mechanism for enforcing the decentralized safety

norms envisioned in dual governance models.

Parallel to agent alignment is the urgent need for robust, trustworthy world models that can serve as the backbone for autonomous decision-making and verify compliance with safety constraints. The limitations of current black-box architectures in safety-critical domains highlight the necessity for systems that can learn reliable world models, reason about causal relationships, and plan complex action sequences [527]. Future research must prioritize the creation of probabilistic world models capable of forecasting long-horizon trajectories and assessing agent competency with high precision [528]. These models must be designed to handle the intrinsic and extrinsic complexities of open-world domains, effectively managing out-of-distribution elements that are absent from controlled training simulations [529]. The integration of such world models with generative priors will enable agents to simulate potential futures, evaluate risks, and execute actions with a degree of predictability that is currently lacking in purely data-driven approaches [530], thus providing the technical substrate for the “Reasonable Machines” concept introduced in the governance discussion.

Explainability stands as the third pillar of this future roadmap, acting as the bridge between opaque model internals and the transparency required for accountability. The opacity of deep generative models poses significant barriers to trust and adoption, necessitating a shift toward systems that possess explicit declarative knowledge representations [531]. Future methodologies must move beyond post-hoc explanations to incorporate “local-first” strategies that aggregate local decision rules into global, interpretable models, ensuring that high accuracy does not come at the cost of comprehensibility [532]. In the realm of autonomous robotics and embodied agents, explainability must extend to communicating perceptual functions and cognitive reasoning processes, such as beliefs, desires, and intentions, to human collaborators [533]. This transparency is essential for calibrating trust in multi-robot systems and ensuring that shared mental models between humans and machines remain accurate during complex operations [534], directly supporting the verification mechanisms central to the dual governance framework.

Moreover, the development of these advanced systems must be grounded in a rigorous framework for responsible AI that addresses fairness, privacy, and accountability throughout the entire lifecycle, reinforcing the legal and ethical dimensions previously discussed. The emergence of generative AI has intensified concerns regarding data provenance and the potential for misuse, calling for comprehensive metrics catalogues that operationalize accountability through process, resource, and product metrics [535]. Future research must also grapple with the preservation of human agency, arguing that safe AI systems should be optimized to prevent the depletion of human autonomy rather than simply aligning with transient human intents [536]. This involves establishing “agency foundations” that include benevolent game theory and mechanistic interpretability to protect against the subtle erosion of human control, ensuring that the “Autonomous Principals” described in governance literature remain truly sovereign.

Finally, the realization of these frontiers will require a paradigm shift in how we evaluate and govern AI systems, closing the loop between technical capability and societal oversight. The community must move toward pluralistic alignment that serves diverse human values rather than a single monolithic objective [537]. This includes developing benchmarks that assess trade-offs and model behavior across different cultural and ethical contexts. As we approach the potential threshold where a “Manhattan Project” effort could accelerate the arrival of AGI, the research community must proactively formulate a “roadmap to the roadmap” to assess our proximity to such transformative capabilities and manage the associated geopolitical and safety risks [538]. By integrating unified alignment, trustworthy world modeling, and deep explainability, the next generation of controllable generative models will evolve from passive tools into collaborative partners capable

of navigating the complexities of the real world while adhering to the strict ethical and safety standards demanded by a dual governance ecosystem.

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